

APPLICATION OF DERIVATIVE

JEE EXERCISES

Exercise J-I

TANGENT AND NORMAL AT A POINT

1. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is :

यदि वक्र $x^2y = c^3$ की चर स्पर्श रेखा, x - अक्ष और y -अक्ष से क्रमशः a एवं b अन्तःखण्ड काटती है तब a^2b का मान है :

- (A) $27c^3$ (B) $\frac{4}{27}c^3$ (C) $\frac{27}{4}c^3$ (D) $\frac{4}{9}c^3$

Ans. C

Sol. $x^2y = c^3$

$$x^2 \frac{dy}{dx} + 2xy = 0 \Rightarrow \frac{dy}{dx} = -\frac{2y}{x}$$

equation of tangent at (x, y)

$$Y - y = -\frac{2y}{x}(X - x)$$

$$Y = 0, \text{ gives } X = \frac{3x}{2} = a$$

$$\text{and } X = 0, \text{ gives } Y = 3y = b$$

$$\text{Now } a^2b = \frac{9x^2}{4} \cdot 3y = \frac{27}{4}x^2y = \frac{27}{4}c^3 \Rightarrow (C)$$

2. At the point $P(a, a^n)$ on the graph of $y = x^n$ ($n \in \mathbb{N}$) in the first quadrant a normal is drawn. The normal intersects the y -axis at the point $(0, b)$. If $\lim_{a \rightarrow 0} b = \frac{1}{2}$, then n equals :

प्रथम चतुर्थांश में वक्र $y = x^n$ ($n \in \mathbb{N}$) पर बिन्दु $P(a, a^n)$ पर अभिलम्ब, y -अक्ष को बिन्दु $(0, b)$ पर प्रतिच्छेद करता है। यदि

$$\lim_{a \rightarrow 0} b = \frac{1}{2} \text{ तब } n \text{ बराबर :}$$

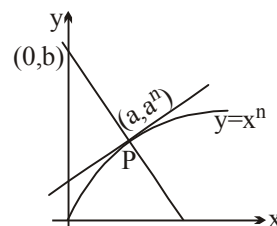
- (A) 1 (B) 3 (C) 2 (D) 4

Ans. C

Sol. $y = x^n$

$$\frac{dy}{dx} = n x^{n-1} = n a^{n-1}$$

$$\text{slope of normal} = -\frac{1}{n a^{n-1}}$$



equation of normal $y - a^n = -\frac{1}{na^{n-1}} (x - a)$

put $x = 0$ to get y-intercept

$$y = a^n + \frac{1}{na^{n-2}}; \quad \text{Hence } b = a^n + \frac{1}{na^{n-2}}$$

$$\lim_{a \rightarrow 0} b = \begin{cases} 0 & \text{if } n < 2 \\ \frac{1}{2} & \text{if } n = 2 \\ \infty & \text{if } n > 2 \end{cases} \Rightarrow (A)$$

3. The normal to the curve $x = a(1 + \cos \theta)$, $y = a \sin \theta$ at $P(\theta)$ always passes through the fixed point :

वक्र $x = a(1 + \cos \theta)$, $y = a \sin \theta$ के बिन्दु $P(\theta)$ पर अभिलम्ब सदैव अचर बिन्दु से होकर जाता है, वह बिन्दु है :

- (A) $(a, 0)$ (B) $(0, a)$ (C) $(0, 0)$ (D) (a, a)

Ans. A

Sol. $x = a(1 + \cos \theta)$, $y = a \sin \theta$

$$\frac{dx}{d\theta} = -a \sin \theta; \quad \frac{dy}{d\theta} = a \cos \theta$$

$$\left(\frac{dy}{dx} \right) = -\frac{\cos \theta}{\sin \theta} \text{ slope of normal} = -\left(\frac{dx}{dy} \right) = \frac{\sin \theta}{\cos \theta}$$

$$y - a \sin \theta = \frac{\sin \theta}{\cos \theta} (x - a - a \cos \theta)$$

$$y \cos \theta - a \sin \theta \cos \theta = x(\sin \theta) - a \sin \theta (1 + \cos \theta)$$

$$x \sin \theta - y \cos \theta = a \sin \theta (1 + \cos \theta - \cos \theta)$$

clearly passes through $(a, 0)$

4. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at $(1, 1)$:

[JEE(Main)2015]

- (A) does not meet the curve again (B) meets the curve again in the second quadrant
(C) meets the curve again in the third quadrant (D) meets the curve again in the fourth quadrant

वक्र $x^2 + 2xy - 3y^2 = 0$ के बिन्दु $(1, 1)$ पर अभिलम्ब :

[JEE(Main)2015]

- (A) वक्र को दोबारा नहीं मिलता (B) वक्र को दोबारा द्वितीय चतुर्थांश में मिलता है
(C) वक्र को दोबारा तृतीय चतुर्थांश में मिलता है (D) वक्र को दोबारा चतुर्थ चतुर्थांश में मिलता है

Ans. D

Sol. $(x + 3y)(x - y) = 0$

Equation of normal at $(1, 1)$ $x + y = 2$
will meet the curve again at $(3, -1)$

5. Consider $f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $x \in \left(0, \frac{\pi}{2}\right)$. A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point :

[JEE(Main)2016]

$f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $x \in \left(0, \frac{\pi}{2}\right)$ पर विचार कीजिए। $y = f(x)$ के बिन्दु $x = \frac{\pi}{6}$ पर खींचा गया अभिलम्ब निम्न बिन्दु से भी

होकर जाता है :

[JEE(Main)2016]

- (A) $\left(\frac{\pi}{6}, 0\right)$ (B) $\left(\frac{\pi}{4}, 0\right)$ (C) $(0, 0)$ (D) $\left(0, \frac{2\pi}{3}\right)$

Ans. D

Sol. $f(x) = \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right)$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right)$$

$$f(x) = \frac{\pi}{4} + \frac{x}{2}$$

$$f'(x) = \frac{1}{2}$$

$$\text{Eq. of normal } y - \frac{\pi}{3} = -2\left(x - \frac{\pi}{6}\right)$$

$$2x + y = \frac{2\pi}{3}$$

6. The normal to the curve $y(x-2)(x-3) = x+6$ at the point where the curve intersects the y-axis passes through the point :

[JEE(Main)2017]

वक्र $y(x-2)(x-3) = x+6$ के उस बिन्दु पर, जहाँ वक्र y-अक्ष को काटती है, खींचा गया अभिलम्ब निम्न में से किस बिन्दु से होकर जाता है :

[JEE(Main)2017]

- (A) $\left(\frac{1}{2}, -\frac{1}{3}\right)$ (B) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (C) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{1}{2}\right)$

Ans. D

Sol. Point (0, 1)

$$y(x-2)(x-3) = x+6$$

$$\left(\frac{dy}{dx}\right)(x-2)(x-3) + (2x-5)y = 1$$

Slope of tangent at (0, 1) = 1

Equation of norm.

$$y-1 = -1(x-y)$$

$$x+y=1$$

ULTIMATE CONCEPT

7. The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates the point(s) is/are given by :

- (A) (-1, 9) (B) (3, -15) (C) (1, -3) (D) None

बिन्दु जहाँ वक्र $y = x^3 - 3x^2 - 7x + 6$ पर बनायी गई स्पर्श रेखायें धनात्मक x-अक्ष को, ऋणात्मक y-अक्ष की तुलना में आधा रेखाखण्ड काटती है :

- (A) (-1, 9) (B) (3, -15) (C) (1, -3) (D) कोई नहीं

Ans. B

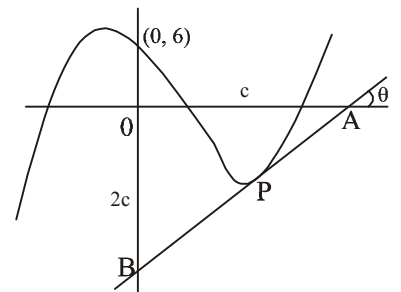
Sol. If OA = a ; OB = 2a $\Rightarrow \tan\theta = 2$

slope of the tangent is 2

$$\Rightarrow \left(\frac{dy}{dx}\right)_{x_1 y_1} = 2 \Rightarrow 3x_1^2 - 6x_1 - 7 = 2 \Rightarrow 3x_1^2 - 2x_1 - 3 = 0$$

$$\Rightarrow x_1 = 3 \text{ or } -1 \text{ (rejected)}$$

$$\Rightarrow (3, -15) \Rightarrow B$$



8. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A meets the curve again at B. If the gradient at B is K times the gradient at A then K is equal to :

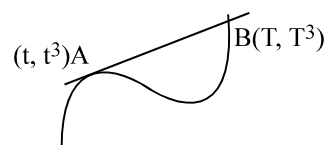
माना C वक्र $y = x^3$ हैं । A पर स्पर्श रेखा वक्र को पुनः बिन्दु B पर मिलती है । यदि B पर प्रवणता, A पर प्रवणता की K गुना है तब K बराबर :

- (A) 4 (B) 2 (C) -2 (D) $\frac{1}{4}$

Ans. A

Sol. $\frac{dy}{dx} = 3x^2 = 3t^2$ at 'A'

$$\therefore 3t^2 = \frac{T^3 - t^3}{T - t} = T^2 + Tt + t^2$$



$$T^2 + Tt - 2t^2 = 0$$

$$(T - t)(T + 2t) = 0 \Rightarrow T = t \text{ or } T = -2t \quad (T = t \text{ is not possible})$$

$$\frac{m_B}{m_A} = \frac{3T^2}{3t^2} = \frac{4t^2}{t^2} \quad (\text{using } T = -2t)$$

$$m_B = 4$$

9. A circle with centre at $(15, -3)$ is tangent to $y = \frac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to :

केन्द्र $(15, -3)$ वाला वृत्त वक्र $y = \frac{x^2}{3}$ को प्रथम चतुर्थोश में किसी बिन्दु पर स्पर्श करता है तब वृत्त की त्रिज्या होगी :

- (A) $5\sqrt{6}$ (B) $8\sqrt{3}$ (C) $9\sqrt{2}$ (D) $6\sqrt{5}$

Ans. D

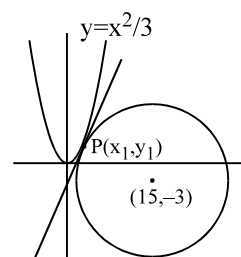
Sol. $y = \frac{x^2}{3} \Rightarrow \left. \frac{dy}{dx} \right|_P = \frac{2x_1}{3}$

$$\therefore m_N = -\frac{3}{2x_1} = \frac{y_1 + 3}{x_1 - 15} \Rightarrow 2x_1y_1 + 6x_1 = -3x_1 + 45$$

Put $y_1 = \frac{x_1^2}{3}$, we get $2x_1 \cdot \frac{x_1^2}{3} + 9x_1 - 45 = 0$; $2x_1^3 + 27x_1 - 135 = 0$

$$2x_1^2(x_1 - 3) + 6x_1(x_1 - 3) + 45(x_1 - 3) = 0$$

$$(x_1 - 3)(2x_1^2 + 6x_1 + 45) = 0 \Rightarrow x_1 = 3$$



10. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to x-axis, is : [AIEEE-2010]

वक्र $y = x + \frac{4}{x^2}$ की स्पर्श रेखा, जो x-अक्ष के समांतर है, का समीकरण है : [AIEEE-2010]

- (A) $y = 0$ (B) $y = 1$ (C) $y = 2$ (D) $y = 3$

Ans. D

Sol. $y = x + \frac{4}{x^2}$

$$\frac{dy}{dx} = 1 - \frac{8}{x^3}$$

Equation of tangent is parallel to x-axis

$$\therefore \frac{dy}{dx} = 0$$

$$\Rightarrow 1 - \frac{8}{x^3} = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$$

$$\text{At, } x = 2, y = 2 + \frac{4}{4} = 3 \Rightarrow y_1 = 3$$

$\therefore$ point is (2, 3)

equation of tangent is :

$$y - y_1 = 0(x - x_1)$$

$$y = 3$$

MINIMUM DISTANCE BETWEEN TWO CURVES

11. The coordinates of the point of the parabola $y^2 = 8x$, which is at minimum distance from the circle $x^2 + (y + 6)^2 = 1$ are :

(A) (2, -4) (B) (18, -12) (C) (2, 4) (D) None of these

परवलय $y^2 = 8x$ पर उस बिन्दु के निर्देशांक जो वृत्त $x^2 + (y + 6)^2 = 1$ से न्यूनतम दूरी पर हैं :

(A) (2, -4) (B) (18, -12) (C) (2, 4) (D) इनमें से कोई नहीं

Ans. A

Sol. Here, the minimum distance occurs along the common normal.

Any normal to

$$y^2 = 8x \text{ at } (2t^2, 4t) \text{ is}$$

$$tx + y = 4t + 2t^3$$

It passes through the centre of circle (0, 6)

$$6 = 4t + 2t^3$$

$$\Rightarrow 2t^3 + 4t - 6 = 0$$

$$\Rightarrow t^3 + 2t - 3 = 0$$

$$\Rightarrow (t - 1)(t^2 + t + 3 \neq 0) (\because D < 0)$$

At $t = 1$, required point is (2, 4)

12. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is :

[AIEEE-2009]

रेखा $y - x = 1$ और वक्र $x = y^2$ के बीच की लघुतम दूरी है :

[AIEEE-2009]

- (A) $\frac{3\sqrt{2}}{5}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{8}$ (D) $\frac{2\sqrt{3}}{8}$

Ans. C

Sol. Point (t^2, t) is on the parabola $x = y^2$

Its distance from $y - x = 1$

$$d(t) = \frac{t^2 - t + 1}{\sqrt{2}}$$

$$d'(t) = \frac{1}{\sqrt{2}} [2t - 1] = 0$$

$$t = \frac{1}{2}$$

$$d''(t) = \frac{2}{\sqrt{2}} > 0$$

$$d(t) \text{ is min at } t = \frac{1}{2}$$

Its value

$$d\left(\frac{1}{2}\right) = \frac{1}{\sqrt{2}} \left(\frac{1}{4} - \frac{1}{2} + 1 \right)$$

$$d\left(\frac{1}{2}\right) = \frac{3\sqrt{2}}{8}$$

13. The lines tangent to the curves $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to :

वक्रों $y^3 - x^2y + 5y - 2x = 0$ तथा $x^4 - x^3y^2 + 5x + 2y = 0$ की मूल बिन्दु पर स्पर्श रेखाएँ θ कोण पर प्रतिच्छेद करती हैं तब θ बराबर :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. Given curve is

$$y^3 - x^2y + 5y - 2x = 0$$

Slope of tangent to curve at $(0, 0) = m_1 = \frac{2}{5}$

Given curve II as

$$x^4 - x^3y^2 + 5x + 2y = 0$$

Slope of tangent to curve at $(0, 0) = m_2 = -\frac{5}{2}$

Since $m_1 \cdot m_2 = -1$

Hence, angle between tangent to the curves is $\frac{\pi}{2}$

14. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$:

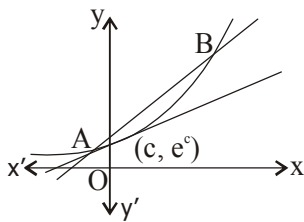
- (A) on the left of $x = c$ (B) on the right of $x = c$
(C) at no point (D) at all points

वक्र $y = e^x$ के बिन्दु (c, e^c) पर खींची गई स्पर्श रेखा बिन्दुओं $(c-1, e^{c-1})$ तथा $(c+1, e^{c+1})$ को मिलाने वाली रेखा को कहाँ प्रतिच्छेद करती है :

- (A) $x = c$ के बाईं ओर (B) $x = c$ के दाईं ओर (C) कहीं भी नहीं (D) सभी बिन्दुओं पर

Ans. A

Sol. Slope of the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$ is equal to $\frac{e^{c+1} - e^{c-1}}{2} > e^c$



tangent to the curve $y = e^x$ will intersect the given line to the left of the line $x = c$.

ORTHOGONALITY OF TWO CURVES

15. If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is :

यदि वक्र $y = 1 - ax^2$ और $y = x^2$ परस्पर लम्बवत् प्रतिच्छेद करते हैं, तो a का मान है :

- (A) $1/2$ (B) $1/3$ (C) 2 (D) 3

Ans. B

Sol. $y = 1 - ax^2 \Rightarrow \frac{dy}{dx} = -2ax$

$$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$$

two curve intersect orthogonally if

$$(-2ax)(2x) = -1$$

$$\Rightarrow 4ax^2 = 1 \dots\dots\dots i$$

$$\Rightarrow x^2 = \frac{1}{1+a} \dots\dots\dots ii$$

from i and ii

$$\frac{4a}{1+a} = 1$$

$$\Rightarrow a = \frac{1}{3}$$

16. If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is :

[JEE(Main)2018]

यदि वक्र $y^2 = 6x$, $9x^2 + by^2 = 16$ समकोण पर प्रतिच्छेद करते हैं, तो b का मान है :

[JEE(Main)2018]

- (A) 4 (B) $\frac{9}{2}$ (C) 6 (D) $\frac{7}{2}$

Ans. B

Sol. $y^2 = 6x$
 $2yy' = 6$

$$y' = \frac{3}{y_1}$$

$$9x^2 + by^2 = 16$$

$$18x + 2byy' = 0$$

$$y' = \frac{-9x_1}{by_1}$$

$$m_1 m_2 = -1$$

$$27x_1 = by_1^2 \dots\dots (i)$$

$$6x_1 = y_1^2 \dots\dots (ii)$$

$$(i) \div (ii)$$

$$b = \frac{9}{2}$$

CASE OF TWO CURVES TOUCHING EACH OTHER

17. Number of positive integral values of 'a' for which the curve $y = a^x$ intersects the line $y = x$ is :

- (A) 0 (B) 1 (C) 2 (D) More than 2

'a' के घनात्मक पूर्णांक मानों की संख्या जब कि वक्र $y = a^x$ रेखा $y = x$ को प्रतिच्छेद करता है :

- (A) 0 (B) 1 (C) 2 (D) 2 से अधिक

Ans. B

Sol. for $0 < a \leq 1$ the line

always cuts $y = a^x$

for $a > 1$ say $a = e$

consider $f(x) = e^x - x$

$$f'(x) = e^x - 1$$

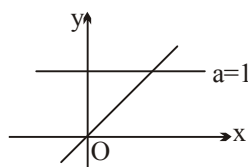
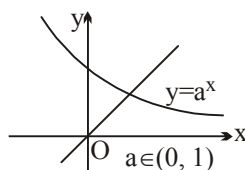
$f'(x) > 0$ for $x > 0$ and $f'(x) < 0$ for $x < 0$

$\therefore f(x)$ is increasing ($\uparrow$) for $x > 0$

and decreasing ($\downarrow$) for $x < 0$

$y = e^x$ always lies above $y = x$ i.e. $e^x - x \geq 1$ for $a > 1$

hence never cuts $a = (0, 1] \Rightarrow$ (B)

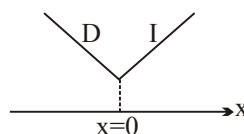


$f(0)$ is the minimum

$$f(x)|_{\min} = 1$$

$$f(x) \geq 1$$

$$e^x - x \geq 1$$



18. The value of k for which the equation $e^{2x} = k\sqrt{x}$ has exactly one solution, is :

k का वह मान जिसके लिए समीकरण $e^{2x} = k\sqrt{x}$ का केवल एक हल होगा :

- (A) $\frac{1}{\sqrt{e}}$ (B) $\sqrt{e}$ (C) $2\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Ans. C

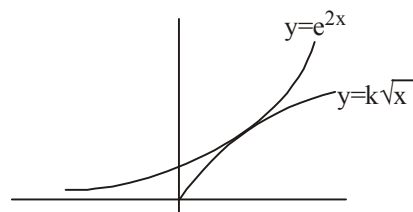
Sol. k can not be negative

$$y = e^{2x}$$

$$\left. \frac{dy}{dx} \right|_P = 2e^{2x_1}$$

$$\therefore \left. \frac{dy}{dx} \right|_P = \frac{k}{2\sqrt{x_1}}$$

$$2e^{2x_1} = \frac{k}{2\sqrt{x_1}} \quad (\text{but } e^{2x_1} = k\sqrt{x_1})$$



$$2k\sqrt{x_1} = \frac{k}{2\sqrt{x_1}}; \quad x_1 = \frac{1}{4}; \quad \therefore \quad e^{\frac{1}{2}} = \frac{k}{2}; \quad k = 2\sqrt{e}$$

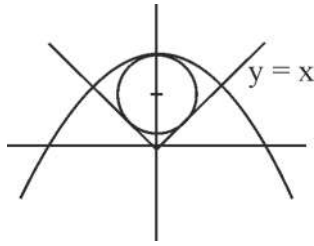
19. The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines, $y = |x|$ is :

[JEE(Main)2017]

न्यूनतम क्षेत्रफल वाले ऐसे वृत्त, जो वक्र $y = 4 - x^2$ तथा रेखाओं $y = |x|$ को स्पर्श करता है, की त्रिज्या है :

- (A) $4(\sqrt{2} - 1)$ (B) $4(\sqrt{2} + 1)$ (C) $2(\sqrt{2} + 1)$ (D) $2(\sqrt{2} - 1)$

Ans. A



Sol.

Center will be $(0, 4 - r)$

Distance of $(0, 4 - r)$ from $L : x - y = 0$

radius = r

$$\left| \frac{4 - r}{\sqrt{2}} \right| = r$$

$$4 - r = \sqrt{2}r$$

$$r = 4(\sqrt{2} - 1)$$

20. At any point of a curve $\sqrt{\frac{\text{sub normal}}{\text{sub tangent}}}$ is equal to :

- (A) the abscissa of that point (B) the ordinate of that point
(C) slope of the tangent at that point (D) slope of the normal at that point

वक्र के किसी बिन्दु पर $\sqrt{\frac{\text{अधोलम्ब}}{\text{अधःस्पर्शी}}}$ बराबर है :

- (A) उस बिन्दु के भुज (B) उस बिन्दु की कोटि
(C) उस बिन्दु पर स्पर्श रेखा की प्रवणता (D) उस बिन्दु पर अभिलम्ब की प्रवणता

Ans. C

Sol.
$$\sqrt{\frac{y \frac{dy}{dx}}{\frac{y}{\frac{dy}{dx}}}} = \sqrt{\left(\frac{dy}{dx}\right)^2} = \left|\frac{dy}{dx}\right|$$

RATE MEASURE

21. Water is poured at the rate of $2\text{ m}^3/\text{sec}$. into a cone of semi-vertical angle 45° . The rate at which periphery of water surface changes when height of the water in the cone is 2 meter, is :

एक शंकु जिसका अर्ध शीर्ष कोण 45° है, में जल $2\text{ m}^3/\text{sec}$ की नियत दर से भरा जा रहा है। जब शंकु में जल की सतह की ऊँचाई 2 m है तब जल की सतह की परिधी के बदलने की दर है :

- (A) 1 m/sec. (B) 2 m/sec. (C) 3 m/sec. (D) 4 m/sec.

Ans. A

Sol. We have

$$\frac{dV}{dt} = 2 \Rightarrow \frac{d}{dt} \left(\frac{1}{3} \pi r^3 \right) = 2 \quad [\text{Here } r = h, \text{ as } \theta = 45^\circ]$$

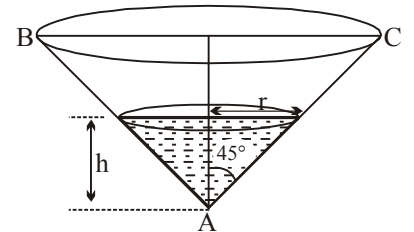
$$\Rightarrow \pi r^2 \frac{dr}{dt} = 2 \Rightarrow \frac{dr}{dt} = \frac{2}{\pi r^2} \quad \dots(1)$$

Now, perimeter = $2\pi r = p$ (let)

$$\Rightarrow \frac{d}{dt} (2\pi r) = 2\pi \left(\frac{2}{\pi r^2} \right) = \frac{4}{r^2} \quad \dots(2) \quad (\text{Using equation (1)})$$

When $h = 2$ meters $\Rightarrow r = 2$ meters

$$\text{Hence } \frac{dp}{dt} = \frac{4}{4} = 1 \text{ m/sec.}$$



22. A point is moving along the curve $y^3 = 27x$. The interval in which the abscissa changes at slower rate than ordinate, is:

वक्र $y^3 = 27x$ पर एक चर बिन्दु किसी अन्तराल में इसप्रकार से गमन करता है कि उसके भुज में परिवर्तन की दर कोटि में परिवर्तन की दर से कम है, तब वह अन्तराल होगा :

- (A) $(-3, 3)$ (B) $(-\infty, \infty)$ (C) $(-1, 1)$ (D) $(-\infty, -3) \cup (3, \infty)$

Ans. C

Sol. $y^3 = 27x \Rightarrow 3y^2 \frac{dy}{dt} = 27 \frac{dx}{dt}$

But $\left| \frac{dx/dt}{dy/dt} \right| < 1 \Rightarrow \frac{y^2}{9} < 1 \Rightarrow -3 < y < 3$ for $y \in (-3, 3)$, $x \in (-1, 1) \Rightarrow (C)$

23. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate of which the thickness of ice decreases, is :

(A) $\frac{1}{36\pi} \text{ cm/min.}$ (B) $\frac{1}{18\pi} \text{ cm/min.}$ (C) $\frac{1}{54\pi} \text{ cm/min.}$ (D) $\frac{5}{6\pi} \text{ cm/min.}$

एक गोलाकार लोहे का गेंद, जिसकी त्रिज्या 10 सेमी है, बर्फ की एक समान मोटाई की परत से ढका है, जो $50 \text{ सेमी}^3/\text{मिनट}$ की दर से पिघल रही है। जब बर्फ की परत की मोटाई 5 सेमी है, वह दर जिससे बर्फ की परत की मोटाई कम हो रही है, है :

(A) $\frac{1}{36\pi} \text{ सेमी/मि.}$ (B) $\frac{1}{18\pi} \text{ सेमी/मि.}$ (C) $\frac{1}{54\pi} \text{ सेमी/मि.}$ (D) $\frac{5}{6\pi} \text{ सेमी/मि.}$

Ans. B

Sol. $V = 4\pi(10+r)^3$, $0 \leq r \leq 15$ $\therefore \frac{dV}{dt} = -50$.

$4\pi(10+r)^2 \frac{dr}{dt} = -50 \Rightarrow \frac{dr}{dt} = \frac{-1}{18\pi}$ (where $r = 5$)

MONOTONOCITY

DETERMINE MONOTONOCITY

24. The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is :

- (A) increasing in its domain (B) decreasing in its domain
(C) decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$ (D) increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$

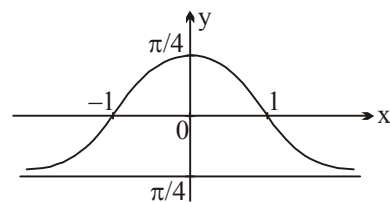
फलन $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ है :

- (A) अपने प्रान्त में वर्धमान (B) अपने प्रान्त में ह्यसमान
(C) $(-\infty, 0)$ में ह्यसमान, एवं $(0, \infty)$ में वर्धमान (D) $(-\infty, 0)$ में वर्धमान और $(0, \infty)$ में ह्यसमान

Ans. D

Sol. put $x^2 = \tan \theta$ to get

$f(x) = \frac{\pi}{4} - \tan^{-1}(x^2)$



$$\Rightarrow f'(x) = -\frac{2x}{1+x^4} \text{ which is greater than zero for } x < 0 \text{ and less than zero for } x > 0$$

$\Rightarrow$ (D)

25. If $f(x) = xe^{x(1-x)}$, then $f(x)$ is :

(A) increasing on $\left(-\frac{1}{2}, 1\right)$

(B) decreasing on $\left[-\frac{1}{2}, 1\right]$

(C) increasing on R

(C) decreasing on R

यदि $f(x) = xe^{x(1-x)}$, तो $f(x)$ एक :

(A) $\left(-\frac{1}{2}, 1\right)$ में वर्द्धमान फलन है

(B) $\left[-\frac{1}{2}, 1\right]$ में ह्रासमान फलन है

(C) R में वर्द्धमान है

(C) R में ह्रासमान है

Ans. A

Sol. $f(x) = xe^{x(1-x)}$

$$\Rightarrow f'(x) = e^{x(1-x)} + x^2(-1)e^{x(1-x)} + x(1-x)e^{x(1-x)}$$

$$= -e^{x(1-x)}(2x^2 - x - 1) = e^{x(1-x)}(x-1)\left(x + \frac{1}{2}\right)$$

$$f'(x) = 0 \text{ at } x = -\frac{1}{2}, 1 \text{ and } f'(x) > 0 \text{ when } -\frac{1}{2} < x < 1$$

$$\Rightarrow f(x) \text{ is increasing function on } \left[-\frac{1}{2}, 1\right]$$

26. A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched?

interval

function

(A) $(-\infty, \infty)$

$$x^3 - 3x^2 + 3x + 3$$

(B) $[2, \infty)$

$$2x^3 - 3x^2 - 12x + 6$$

(C) $\left(-\infty, \frac{1}{3}\right]$

$$3x^2 - 2x + 1$$

(D) $(-\infty, -4]$

$$x^3 + 6x^2 + 6$$

नीचे दिये गए फलनों को उनके सामने दिये गये अन्तरालों से मिलान किया गया है, जिनमें फलन वर्धमान हैं। निम्न युग्मों में से कौन से युग्म का मिलान ठीक नहीं है?

अन्तराल	फलन
(A) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(B) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(C) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(D) $(-\infty, -4]$	$x^3 + 6x^2 + 6$

Ans. C

Sol. Let $f_1(x) = x^3 + 6x^2 + 6 \Rightarrow f_1'(x) = 3x(x+4)$
Increasing in $(-\infty, -4]$

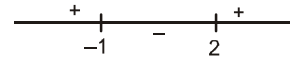
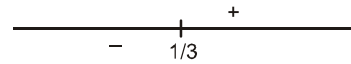
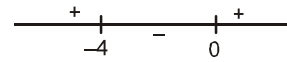
Let $f_2(x) = 3x^2 - 2x + 1 \Rightarrow f_2'(x) = 6x - 2$

Not increasing in $\left(-\infty, \frac{1}{3}\right]$

Let $f_3(x) = 2x^3 - 3x^2 - 12x + 6 \Rightarrow f_3'(x) = 6x^2 - 6x - 12$
 $= 6(x+1)(x-1)$

Increasing in $[2, \infty)$

Let $f_4(x) = x^3 - 3x^2 + 3x + 3 \Rightarrow f'(x) = 3(x-1)^2 \geq 0$
Increasing in $(-\infty, \infty)$



27. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in :

फलन $f(x) = \tan^{-1}(\sin x + \cos x)$, जिस अन्तराल में वर्धमान फलन है वह है :

(A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

Ans. B

Sol. $f(x) = \tan^{-1}(\sin x + \cos x)$

$$f'(x) = \frac{1 \times (\cos x - \sin x)}{1 + (\sin x + \cos x)^2} > 0$$

$$\cos x - \sin x > 0 \quad \cos x > \sin x$$

$$\sin x < \cos x \quad \tan x < 1$$

$$x < \frac{\pi}{4}$$

$$x \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$$

ALWAYS INCREASING/ ALWAYS DECREASING BASED QUESTIONS

28. The function $f : [a, \infty) \rightarrow \mathbb{R}$ where $\mathbb{R}$ denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided :

यदि फलन $f : [a, \infty) \rightarrow \mathbb{R}$ जहाँ $\mathbb{R}$ दिये गये प्रान्त के सापेक्ष परिसर को दर्शाता है इस प्रकार है कि $f(x) = 2x^3 - 3x^2 + 6$ तब $f(x)$ प्रतिलोमीय होगा यदि :

- (A) $a \geq 1$ (B) $a \geq 0$ (C) $a \leq 0$ (D) $a \leq 1$

Ans. A

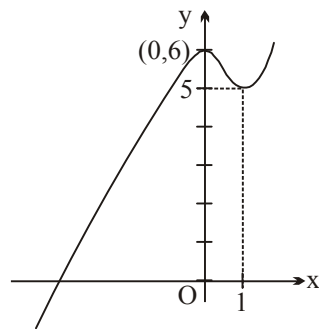
Sol. $f(x) = 2x^3 - 3x^2 + 6$

$$f'(x) = 6x^2 - 6x = 6(x^2 - x) = 0 \quad \text{gives } x = 0 \text{ or } x = 1$$

for inverse to exist function must be one one onto

hence Domain is $[1, \infty)$

Hence $a \geq 1$



29. Let $f(x) = x - \frac{1}{x}$ then which one of the following statement is true :

- (A) Function is invertible if defined from $\mathbb{R} - \{0\} \rightarrow \mathbb{R}$.
 (B) $f(x_1) > f(x_2)$, $\forall x_1 > x_2$ and $x_1, x_2 \neq 0$.
 (C) Graph of the function has exactly one asymptote.
 (D) Function is one-one in every continuous interval $[a, b]$ defined on one side of origin.

माना $f(x) = x - \frac{1}{x}$ तब निम्न में से कौनसा कथन सत्य है :

- (A) फलन प्रतिलोमीय है यदि $f, \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ पर परिभाषित हो
 (B) $f(x_1) > f(x_2)$, $\forall x_1 > x_2$ और $x_1, x_2 \neq 0$.
 (C) वक्र की केवल एक अनन्तस्पर्शी रेखा है
 (D) मूल बिन्दु के एक और परिभाषित प्रत्येक सतत् अन्तराल $[a, b]$ में फलन एकैकी है।

Ans. D

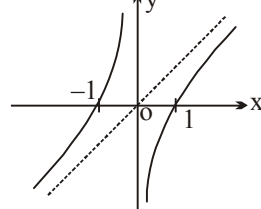
Sol. note that $x = 0$ and $y = x$ are the two asymptote $\Rightarrow$ (C) is wrong

A is wrong as f is many one

$$f(1) = f(-1) = 0$$

$$f'(x) = 1 + \frac{1}{x^2} > 0 \quad \Rightarrow \quad f \text{ is increasing}$$

$$x \rightarrow 0^+; \quad f \rightarrow -\infty \quad x \rightarrow \infty, \quad f \rightarrow \infty$$



$$x \rightarrow 0^-; \quad f \rightarrow +\infty \quad x \rightarrow -\infty, \quad f \rightarrow -\infty$$

30. If the function $f(x) = 2x^2 - kx + 5$ is increasing in $[1, 2]$, then 'k' lies in the interval :

यदि $f(x) = 2x^2 - kx + 5$ अन्तराल $[1, 2]$ में वर्धमान है तब 'k' :

- (A) $(-\infty, 4)$ (B) $(4, \infty)$ (C) $(-\infty, 8]$ (D) $(8, \infty)$

Ans. A

Sol. Given, $f(x) = 2x^2 - kx + 5$ Now $f'(x) > 0$ for $1 \leq x \leq 2$

$$f'(x) = 4x - k, \quad f''(x) = 4 \Rightarrow f''(x) > 0 \text{ for all } x \in [1, 2]$$

$$\Rightarrow f'(x) \text{ is an increasing function in } [1, 2] \Rightarrow f'(x) > 0$$

$$\therefore f'(1) \text{ is the least value of } f'(x) \text{ in } [1, 2] \Rightarrow f'(1) > 0 \Rightarrow 4 - k > 0$$

$$\Rightarrow k < 4 \Rightarrow k \in (-\infty, 4)$$

31. Consider the following behaviours of function in $(-1, 1)$

I. Increasing II. Decreasing III. Continuity IV. Derivability

Which one of the following four functions exhibits atleast three of the four mentioned above? (where $[]$ denotes greatest integer function)

$(-1, 1)$ में माना फलन की निम्न प्रकृति पर विचार कीजिये

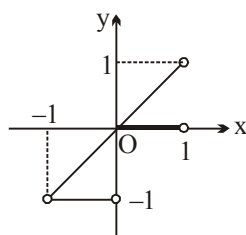
I. वर्धमान II. ह्रसमान III. सांतत्यता IV. अवकलनीयता

तब निम्न में से कौनसा फलन उपरोक्त चार में से कम से कम तीन गुणधर्म रखता है? (जहाँ $[]$ महत्तम पूर्णांक फलन है)

- (A) $\max \{x, x^3\}$ (B) $\max \{x, x^2\}$ (C) $\max \{x, |x|\}$ (D) $\max \{x, [x]\}$

Ans. D

Sol. $\max. \{x, [x]\}$ in $(-1, 1)$ is x as $[x] \leq x$



32. Consider the following statements S and R

S : Both $\sin x$ & $\cos x$ are decreasing functions in the interval $(\pi/2, \pi)$.

R : If a differentiable function decreases in an interval (a, b) , then its derivative also decreases in (a, b) .

Which of the following is true?

- (A) both S and R are wrong
(B) both S and R are correct, but R is not the correct explanation for S
(C) S is correct and R is the correct explanation for S
(D) S is correct and R is wrong

निम्नलिखित कथनों S तथा R के लिये

S : $\sin x$ तथा $\cos x$ दोनों फलन अन्तराल $(\pi/2, \pi)$ में घटसमान है।

R : यदि कोई फलन अन्तराल (a, b) में घटसमान है तो उसका अवकलज भी अन्तराल (a, b) घटसमान होगा तो निम्न में से कौनसा सत्य है?

- (A) S तथा R दोनों असत्य है
 (B) S तथा R दोनों सत्य है परन्तु R कथन S की सही व्याख्या नहीं करता
 (C) S सत्य है तथा R भी सत्य है एवं कथन S की सही व्याख्या है
 (D) S सत्य है परन्तु R असत्य है

Ans. D

Sol. Since, it is true that $\sin x$ and $\cos x$ is decreasing in the interval $\left(\frac{\pi}{2}, \pi\right)$.

Now, let $f(x) = \sin x$, $x \in \left(\pi, \frac{3\pi}{2}\right)$, $f(x)$ decrease from 0 to -1 but $f'(x) = \cos x$ increases from -1 to 0.

CRITICAL AND STATIONARY POINTS

33. Number of critical points of the function, $f(x) = \frac{2}{3} \sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2} \cos 2t - \sqrt{t}\right) dt$ which lie in the interval $[-2\pi, 2\pi]$ is :

फलन $f(x) = \frac{2}{3} \sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2} \cos 2t - \sqrt{t}\right) dt$ के अन्तराल $[-2\pi, 2\pi]$ में क्रान्तिक बिन्दुओं की संख्या है :

- (A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. note that f is defined for $x > 0$

$$f'(x) = \frac{1}{2} \cos 2x = 0 \Rightarrow x = n\pi \pm \frac{\pi}{4} \Rightarrow B]$$

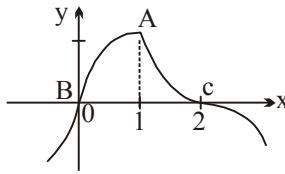
34. Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \leq 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, then the number of critical points on the graph of the function is :

यदि $f(x) = \begin{cases} x^{3/5} & \text{if } x \leq 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, तब फलन $f(x)$ के क्रान्तिक बिन्दुओं की संख्या है :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. A, B, C are the 3 critical points of $y = f(x)$



$f''(x) = 0$ for $x = 2$ and fails to exist at $x = 0$

35. Let a function f be defined as $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if } x > -1 \\ x^2 & \text{if } x \leq -1 \end{cases}$, Then the number of critical point(s) on the graph of this function is/are :

माना $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if } x > -1 \\ x^2 & \text{if } x \leq -1 \end{cases}$, तब फलन के आरेख में क्रांतिक बिन्दुओं की संख्या है :

(A) 4

(B) 3

(C) 2

(D) 1

Ans. A

Sol. $f(x) = \begin{cases} \frac{x-1}{1+x^2} & \text{if } x \geq 1 \\ \frac{1-x}{1+x^2} & \text{if } -1 < x < 1 \\ x^2 & \text{if } x \leq -1 \end{cases}$

$f'(x) = 0$ gives $x = \sqrt{2} + 1$ or

$1 - \sqrt{2}$. The function has a continuity

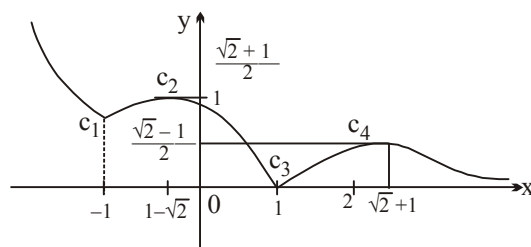
at $x = -1$. As $x \rightarrow \infty$, $f(x) \rightarrow 0$.

The graph is as shown.

Note that critical points are those where $dy/dx = 0$ if it exists or dy/dx is non existent.

The points $x = -1, 1 - \sqrt{2}, 1$ and

$1 + \sqrt{2}$ are the four critical points on the graph of this function $\Rightarrow$ A



36. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then :

(A) f has exactly one stationary point

(B) f must have no stationary point

(C) f must have exactly 2 stationary points

(D) f has either 0 or 2 stationary points

यदि f , तीन घात का बहुपदीय फलन है और $f''(x) \neq 0$ किसी भी स्थिर बिन्दु (stationary point) के लिए तब :

- (A) f का केवल एक स्थिर बिन्दु होगा (B) f का कोई स्थिर बिन्दु नहीं होगा
(C) f के दो स्थिर बिन्दु होंगे (D) f के 0 या 2 स्थिर बिन्दु होंगे

Ans. D

Sol. The derivative of a degree 3 polynomial is a quadratic. This must have either 0, 1 or 2 roots. If this has precisely one root, then this must be repeated. Hence we have $f'(x) = m(x - \alpha)^2$, where α is repeated root and $m \in \mathbb{R}$. So our original function f has a critical point at $x = \alpha$.

Also $f''(x) = 2m(x - \alpha)$, in which case $f''(\alpha) = 0$. But we are told that the 2nd derivative is non zero at critical point. Hence there must be either 0 or 2 critical point.

37. The set of all values of 'a' for which the function, $f(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a - 1)x + \sin 1$ does not possess critical points is :

'a' के मानों का समुच्चय जिसके लिए $f(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a - 1)x + \sin 1$ के कोई क्रांतिक बिन्दु नहीं

है :

- (A) $[1, \infty)$ (B) $(0, 1) \cup (1, 4)$ (C) $(-2, 4)$ (D) $(1, 3) \cup (3, 5)$

Ans. B

Sol. $f(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a - 1)x + \sin 1$

$$\Rightarrow f(x) = (a - 1)(a - 2) \cos \frac{x}{2} + (a - 1)x + \sin 1$$

$$\Rightarrow f'(x) = -\frac{1}{2}(a - 1)(a - 2) \sin \frac{x}{2} + (a - 1)$$

$$\Rightarrow f'(x) = (a - 1) \left[1 - \frac{(a - 2)}{2} \sin \frac{x}{2} \right]$$

If $f(x)$ does not possess critical points, then $f'(x) = 0$ for any $x \in \mathbb{R}$

$$\Rightarrow (a - 1) \left[1 - \frac{(a - 2)}{2} \sin \frac{x}{2} \right] = 0 \text{ for any } x \in \mathbb{R}$$

$$\Rightarrow a = 1 \text{ and } 1 - \left(\frac{a - 2}{2} \right) \sin \frac{x}{2} = 0$$

must not have any solution in $\mathbb{R}$.

$$\Rightarrow a = 1 \text{ and } \sin \frac{x}{2} = \frac{2}{a-2} \text{ is not solvable in } \mathbb{R}.$$

$$\Rightarrow a = 1 \text{ and } \left| \frac{2}{a-2} \right| > 1 \text{ [For } a = 2, f(x) = x + \sin 1 \therefore f'(x) = 1 = 0]$$

$$\Rightarrow a = 1 \text{ and } |a - 2| < 2 \Rightarrow a = 1 \text{ and } -2 < a - 2 < 2$$

$$\Rightarrow a = 1 \text{ and } 0 < a < 4 \Rightarrow a \in (0, 1) \cup (1, 4).$$

PROVING INEQUALITIES

38. For all $x \in (0, 1)$:

सभी $x \in (0, 1)$ के लिये :

(A) $e^x < 1 + x$ (B) $\log_e(1 + x) < x$ (C) $\sin x > x$ (D) $\log_e x > x$

Ans. B

Sol. $y = e^x \Rightarrow \frac{dy}{dx} = e^x$

Then equation of the tangent at $x = 0$ is $y - 1 = 1(x - 0)$ or $y = x + 1$

Graph of $y = e^x$ always lies above the graph of $y = 1 + x$,

Hence, $e^x > 1 + x \Rightarrow x > \log_e(1 + x)$. Hence, b is true.

39. The true set of real values of x for which the function, $f(x) = x \ln x - x + 1$ is positive is :

(A) $(1, \infty)$ (B) $(1/e, \infty)$ (C) $[e, \infty)$ (D) $(0, 1)$ and $(1, \infty)$

x के सभी मानों के समुच्चय जिसके लिये फलन $f(x) = x \ln x - x + 1$ का मान धनात्मक है :

(A) $(1, \infty)$ (B) $(1/e, \infty)$ (C) $[e, \infty)$ (D) $(0, 1)$ तथा $(1, \infty)$

Ans. D

Sol. We have $f(x) = \log_e x - x + 1, x > 0$

$$\therefore f(x) = 1 + \log_e x - 1 = \log_e x$$

$$\therefore f(x) < 0 \text{ if } 0 < x < 1$$

$$f(x) > 0 \text{ if } x > 1$$

$$\text{also } \lim_{x \rightarrow 0^+} (x \log_e x - x + 1) = 1$$

So in $(0, 1)$, $f(x)$ decreases from 1 to 0 and in $(1, \infty)$ increase from 0 $\rightarrow \infty$

Hence $f(x) > 0$ for $x \in (0, 1) \cup (1, \infty)$

40. The set of all x for which $\ln(1+x) \leq x$ is equal to :

- (A) $x > 0$ (B) $x > -1$ (C) $-1 < x < 0$ (D) null set

$\ln(1+x) \leq x$ के लिए x के समस्त मानों का समुच्चय है :

- (A) $x > 0$ (B) $x > -1$ (C) $-1 < x < 0$ (D) रिक्त समुच्चय

Ans. B

Sol. $\ln(1+x) - x \leq 0$

Let $f(x) = \ln(1+x) - x$, {Domain is $(-1, \infty)$ }

$$f(x) = \frac{1}{x+1} - 1 = \frac{-x}{x+1} \quad \begin{array}{c} + \quad - \\ -1 \quad 0 \end{array}$$

$$\Rightarrow f(x) \leq f(0) \quad \forall x \in (-1, \infty) \Rightarrow f(x) \leq 0 \quad \forall x \in (-1, \infty).$$

ROLLE'S THERORM

41. The equation $\sin x + x \cos x = 0$ has at least one root in :

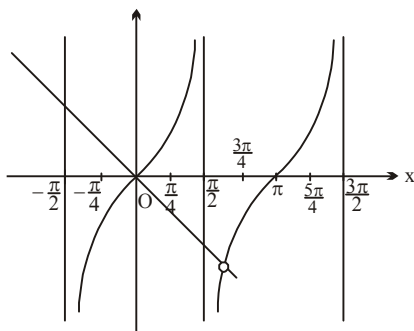
समीकरण $\sin x + x \cos x = 0$ का कम से कम एक मूल निम्न अन्तराल में है :

- (A) $\left(-\frac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

Ans. B

Sol. Let $f(x) = \sin x + x \cos x$

$$\text{consider } g(x) = \int_0^x (\sin t + t \cos t) dt = t \sin t \Big|_0^x = x \sin x$$



$g(x) = x \sin x$ which is differentiable

now $g(0) = 0$ and $g(\pi) = 0$, using Rolles Theorem

hence $\exists$ atleast one $x \in (0, \pi)$ where $g'(x) = 0$

i.e. $x \cos x + \sin x = 0$ for atleast one $x \in (0, \pi)$

Alter: Draw graph of $y = \tan x$ & $y = -x$ Ans. $\Rightarrow$ (B)

42. Let $f(x) = (x-4)(x-5)(x-6)(x-7)$ then :
- (A) $f'(x) = 0$ has four roots
(B) Three roots of $f'(x) = 0$ lie in $(4, 5) \cup (5, 6) \cup (6, 7)$
(C) The equation $f'(x) = 0$ has only one real root
(D) Three roots of $f'(x) = 0$ lie in $(3, 4) \cup (4, 5) \cup (5, 6)$

यदि $f(x) = (x-4)(x-5)(x-6)(x-7)$ हो, तो :

- (A) $f'(x) = 0$ के चार मूल हैं
(B) $f'(x) = 0$ के तीन मूल $(4, 5) \cup (5, 6) \cup (6, 7)$ में स्थित हैं
(C) समीकरण $f'(x) = 0$ का केवल एक वास्तविक मूल है
(D) $f'(x) = 0$ के लिए तीन मूल $(3, 4) \cup (4, 5) \cup (5, 6)$ में स्थित हैं

Ans. B

Sol. $f(4) = f(5) = f(6) = f(7) = 0$

By Rolle's theorem on interval

$[4, 5], [5, 6], [6, 7]$ we have

$f'(x) = 0$ for at least once in each intervals $(4, 5), (5, 6), (6, 7)$.

43. Let a, b, c, d are non-zero real numbers such that $6a + 4b + 3c + 3d = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has :

- (A) atleast one root in $[-2, 0]$ (B) atleast one root in $[0, 2]$
(C) atleast two roots in $[-2, 2]$ (D) no root in $[-2, 2]$

यदि a, b, c, d अशून्य वास्तविक संख्याएँ इस प्रकार है कि $6a + 4b + 3c + 3d = 0$, तब समीकरण $ax^3 + bx^2 + cx + d = 0$ के

- (A) अन्तराल $[-2, 0]$ में कम से कम एक मूल है (B) अन्तराल $[0, 2]$ में कम से कम एक मूल है
(C) अन्तराल $[-2, 2]$ में कम से कम दो मूल है (D) $[-2, 2]$ में कोई मूल नहीं है

Ans. B

Sol. We have $ax^3 + bx^2 + cx + d = 0$

$$\text{Let } f(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx + e$$

$$\therefore f(0) = e$$

$$f(2) = 4a + \frac{8b}{3} + 2c + 2d + e = \frac{(12a + 8b + 6c + 6d)}{3} + e = \frac{2}{3}(6a + 4b + 3c + 3d) + e = 0 + e$$

$$\Rightarrow f(2) = e$$

$\therefore$ By Rolle's theorem, there exist atleast one value of $x \in (0, 2)$ such that $f'(x) = 0$

$\Rightarrow$ The equation $ax^3 + bx^2 + cx + d = 0$ has atleast one real root in $[0, 2]$

44. If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$; $a_1 \neq 0$, $n \geq 2$, has a positive root $x = \alpha$, then the equation $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root, which is -
 (A) greater than α (B) smaller than α
 (C) greater than or equal to α (D) equal to α

यदि समीकरण $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$; $a_1 \neq 0$, $n \geq 2$ का एक धनात्मक मूल $x = \alpha$ है, तो समीकरण $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ का एक धनात्मक मूल होगा जो कि :

- (A) α से बड़ा है (B) α से छोटा है
 (C) α से बड़ा है या बराबर है (D) α के बराबर है

Ans. B

Sol. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x$

Now for $x = 0$

$$f(0) = 0$$

And as a is one root

$$f(a) = 0$$

Hence $f'(x)$ will have one root in $(0, a)$

$$\text{As } f'(x) = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$$

Then root of $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ is less than a

45. If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in]0, 1[$ **[JEE(Main)2014]**

यदि f तथा g , $[0, 1]$ में अवकलनीय फलन है जो $f(0) = 2 = g(1)$, $g(0) = 0$ और $f(1) = 6$ को संतुष्ट करते हैं, तो किसी $c \in]0, 1[$ के लिए : **[JEE(Main)2014]**

- (A) $f(c) = g'(c)$ (B) $f(c) = 2g'(c)$ (C) $2f(c) = g'(c)$ (D) $2f(c) = 3g'(c)$

Ans. B

Sol. Consider $f(x) - 2g(x) = h(x)$

Then, $h(x)$ is continuous and differentiable in $[0, 1]$

$$\text{Also } h(0) = 2 \text{ \& } h(1) = 2$$

Hence $h(x)$ satisfies conditions of Rolle's Theorem in $(0, 1)$

Thus, There exist a ' c ' such that $h'(c) = 0$ where $c \in (0, 1)$

$$\Rightarrow f'(c) = 2g'(c)$$

LMVT

46. Let $f(x) = x + \sqrt{x}$ on $[1, 4]$. The mean value theorem says that there must be some number ' c ' between 1 and 4 so that $f'(c)$ is equal to the average slope of $f(x)$ on $[1, 4]$. the number ' c ' must be :

यदि $f(x) = x + \sqrt{x}$ अन्तराल $[1, 4]$ में मध्यमान प्रमेय द्वारा कम से कम एक 'c' $(1, 4)$ में इस प्रकार से होगा की $f'(c)$ अन्तराल $[1, 4]$ में $f(x)$ की औसत प्रवणता के बराबर है तो 'c' होना चाहिये :

- (A) $\frac{5}{2}$ (B) $\frac{9}{4}$ (C) $\frac{11}{4}$ (D) 3

Ans. B

Sol. Hint: $f'(c) = \frac{f(4) - f(1)}{4 - 1}$

$$1 + \frac{1}{2\sqrt{c}} = \frac{6-2}{3} = \frac{4}{3} \Rightarrow \frac{1}{2\sqrt{c}} = \frac{1}{3} \Rightarrow \sqrt{c} = \frac{3}{2} \Rightarrow c = \frac{9}{4}$$

47. Let f be differentiable for all x . If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then :

- (A) $f(6) \geq 8$ (B) $f(6) < 8$ (C) $f(6) < 5$ (D) $f(6) = 5$

माना f सभी x के लिए अवकलनीय है। यदि $f(1) = -2$ तथा $x \in [1, 6]$ के लिए $f'(x) \geq 2$, तो :

- (A) $f(6) \geq 8$ (B) $f(6) < 8$ (C) $f(6) < 5$ (D) $f(6) = 5$

Ans. A

Sol. By LMVT

$$\frac{f(6) - f(1)}{6 - 1} = f'(x), \quad x \in (1, 6)$$

$$\text{and (तथा) } f'(x) \geq 2 \Rightarrow \frac{f(6) - f(1)}{6 - 1} \geq 2$$

$$\Rightarrow \frac{f(6) + 2}{5} \geq 2 \Rightarrow f(6) \geq 8.$$

48. A value of C for which the conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is :

C का वह मान जिसके लिये अन्तराल $[1, 3]$ में फलन $f(x) = \log_e x$ के लिये माध्यमान प्रमेय सत्य है :

- (A) $2\log_3 e$ (B) $\frac{1}{2} \log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

Ans. A

Sol. $\frac{f(3) - f(1)}{3 - 1} = f'(c), 1 < c < 3$

$$\frac{\log_e^3 - 0}{2} = \frac{1}{c}$$

$$c = \frac{2}{\log_3 e} = 2 \log_3 e$$

MAXIMA & MINIMA

MAXIMA AND MINIMA AT A POINT

49. Difference between the greatest and the least values of the function $f(x) = x(\ln x - 2)$ on $[1, e^2]$ is :

फलन $f(x) = x(\ln x - 2)$ के अन्तराल $[1, e^2]$ में अधिकतम और न्यूनतम मानों का अन्तर है :

- (A) 2 (B) e (C) e^2 (D) 1

Ans. B

Sol. $y = x(\ln x - 2)$

$$y' = x \left(\frac{1}{x} \right) + (\ln x - 2) = \ln x - 1$$

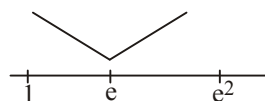
$$\frac{dy}{dx} = \ln x - 1 = 0 \quad \Rightarrow \quad x = e$$

now $f(1) = -2$

$f(e) = -e$ (least)

$f(e^2) = 0$ (greatest)

$\therefore$ difference $= 0 - (-e) = e$



50. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?

- (A) f is not differentiable at $x = 0$ (B) f is monotonic
(C) f is even (D) f has an extremum

फलन $f(x) = \cos^{-1}(2x^2 - 1)$ के लिए निम्न में से कौनसा सत्य नहीं है :

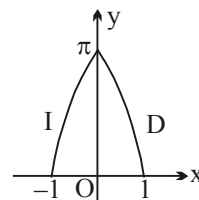
- (A) f , $x = 0$ पर अवकलनीय नहीं है (B) f एकदिष्ट है
(C) f समफलन है (D) f का एक चरम मान है

Ans. B

Sol. $\cos^{-1}(2x^2 - 1) = \begin{cases} 2 \cos^{-1} x & \text{for } x \in [0, 1] \\ 2\pi - 2 \cos^{-1} x & \text{for } x \in [-1, 0] \end{cases}$

f is increasing in $(-1, 0)$ and decreasing in $(0, 1)$

$\Rightarrow$ non monotonic $\Rightarrow$ (B) is wrong



51. The minimum value of $f(x, y) = x^2 - 4x + y^2 + 6y$ when x and y are subjected to the restrictions $0 \leq x \leq 1$ and $0 \leq y \leq 1$, is :

$f(x, y) = x^2 - 4x + y^2 + 6y$ का न्यूनतम मान जहाँ $0 \leq x \leq 1$ तथा $0 \leq y \leq 1$ है

- (A) -1 (B) -2 (C) -3 (D) -5

Ans. C

Sol. We have $f(x, y) = x^2 + y^2 - 4x + 6y$

Let $(x, y) = (\cos \theta, \sin \theta)$, then $\theta \in [0, \pi/2]$ and

$$f(x, y) = f(\theta) = \cos^2 \theta + \sin^2 \theta - 4 \cos \theta + 6 \sin \theta$$

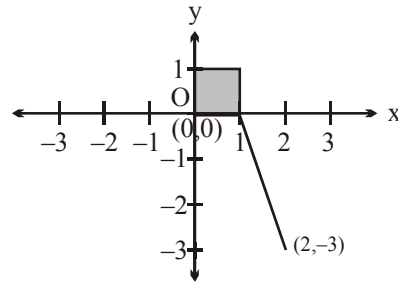
$$f'(\theta) = 6 \cos \theta + 4 \sin \theta > 0 \quad \forall \theta \in [0, \pi/2]$$

$\therefore f'(\theta)$ is strictly increasing in $[0, \pi/2]$

$$\therefore f(\theta)_{\min} = f(0) = 1 - 4 + 0 = -3 \text{ Ans.}]$$

Alternatively: $f(x, y) = (x-2)^2 + (y+3)^2 - 13$.

So, to minimise f over the unit square is the same as minimising the distance from a point in this square to the point of coordinates $(2, -3)$.



The closest point in the square to $(2, -3)$ is then $(1, 0)$.

So $f(1, 0) = -3$ must be the minimum.

52. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ has :

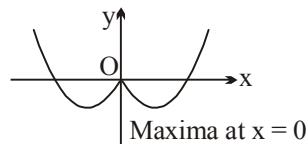
- (A) a maxima whenever $a > 0, b > 0$ (B) a maxima whenever $a > 0, b < 0$
(C) minima whenever $a > 0, b > 0$ (D) neither a maxima nor minima whenever $a > 0, b < 0$

माना $f(x) = ax^2 - b|x|$, जहाँ a तथा b अचर है तब $x = 0$ पर फलन $f(x)$:

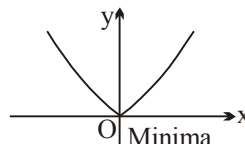
- (A) एक उच्चिष्ठ मान देगा जबकि $a > 0, b > 0$ (B) एक उच्चिष्ठ मान देता जबकि $a > 0, b < 0$
(C) एक निम्निष्ठ मान देगा जबकि $a > 0, b > 0$ (D) न तो उच्चिष्ठ न ही निम्निष्ठ देगा जबकि $a > 0, b < 0$

Ans. A

Sol. for $a > 0, b > 0$



for $a > 0, b < 0$



53. If a function f is continuous for all x and if f has a relative maximum at $(-1, 4)$ and a relative minimum at $(3, -2)$, then which of the following statements must be true?

- (A) The graph of f has a point of inflection somewhere between $x = -1$ and $x = 3$.
(B) $f'(-1) = 0$.

(C) The graph of f has a horizontal tangent line at $x = 3$.

(D) The graph of f intersect both co-ordinate axes.

यदि फलन f सभी x के लिये सतत है तथा f का स्थानीय उच्चिष्ठ $(-1, 4)$ पर है तथा स्थानीय निम्नष्ठ $(3, -2)$ पर है तो निम्न में से कौनसे कथन सत्य होना चाहिये?

(A) f का आरेख पर $x = -1$ एवं $x = 3$ के मध्य कही पर एक नति परिवर्तन बिन्दु होगा

(B) $f'(-1) = 0$

(C) f का आरेख में $x = 3$ पर स्पर्श रेखा क्षैतिज होगी

(D) f का आरेख दोनों निर्देशांक अक्षों को प्रतिच्छेद करता है

Ans. D

Sol. Because f is continuous for all x , the intermediate value theorem implies that the graph of f must intersect the x -axis. The graph must also intersect the y -axis since f is defined for all x , in particular, at $x = 0$. As f need not be differentiable. Hence A, B and C need not be correct.

54. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals :

यदि फलन $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, (जहाँ $a > 0$) के बिन्दु p तथा q पर क्रमशः उच्चिष्ठ एवं निम्निष्ठ हों तथा $p^2 = q$ हो, तो a का मान होगा :

(A) $1/2$

(B) 3

(C) 1

(D) 2

Ans. D

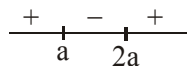
Sol. $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, $a > 0$

$$f'(x) = 6x^2 - 18ax + 12a^2$$

$$= 6(x^2 - 3ax + 2a^2)$$

$$= 6(x - a)(x - 2a)$$

$$p = a, \quad q = 2a$$



$$\Rightarrow a^2 = 2a$$

$$\Rightarrow a = 0 \text{ (rejected) or } a = 2$$

$$a = 2$$

55. If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is :

यदि p तथा q इस प्रकार से धनात्मक वास्तविक संख्या है कि $p^2 + q^2 = 1$, तब $(p + q)$ का अधिकतम मान होगा :

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\sqrt{2}$

Ans. D

Sol. Using A.M. $\geq$ G.M.

$$\frac{p^2 + q^2}{2} \geq p \cdot q$$

$$\Rightarrow pq \leq \frac{1}{2}$$

$$\Rightarrow (p + q)^2 = p^2 + q^2 + 2pq$$

$$\Rightarrow (p + q) \leq \sqrt{2}$$

56. Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then which one of the following holds? **[AIEEE-2008]**

(A) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$

(B) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(C) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(D) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$

माना त्रिघात बहुपद $x^3 - px + q$ के तीन विभिन्न वास्तविक मूल हैं जबकि $p > 0$ तथा $q > 0$ हैं। तो निम्न में से कौनसा एक सत्य है? **[AIEEE-2008]**

(A) त्रिपद का न्यूनतम मान $-\sqrt{\frac{p}{3}}$ पर है तथा अधिकतम मान $\sqrt{\frac{p}{3}}$ पर है

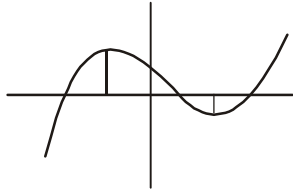
(B) त्रिपद का दोनों $\sqrt{\frac{p}{3}}$ तथा $-\sqrt{\frac{p}{3}}$ पर न्यूनतम मान है

(C) त्रिपद का दोनों $\sqrt{\frac{p}{3}}$ तथा $-\sqrt{\frac{p}{3}}$ पर अधिकतम मान है

(D) त्रिपद का $\sqrt{\frac{p}{3}}$ पर न्यूनतम मान तथा $-\sqrt{\frac{p}{3}}$ पर अधिकतम मान है

Ans. D

Sol. Graph of $y = x^3 - px + q$ cuts x-axis at three distinct points



$$\frac{dy}{dx} = 0 \Rightarrow x = \pm \sqrt{\frac{p}{3}}$$

$$\text{Maxima at } x = -\sqrt{\frac{p}{3}}, \text{ minima at } x = \sqrt{\frac{p}{3}}$$

57. Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$:

[AIEEE-2009]

- (A) $P(-1)$ is the minimum and $P(1)$ is the maximum of P
- (B) $P(-1)$ is not minimum but $P(1)$ is the maximum of P
- (C) $P(-1)$ is the minimum but $P(1)$ is not the maximum of P
- (D) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P

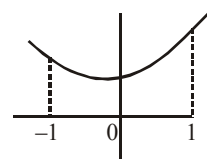
माना $P(x) = x^4 + ax^3 + bx^2 + cx + d$ के लिए $x = 0$ ही अकेला $P'(x) = 0$ का वास्तविक मूल है। यदि $P(-1) < P(1)$ है तब अंतराल $[-1, 1]$ में :

[AIEEE-2009]

- (A) P का न्यूनतम $P(-1)$ और अधिकतम $P(1)$ है
- (B) P का न्यूनतम $P(-1)$ नहीं है परन्तु अधिकतम $P(1)$ है
- (C) P का न्यूनतम $P(-1)$ है परन्तु अधिकतम $P(1)$ नहीं है
- (D) न तो P का न्यूनतम $P(-1)$ है और न ही P का अधिकतम $P(1)$ है

Ans. B

Sol. Graph of $P(x)$ under given conditions. It is clear that $P(x)$ has max. at 1 but not minimum at -1.



58. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} k - 2x, & \text{if } x \leq -1 \\ 2x + 3, & \text{if } x > -1 \end{cases}$$

If f has a local minimum at $x = -1$, then a possible value of k is :

[AIEEE-2010]

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ निम्न रूप में परिभाषित है

$$f(x) = \begin{cases} k - 2x, & \text{if } x \leq -1 \\ 2x + 3, & \text{if } x > -1 \end{cases} \quad \text{यदि फलन } f \text{ का स्थानीय निम्नतम } x = -1 \text{ पर है, तो } k \text{ का संभावित मान है :}$$

[AIEEE-2010]

- (A) 1
- (B) 0
- (C) $-\frac{1}{2}$
- (D) -1

Ans. D

Sol. f has a local minimum at $x = -1$

$$\therefore \lim_{x \rightarrow -1} f(x) \geq f(-1)$$

$$k + 2 \leq 1$$

$$k \leq -1$$

$$\therefore k = -1$$

59. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t dt$. Then f has :

[AIEEE-2011]

(A) local minimum at π and local maximum at 2π

(B) local maximum at π and local minimum at 2π

(C) local maximum at π and 2π

(D) local minimum at π and 2π

$x \in \left(0, \frac{5\pi}{2}\right)$, के लिए $f(x) = \int_0^x \sqrt{t} \sin t dt$ को परिभाषित कीजिए। तो f का :

[AIEEE-2011]

(A) स्थानीय निम्नतम मान π पर तथा स्थानीय उच्चतम मान 2π पर है

(B) स्थानीय उच्चतम मान π पर तथा स्थानीय निम्नतम मान 2π पर है

(C) स्थानीय उच्चतम मान π तथा 2π पर है

(D) स्थानीय निम्नतम मान π तथा 2π पर है

Ans. B

Sol. $f'(x) = \sqrt{x} \sin x$

$f(\pi)$ and $f(2\pi)$ are 0.

$$f'(x) \quad \begin{array}{c} + \quad \quad - \quad \quad + \\ \hline \pi \quad \quad 2\pi \end{array}$$

$\Rightarrow$ local maximum at $x = \pi$ and local minimum at $x = 2\pi$

60. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$ then :

[JEE(Main)2014]

यदि $x = -1$ तथा $x = 2$, $f(x) = \alpha \log |x| + \beta x^2 + x$ के चरमबिन्दु है तो :

[JEE(Main)2014]

$$(A) \alpha = 2, \beta = -\frac{1}{2} \quad (B) \alpha = 2, \beta = \frac{1}{2} \quad (C) \alpha = -6, \beta = \frac{1}{2} \quad (D) \alpha = -6, \beta = -\frac{1}{2}$$

Ans. A

Sol. $f(x) = \alpha \ln|x| + \beta x^2 + x$

$$(1). \quad f'(x) = \frac{\alpha}{x} + 2\beta x + 1.$$

$$= \frac{2\beta x^2 + x + \ell}{x}$$

Since $x = -1, 2$ are extreme points $\Rightarrow f'(x) = 0$ at these points.

Hence अतः $2\beta - 1 + \alpha = 0$

$$8\beta + 2 + \alpha = 0$$

$$\underline{\hspace{1.5cm}}$$

$$-6\beta - 3 = 0 \Rightarrow \beta = -\frac{1}{2} \quad \& \quad \alpha = 2.$$

61. Let $f(x)$ be a polynomial of degree four having extreme values at $x = 1$ and $x = 2$. If $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$

is equal to :

[JEE(Main)2015]

माना $f(x)$ घात 4 का एक बहुपद है जिसके $x = 1$ तथा $x = 2$ पर चरम मान है। यदि $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$ है, तो $f(2)$ बराबर है :

[JEE(Main)2015]

(A) -8

(B) -4

(C) 0

(D) 4

Ans. C

Sol. $f(x) = ax^4 + bx^3 + 2x^2$

$$f'(1) = 0$$

$$4a + 3b + 4 = 0$$

$$f'(2) = 0$$

$$32a + 12b + 8 = 0$$

$$\underline{\hspace{1.5cm}}$$

$$a = \frac{1}{2}, b = -2$$

$$f(x) = \frac{x^4}{2} - x^3 + 2x^2$$

$$f(2) = 8 - 16 + 8 = 0$$

62. Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$, $x \in \mathbb{R} - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of $h(x)$

is:

[JEE(Main)2018]

माना $f(x) = x^2 + \frac{1}{x^2}$ तथा $g(x) = x - \frac{1}{x}$, $x \in \mathbb{R} - \{-1, 0, 1\}$ हैं। यदि $h(x) = \frac{f(x)}{g(x)}$ है, तो $h(x)$ का स्थानीय न्यूनतम मान

है:

[JEE(Main)2018]

(A) $-2\sqrt{2}$

(B) $2\sqrt{2}$

(C) 3

(D) -3

Ans. B

Sol.
$$h(x) = \frac{x^2 + \frac{1}{x^2}}{1 - \frac{1}{x}}$$

$$h(x) = \frac{t^2 + 2}{t} = t + \frac{2}{t} \text{ where } t = x - \frac{1}{x}$$

$$h'(x) = \left(1 - \frac{2}{t^2}\right) \left(\frac{dt}{dx}\right)$$

$$h'(x) = \left(\frac{t^2 - 2}{t}\right) \left(1 + \frac{1}{x^2}\right)$$

Minima at $t = \sqrt{2}$

$$h(x)|_{\text{Min}} = \frac{2+2}{\sqrt{2}} = 2\sqrt{2}$$

63. (a) The total number of local maxima and local minima of the function $f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$ is :

(a) फलन $f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$ के कुल स्थानीय उच्चिष्ठ तथा स्थानीय निम्निष्ठ की संख्या है :

(A) 0

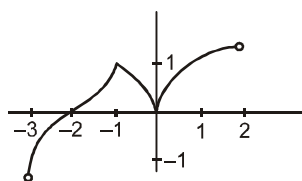
(B) 1

(C) 2

(D) 3

Ans. C

Sol.
$$f(x) = \begin{cases} (2+x)^2, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$$



one point of local maxima at $x = -1$

one point of local minima at $x = 0$

∴ (C) is the correct option

GEOMETRY BASED QUESTIONS ON MAXIMA AND MINIMA

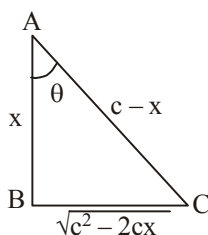
64. The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :

यदि किसी समकोण त्रिभुज में कर्ण तथा अन्य एक भुजा की लम्बाई का योग दिया गया है। तब त्रिभुज का क्षेत्रफल अधिकतम होगा यदि इनके मध्य का कोण है :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol. $A = \frac{x\sqrt{c^2 - 2cx}}{2}$
 $f(x) = 4A^2 = x^2(c^2 - 2cx)$



65. The length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2 + y^2 + 12x - 20y + 120 = 0$, is :

न्यूनतम पथ की लम्बाई जिसका प्रारम्भिक बिन्दु (2, 5) है तथा x-अक्ष को स्पर्श करता है तथा वृत्त $x^2 + y^2 + 12x - 20y + 120 = 0$ के किसी बिन्दु पर अन्त होता है :

- (A) 13 (B) $4\sqrt{10}$ (C) 15 (D) $6 + \sqrt{89}$

Ans. A

Sol. Circle with centre (-6, 10) and radius $= \sqrt{36 + 100 - 120} = 4$

Now let (a, 0) be a point on the x-axis.

If y is the distance from A to P and P to M

$$y = \sqrt{(a-2)^2 + 25} + \sqrt{(a+6)^2 + 100} - 4$$

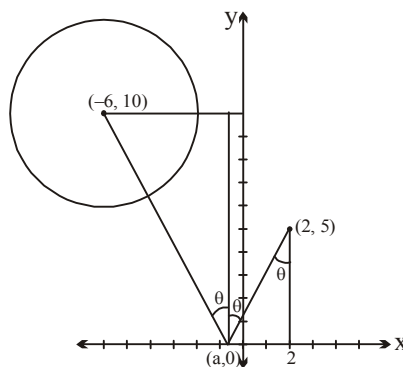
$$\frac{dy}{da} = \frac{2(a-2)}{2\sqrt{(a-2)^2 + 25}} + \frac{2(a+6)}{2\sqrt{(a+6)^2 + 100}}$$

$$\frac{dy}{da} \text{ can be zero only if } a-2 > 0$$

$$\text{and } a+6 < 0 \text{ not possible}$$

$$\text{or } a-2 < 0 \text{ and } a+6 > 0$$

$$\text{hence } a \in (-6, 2)$$



solving $\frac{dy}{da} = 0$, gives $a = 10$ (rejected) or $a = -\frac{2}{3}$

$$\begin{aligned} \text{hence } y_{\min} &= \sqrt{\frac{64}{9} + 25} + \sqrt{\frac{256}{9} + 100} - 4 \\ &= \frac{17}{3} + \frac{\sqrt{1156}}{3} - 4 = \frac{17}{3} + \frac{34}{3} - 4 = 17 - 4 = 13 \text{ Ans.} \end{aligned}$$

$$\text{Aliter: } \frac{a+2}{5} = \frac{6-a}{10} \Rightarrow 10a + 20 = 30 - 5a \Rightarrow 15a = 10 \Rightarrow a = \frac{2}{3}$$

Hence $P\left(\frac{-2}{3}, 0\right)$ for find $y_{\min}$.

66. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x . The maximum area enclosed by the park is :

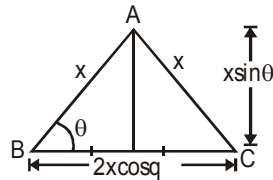
एक त्रिभुजाकार पार्क दो तरफ से बाड़े (fence) द्वारा घिरा है तथा तीसरी ओर एक नदी के सीधे किनारे द्वारा घिरा है। यदि बाड़े द्वारा घिरी भुजाओं की समान लम्बाई x है, तो पार्क का अधिकतम क्षेत्रफल है :

- (A) $\sqrt{\frac{x^3}{8}}$ (B) $\frac{1}{2}x^2$ (C) πx^2 (D) $\frac{3}{2}x^2$

Ans. B

Sol. A triangular park

$$\begin{aligned} \Delta &= \frac{1}{2} (2x \cos \theta) (x \sin \theta) \\ &= \frac{1}{2} x^2 \sin 2\theta \end{aligned}$$



$$\Delta_{\max.} = \frac{x^2}{2}$$

67. A right triangle is drawn in a semicircle of radius $\frac{1}{2}$ with one of its legs along the diameter. The maximum area of the triangle is :

$\frac{1}{2}$ त्रिज्या वाले अर्द्धवृत्त में एक समकोण त्रिभुज बनाया जाता है इसप्रकार की उसका आधार व्यास पर है तब त्रिभुज का अधिकतम क्षेत्रफल है :

- (A) $\frac{1}{4}$ (B) $\frac{3\sqrt{3}}{32}$ (C) $\frac{3\sqrt{3}}{16}$ (D) $\frac{1}{8}$

Ans. B

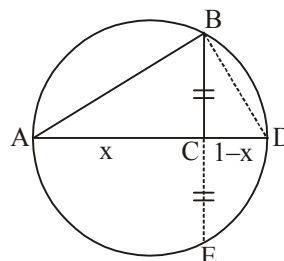
Sol. $(BC)(CE) = x(1-x)$ (property of circle) $[AC = x ; CD = 1-x]$ [12 & 13th test (09-10-2005)]

but $BC = CE$

$$\therefore BC = \sqrt{x(1-x)}$$

$$A = \frac{x\sqrt{x-x^2}}{2}. \text{ Now maximise } A.$$

$$A_{\max} \text{ occurs at } x = \frac{3}{4}$$



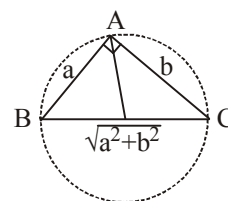
68. Two sides of a triangle are to have lengths 'a' cm & 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is :

किसी त्रिभुज की दो भुजाओं की लम्बाइयाँ 'a' cm तथा 'b' cm है। यदि त्रिभुज का क्षेत्रफल अधिकतम है तब भुजाओं 'a' तथा 'b' को मिलने वाले शीर्ष से डाली गई माध्यिका की लम्बाई है :

- (A) $\frac{1}{2}\sqrt{a^2+b^2}$ (B) $\frac{2a+b}{3}$ (C) $\sqrt{\frac{a^2+b^2}{2}}$ (D) $\frac{a+2b}{3}$

Ans. A

Sol. $A = (1/2)ab \sin\theta \Rightarrow A$ is maximum if $\theta = 90^\circ$



69. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is : [AIEEE-2012]

एक गोलाकार गुब्बारे में 4500π घन मी. हीलियम गैस भरी गई है। यदि गुब्बारे में छेद के कारण 72π घन मी. प्रति मिनट की दर से गैस रिस रही है, तो गैस रिसने के प्रारम्भ होने के 49 मिनट बाद, जिस दर से गुब्बारे की त्रिज्या घट रही है (मीटर प्रति मिनट में), हैं :

[AIEEE-2012]

- (A) $9/2$ (B) $9/7$ (C) $7/9$ (D) $2/9$

Ans. D

Sol. $V = \frac{4}{3} \pi r^3$

Initially $r = 4500 \pi$, $r = r_0$

$$4500 \pi = \frac{4}{3} \pi r_0^3 \Rightarrow \boxed{r_0 = 15\text{m}}$$

Now $\frac{dV}{dt} = \frac{4}{3} \pi (3r^2) \frac{dr}{dt}$

$$-72 \pi = 4 \pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{-18}{r^2} \dots\dots(i)$$

$$\int r^2 dr = - \int 18 dt \Rightarrow \frac{r^3}{3} = -18t + C$$

At $t = 0$, $r = 15 \text{ m}$

$$\text{So, } \frac{(15)^3}{3} = -18(0) + C \Rightarrow C = 1125$$

$$\Rightarrow r^3 = -54t + 3375 \dots\dots(ii)$$

At time $t = 49 \text{ min}$ $r = 9 \text{ m}$

from eq. (i)

$$\left(\frac{dr}{dt} \right)_{t=49} = \frac{-18}{(9)^2} = -2/9$$

(Negative sign shows decrement in radii)

70. A wire of length 2 units is cut into two parts which are bent respectively to form a square of side = x units and a circle of radius = r units. If the sum of the areas of the square and the circle so formed is minimum, then :

[JEE(Main)2016]

2 इकाई लम्बी एक तार को दो भागों में काट कर उन्हें क्रमशः x इकाई भुजा वाले वर्ग तथा r इकाई त्रिज्या वाले वृत्त के रूप में मोड़ा जाता है। यदि बनाये गये वर्ग तथा वृत्त के क्षेत्रफलों का योग न्यूनतम है, तो :

[JEE(Main)2016]

- (A) $x = 2r$ (B) $2x = r$ (C) $2x = (\pi + 4)r$ (D) $(4 - \pi)x = \pi r$

Ans. A

Sol. 

Perimeter of square = $4x$ and circumference of circle = $2 - 4x$

$$2\pi r = (2 - 4x)$$

$$A = x^2 + \pi \left(\frac{2 - 4x}{2\pi} \right)^2$$

$$\frac{dA}{dx} = 0$$

$$2x = \frac{2}{\pi}(2 - 4x)$$

$$2x = \frac{2(2\pi r)}{\pi}$$

$$x = 2r$$

71. Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower-bed, is : [JEE(Main)2017]

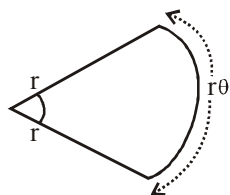
एक फूलों की बग़ीचा, जो एक वृत्त के त्रिज्य खंड के रूप में है, की घेराबंदी करने के लिए बीस मीटर तार उपलब्ध है। तो फूलों की बग़ीचा का अधिकतम क्षेत्रफल (वर्ग मी. में) है :

[JEE(Main)2017]

- (A) 25 (B) 30 (C) 12.5 (D) 10

Ans. A

Sol.



$$r + r + r\theta = 20$$

$$\theta = \frac{20 - 2r}{r}$$

$$A = \frac{1}{2}r^2\theta$$

$$A = \frac{r(20 - r)}{r}$$

$$\frac{dA}{dr} = 0 \Rightarrow r = 5$$

$$\Rightarrow \theta = 2$$

$$\text{Area} = \frac{1}{2}r^2\theta = 25$$

POINT OF INFLECTION & CONCAVITY OF GRAPH

72. If the function $f(x) = x^4 + bx^2 + 8x + 1$ has a horizontal tangent and a point of inflection for the same value of x then the value of b is equal to :

यदि $f(x) = x^4 + bx^2 + 8x + 1$ की द्वैतिज स्पर्श रेखा और नति परिवर्तन (inflection point) बिन्दु दोनों एक ही बिन्दु x पर मिलते हैं तब $b =$

- (A) -1 (B) 1 (C) 6 (D) -6

Ans. D

Sol. $f'(x) = 0$ and $f''(x) = 0$ for the same $x = x_1$ (say)

$$\text{now } f'(x) = 4x^3 + 2bx + 8$$

$$f'(x_1) = 2[2x_1^3 + bx_1 + 4] = 0 \dots(1)$$

$$f''(x_1) = 2[6x_1^2 + b] = 0 \dots(2)$$

$$\text{from (2) } b = -6x_1^2$$

substituting this value of b in (1)

$$2x_1^3 + (-6x_1^2) + 4 = 0 \Rightarrow 4x_1^3 = 4 \Rightarrow x_1 = 1. \text{ Hence } b = -6$$

73. The function $f(x) = 2x^3 + \alpha x^2 + \beta x + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{R}$ has local minimum at $P(\log_3 t^2, f(\log_3 t^2))$ and local maximum at $Q(\log_3 t, f(\log_3 t))$. If $R\left(\frac{5}{2}, f\left(\frac{5}{2}\right)\right)$ is the point of inflection, then 't' is equal to :

यदि फलन $f(x) = 2x^3 + \alpha x^2 + \beta x + \gamma$, जहाँ $\alpha, \beta, \gamma \in \mathbb{R}$ बिन्दु $P(\log_3 t^2, f(\log_3 t^2))$ और $Q(\log_3 t, f(\log_3 t))$ पर

क्रमशः स्थानीय निम्नष्ट और स्थानीय उच्चिष्ट मान देता है। यदि $R\left(\frac{5}{2}, f\left(\frac{5}{2}\right)\right)$ फलन का नतिपरिवर्तन बिन्दु (Point of inflection)

है तब 't' =

- (A) $3^{2/5}$ (B) $3^{1/5}$ (C) $3^{5/3}$ (D) $3^{3/5}$

Ans. C

Sol. We have $f'(x) = 6x^2 + 2\alpha x + \beta \Rightarrow f'(\log_3 t^2) = 0$ and $f'(\log_3 t) = 0$

$$\text{Now } \log_3 t^2 + \log_3 t = -\frac{2\alpha}{6} \quad (\text{sum of the roots})$$

$$\Rightarrow 3 \log_3 t = \frac{-\alpha}{3} \dots(i)$$

$$\text{Also, } f''(x) = 12x + 2\alpha$$

$$f''\left(\frac{5}{2}\right) = 0 \Rightarrow 12\left(\frac{5}{2}\right) + 2\alpha = 0 \Rightarrow \alpha = -15$$

$$\therefore 3 \log_3 t = \frac{15}{3} \Rightarrow t = 3^{5/3}$$

74. If the point (1, 3) serves as the point of inflection of the curve $y = ax^3 + bx^2$ then the value of 'a' and 'b' are :
 (A) $a = 3/2$ and $b = -9/2$ (B) $a = 3/2$ and $b = 9/2$
 (C) $a = -3/2$ and $b = -9/2$ (D) $a = -3/2$ and $b = 9/2$

यदि बिन्दु (1, 3) वक्र $y = ax^3 + bx^2$ पर नति परिवर्तन की तरह कार्य करता है, तो 'a' और 'b' का मान है :

- (A) $a = 3/2$ तथा $b = -9/2$ (B) $a = 3/2$ तथा $b = 9/2$
 (C) $a = -3/2$ तथा $b = -9/2$ (D) $a = -3/2$ तथा $b = 9/2$

Ans. D

Sol. $y = ax^3 + bx^2$
 $y' = 3ax^2 + 2bx$
 $y'' = 6ax + 2b$

for point of inflection $y'' = 0$

$$x = \frac{-b}{3a}$$

$$3a + b = 0 \quad \dots(i) \quad (\text{as } x = 1)$$

point satisfy the curve also, so

$$3 = a + b \quad \dots(ii)$$

from (i) & (ii)

$$a = -\frac{3}{2}, \quad b = \frac{9}{2}$$

SUBJECTIVE / INTEGER TYPE QUESTIONS

TANGENT AND NORMAL

TANGENT AND NORMAL AT A POINT

1. Find the equation of the normal to the curve $y = (1+x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$.

वक्र $y = (1+x)^y + \sin^{-1}(\sin^2 x)$ के बिन्दु $x = 0$ पर अभिलम्ब का समीकरण ज्ञात कीजिये।

Ans. $x + y - 1 = 0$

Sol. Given curve is $y = (1+x)^y + \sin^{-1}(\sin^2 x)$

$$y = e^{y \log(1+x)} + \sin^{-1}(\sin^2 x)$$

$$\Rightarrow \frac{dy}{dx} = e^{y \log(1+x)} \left(\frac{y}{1+x} + \log(1+x) \frac{dy}{dx} \right) + \frac{\sin 2x}{\sqrt{1-\sin^4 x}}$$

when $x = 0$, then $y = 1$.

So, the point is $(0, 1)$

$$\left(\frac{dy}{dx} \right)_{(0,1)} = 1$$

So, slope of the normal $= -1$

Hence, the equation of the normal is

$$y - 1 = -1(x - 0) = -x$$

$$\Rightarrow x + y = 1$$

2. Find the point of intersection of the tangents drawn to the curve $x^2 y = 1 - y$ at the points where it is intersected by the curve $xy = 1 - y$.

वक्र $x^2 y = 1 - y$ एवं वक्र $xy = 1 - y$ के प्रतिच्छेदन बिन्दुओं पर वक्र $x^2 y = 1 - y$ पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु ज्ञात कीजिये।

Ans. $(0, 1)$

Sol. Points of intersection of curve $x^2 y = xy$ are

$$(0, 1), (1, \frac{1}{2})$$

The equation of tangent at $(0, 1)$

$$2xy + x^2 y' = -y'$$

$$y' = 0$$

equation is $y - 1 = 0 \Rightarrow y = 1$

equation of tangent at $(1, \frac{1}{2})$ is $2xy + x^2y' = -y'$

$$1 + y' = -y' \Rightarrow y' = -\frac{1}{2}$$

$$y - \frac{1}{2} = -\frac{1}{2}(x - 1)$$

Put $y = 1$ $\frac{1}{2} = -\frac{x}{2} + \frac{1}{2}$

$$x = 0$$

Point of intersection of tangent is $(0, 1)$

3. Find all the lines that pass through the point $(1, 1)$ and are tangent to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$.

बिन्दु $(1, 1)$ से गुजरने वाले सभी रेखाएँ ज्ञात कीजिये जो एक वक्र पर स्पर्श रेखाएँ हैं, जिसका प्राचलिक समीकरण $x = 2t - t^2$ तथा $y = t + t^2$ से दिया जाता है।

Ans. $x = 1$ when $t = 1$, $m \rightarrow \infty$; $5x - 4y = 1$ if $t \neq 1$, $t = 1/3$

Sol. Let the tangent at (α, β)

Now, $\frac{dy}{dx} = \frac{2t+1}{2-2t}$

$$\Rightarrow \frac{\beta-1}{\alpha-1} = \frac{2t+1}{2-2t}$$

$$\Rightarrow \frac{t+t^2-1}{2t-t^2-1} = \frac{2t+1}{2-2t}$$

$$\Rightarrow \frac{t^2+t-1}{t^2-2t+1} = \frac{2t+1}{2(t-1)}$$

$$\Rightarrow \frac{t^2+t-1}{(t-1)^2} = \frac{2t+1}{2(t-1)}$$

$$\frac{t^2+t-1}{(t-1)} = \frac{2t+1}{2}$$

$$\Rightarrow 2t^2 + 2t - 2 = 2y^2 - t - 1$$

$$\Rightarrow 2t - 2 = -t - 1$$

$$\Rightarrow 3t = 1$$

$$\Rightarrow t = \frac{1}{3}$$

$$\text{Now, } \frac{dy}{dx} = \frac{2t+1}{2-2t} = \frac{2+3}{6-2} = \frac{5}{4}$$

Thus, the equation of the tangent is

$$y - 1 = \frac{5}{4}(x - 1)$$

$$\Rightarrow 4y - 4 = 5(x - 1)$$

$$\Rightarrow 5x - 4y = 1$$

4. A function is defined parametrically by the equations

$$f(t) = x = \begin{cases} 2t + t^2 \sin \frac{1}{t} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases} \quad \text{and } g(t) = y = \begin{cases} \frac{1}{t} \sin t^2 & \text{यदि } t \neq 0 \\ 0 & \text{यदि } t = 0 \end{cases}$$

Find the equation of the tangent and normal at the point for $t = 0$ if exist.

एक फलन को प्राचलिक समीकरणों से परिभाषित किया गया

$$f(t) = x = \begin{cases} 2t + t^2 \sin \frac{1}{t} & \text{यदि } t \neq 0 \\ 0 & \text{यदि } t = 0 \end{cases} \quad \text{तथा } g(t) = y = \begin{cases} \frac{1}{t} \sin t^2 & \text{यदि } t \neq 0 \\ 0 & \text{यदि } t = 0 \end{cases}$$

समीकरण के $t = 0$ पर स्पर्श रेखा एवं अभिलम्ब का समीकरण लिखिए (यदि अस्तित्व में हो)

Ans. T : $x - 2y = 0$; N : $2x + y = 0$

Sol. At $t = 0$ the point is origin

$$\frac{dx}{dt} = \lim_{t \rightarrow 0} \frac{2t + t^2 \sin 1/t - 0}{t} = 2$$

$$\frac{dy}{dt} = \lim_{t \rightarrow 0} \frac{\frac{1}{t} \sin t^2}{t} = 1$$

$$\frac{dy}{dx} = \frac{1}{2}$$

$$\text{equation of tangent is } y - 0 = \frac{1}{2}(x - 0)$$

$$\text{equation of normal is } y - 0 = -2(x - 0)$$

5. Prove that the segment of the normal to the curve $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ contained between the co-ordinate axes is equal to $2a$.

सिद्ध कीजिये कि वक्र $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ पर अभिलम्ब द्वारा अक्षों के मध्य रेखा खण्ड $2a$ है।

Ans.

Sol.
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3a \cos^2 t \sin t}{2a \cos t + a \cos^3 t - 2a \sin^2 t \cos t}$$

$$= \frac{3a \cos^2 t \sin t}{2a \cos t (1 - \sin^2 t) + a \cos^3 t}$$

$$= \frac{3a \cos^2 t \sin t}{2a \cos^2 t + a \cos^3 t}$$

$$= \frac{3a \cos^2 t \sin t}{3a \cos^3 t}$$

$$\frac{dy}{dx} = \tan t, \text{ equation of normal is}$$

$$y + a \cos^3 t = -\frac{\cot t}{\sin t} (x - 2a \sin t - a \sin t \cos^2 t)$$

$$\therefore y \sin t + a \cos^3 t \sin t = -x \cos t + 2a \sin t \cot t + a \sin t \cos^3 t$$

$$x \cot t + y \sin t = 2a \sin t \cos t$$

It meets axis at $A(2a \sin t, 0)$ and $B(0, 2a \cos t)$

Then $AB = 2a$.

6. Determine a differentiable function $y = f(x)$ which satisfies $f'(x) = [f(x)]^2$ and $f(0) = -\frac{1}{2}$. Find also the equation of the tangent at the point where the curve crosses the y-axis.

अवकलनीय फलन $y = f(x)$ ज्ञात कीजिये यदि $f'(x) = [f(x)]^2$ तथा $f(0) = -\frac{1}{2}$ है। साथ ही उस बिन्दु पर स्पर्श रेखा बताइए जहाँ वक्र y -अक्ष को प्रतिच्छेद करता है।

Ans. $f(x) = -\frac{1}{x+2}$; $x - 4y = 2$

Sol.

7. The curve $y = ax^3 + bx^2 + cx + 5$, touches the x -axis at $P(-2, 0)$ & cuts the y -axis at a point Q where its gradient is 3. Find a , b , c .

वक्र $y = ax^3 + bx^2 + cx + 5$, x -अक्ष को $P(-2, 0)$ पर स्पर्श करता है तथा y -अक्ष को उस बिन्दु Q पर प्रतिच्छेद करता है जहाँ उसका ढाल 3 है। a , b , c का मान ज्ञात कीजिये।

Ans. $a = -1/2$; $b = -3/4$; $c = 3$

Sol. $y(x) = ax^3 + bx^2 + cx + 5$ is passing through $(-2, 0)$ then $8a - 4b + 2c = 5$ _____(1)

$y'(x) = 3ax^2 + 2bx + c$ touches x -axis at $(-2, 0)$

$12a - 4b + c = 0$ _____(2)

again, for $x = 0$, $y(x) = 5$

$c = 3$ _____(3)

Solving eq. (1), (2) and (3) $a = -\frac{1}{2}$, $b = -\frac{3}{4}$

ULTIMATE CONCEPT

8. The tangent to $y = ax^2 + bx + \frac{7}{2}$ at $(1, 2)$ is parallel to the normal at the point $(-2, 2)$ on the curve $y = x^2 + 6x + 10$. Find the value of a and b .

$y = ax^2 + bx + \frac{7}{2}$ के $(1, 2)$ पर स्पर्श रेखा वक्र $y = x^2 + 6x + 10$ के बिन्दु $(-2, 2)$ पर अभिलम्ब के समानान्तर है। a तथा b के मान ज्ञात कीजिये।

Ans. $a = 1$, $b = \frac{-5}{2}$

Sol. $2 = a + b + \frac{7}{2}$ _____(1)

$$\frac{dy}{dx} = 2x + 6 \text{ at } -2, 2$$

$$\Rightarrow \frac{dy}{dx} = 2ax + b$$

$$\Rightarrow m = 2a + b$$

$$2a + b = -\frac{1}{2}$$
 _____(2)

from (1) and (2) $a = 1, b = -\frac{5}{2}$

9. Find all the tangents to the curve $y = \cos(x + y), -2\pi \leq x \leq 2\pi$, that are parallel to the line $x + 2y = 0$.

वक्र $y = \cos(x + y), -2\pi \leq x \leq 2\pi$ पर सभी स्पर्श रेखाओं का समीकरण लिखिए जो रेखा $x + 2y = 0$ के समान्तर है।

Ans. $x + 2y = \pi/2$ & $x + 2y = -3\pi/2$

Sol. Slope $= -\frac{1}{2}$

$$\frac{dy}{dx} = -\sin(x + y) (1 + y')$$

$$\Rightarrow -\frac{1}{2} = -\sin(x + y) \left(1 - \frac{1}{2}\right)$$

$$\Rightarrow \sin(x + y) = 1$$

$$\Rightarrow \cos(x + y) = 0 \Rightarrow y = 0$$

$$\cos x = 0$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, -\frac{\pi}{2}, -\frac{3\pi}{2}$$

$$\sin x = 1 \text{ is possible for } x = \frac{\pi}{2} \text{ or } -\frac{3\pi}{2}$$

$$\text{Equation are : } y - 0 = -\frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

and $y - 0 = -\frac{1}{2} \left(x + \frac{3\pi}{2} \right)$

10. Tangent at a point P_1 [other than $(0, 0)$] on the curve $y = x^3$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 and so on. Show that the abscissae of $P_1, P_2, P_3, \dots, P_n$, form a GP. Also find the ratio $\frac{\text{area}(P_1 P_2 P_3)}{\text{area}(P_2 P_3 P_4)}$.

वक्र $y = x^3$ के बिन्दु P_1 [(0, 0) के अलावा] पर स्पर्श रेखा वक्र को P_2 पर मिलती है तथा P_2 पर स्पर्श रेखा वक्र को P_3 पर मिलती है, इस प्रकार आगे क्रम चलता है। सिद्ध कीजिये की $P_1, P_2, P_3, \dots, P_n$ के भुज गुणोत्तर श्रेणी में है। साथ ही अनुपात $\frac{\text{area}(P_1 P_2 P_3)}{\text{area}(P_2 P_3 P_4)}$ भी ज्ञात करें।

Ans. 1/16

Sol. Let any point P_1 on

$$y = x^3 \text{ be } (h, h^3).$$

Then tangent at P_1 is

$$y - h^3 = 3h^2(x - h) \quad \text{_____ (1)}$$

It meets $y = x^3$ at P_2 .

On putting the value of y in Eq. (1)

$$x^3 - h^3 = 3h^2(x - h)$$

$$\Rightarrow (x - h)(x^2 + xh + h^2) = 3h^2(x - h)$$

$$\Rightarrow x^2 + xh + h^2 = 3h^2 \text{ or } x = h$$

$$\Rightarrow x^2 + xh - 2h^2 = 0$$

$$\Rightarrow (x - h)(x + 2h) = 0$$

$$\Rightarrow x = h \text{ or } x = -2h$$

Therefore, $x = -2h$ is the point P_2 ,

Which implies $y = -8h^3$

Hence, point $P_2 \equiv (-2h, -8h^3)$

Again, tangent at P_2 is

$$y + 8h^3 = 3(-2h)^2(x + 2h)$$

It meets $y = x^3$ at P_3

$$\Rightarrow x^3 + 8h^3 = 12h^2(x + 2h)$$

$$\Rightarrow x^2 - 2hx - 8h^2 = 0$$

$$\Rightarrow (x + 2h)(x - 4h) = 0$$

$$\Rightarrow x = 4h$$

$$\Rightarrow y = 64h^3$$

Therefore, $P_3 \equiv (4h, 64h^3)$

Similarly, we get $P_4 \equiv (-8h, -8^3 h^3)$

Hence, the abscissa are $h, -2h, 4h, -8h, \dots$ which form a GP.

Let $D' = \Delta P_1 P_2 P_3$ and $D'' = \Delta P_2 P_3 P_4$

$$\frac{D'}{D''} = \frac{\Delta P_1 P_2 P_3}{\Delta P_2 P_3 P_4} = \frac{\frac{1}{2} \begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}{\frac{1}{2} \begin{vmatrix} -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \\ -8h & -512h^3 & 1 \end{vmatrix}}$$

$$= \frac{\frac{1}{2} \begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}{\frac{1}{2} \times (-2) \times (-8) \begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}$$

$$\Rightarrow C_2 \frac{D'}{D''} = \frac{1}{16} \text{ which is the required ratio.}$$

11. Find the gradient of the line passing through the point (2, 8) and touching the curve $y = x^3$.

बिन्दु (2, 8) से गुजरने वाली रेखा जो वक्र $y = x^3$ को स्पर्श करती है, की प्रवणता ज्ञात कीजिये।

Ans. 3, 12

Sol. $y' = 3x^2$

$$3x_1^2 = \frac{x_1^3 - 8}{x_1 - 2} \Rightarrow x_1 = -1 \text{ or } 2$$

$$y' = 3 \text{ or } y' = 12$$

12. The graph of a certain function f contains the point (0, 2) and has the property that for each number ' p ' the line tangent to $y = f(x)$ at $(p, f(p))$ intersect the x-axis at $p + 2$. Find $f(x)$.

किसी फलन f के आरेख पर बिन्दु (0, 2) स्थित है तथा उसका एक गुणधर्म है कि प्रत्येक संख्या ' p ' के लिये $y = f(x)$ पर बिन्दु $(p, f(p))$ पर स्पर्श रेखा x-अक्ष को $p + 2$ पर प्रतिच्छेद करती है। $f(x)$ ज्ञात कीजिए।

Ans. $2e^{-x/2}$

Sol. point $(p, f(p))$ $\text{slop} = f'(p)$

$$\text{Eq. : } y - f(p) = f'(p)(x - p)$$

$$y = 0 \quad x = p + 2 \quad -f(p) = f'(p)(p + 2 - p)$$

$$-f(p) = f'(p) \cdot 2$$

$$-\int \frac{f'(p)}{f(p)} \cdot dp^2 \quad \int \frac{1}{2} dp$$

$$-\ln f(p) = \frac{p}{2} + c$$

$$-\ln 2 = c$$

$$c = -\ln 2$$

$$-\ln f(p) = \frac{p}{2} - \ln 2$$

$$\ln 2 - \ln f(p) = \frac{p}{2}$$

MINIMUM DISTANCE BETWEEN TWO CURVES

13. Let $P(a, b)$ be a point on the parabola $y = 4x - x^2$ and is the point nearest to the point $A(-1, 4)$. Find $(a + b)$.

यदि वक्र $y = 4x - x^2$ पर बिन्दु $P(a, b)$ इस प्रकार से है कि यह $A(-1, 4)$ से न्यूनतम दूरी पर है तो $(a + b)$ ज्ञात कीजिये।

Ans. 4

Sol. Let the point P be $a = x_1$ and $b = 4x_1 - x_1^2$

$$\frac{4x_1 - x_1^2 - 4}{x_1 + 1} = - \frac{1}{\left(\frac{dy}{dx}\right)_{x=x_1}} = - \frac{1}{4 - 2x_1}$$

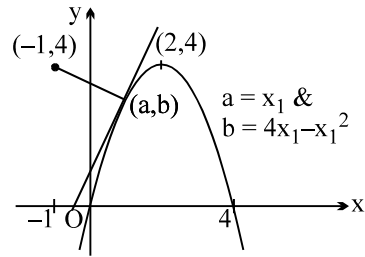
$$(2x_1 - 4)(4x_1 - x_1^2 - 4) = x_1 + 1$$

$$-2x_1^3 + 12x_1^2 - 25x_1 + 15 = 0$$

$$2x_1^3 - 12x_1^2 + 25x_1 - 15 = 0$$

$$\therefore x_1 = 1 ; \quad y_1 = 4 - 1 = 3$$

$$\therefore (a, b) = (1, 3) \Rightarrow a + b = 4$$

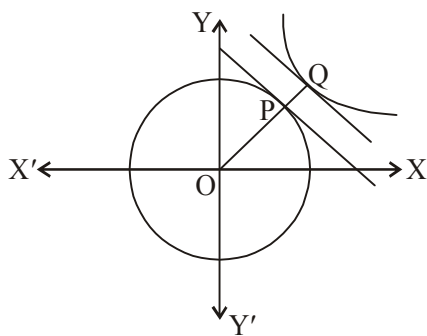


14. Find the shortest distance between curves $xy = 9$ and $x^2 + y^2 = 1$

वक्रों $xy = 9$ और $x^2 + y^2 = 1$ के बीच न्यूनतम दूरी ज्ञात कीजिए।

Ans. $3\sqrt{2} - 1$

Sol.



Given curves are $x^2 + y^2 = 1$ and $xy = 9$

Since both the tangents at P and Q are parallel, so their slopes are same

$$-\frac{x}{y} = -\frac{9}{x^2}$$

$$\Rightarrow x^3 = 9y$$

$$\Rightarrow x^3 = 9 \cdot \frac{9}{x} = \frac{81}{x}$$

$$\Rightarrow x^4 = 81 \Rightarrow x = 3$$

when $x = 3$, then $y = 3$

Thus, the point $Q = (3, 3)$

Therefore, the shortest distance

$$= PQ$$

$$= OQ - OP$$

$$= \sqrt{(9+9)} - \sqrt{2}$$

$$= 3\sqrt{2} - \sqrt{2}$$

$$= 2\sqrt{2}$$

15. Find the point on hyperbola $3x^2 - 4y^2 = 72$ which is nearest to the straight line $3x + 2y + 1 = 0$

अतिपरवलय $3x^2 - 4y^2 = 72$ पर वह बिन्दु ज्ञात कीजिए जो कि सरल रेखा $3x + 2y + 1 = 0$ से न्यूनतम दूरी पर है।

Ans. $(-6, 3)$

Sol. $3x^2 - 4y^2 = 72$

$$m = -\frac{3}{2}$$

$$6x - 8y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{3x}{4y}$$

$$\frac{3x_0}{4y_0} = -\frac{3}{2}$$

$$x_0 = -2y_0$$

$$8y_0^2 = 72$$

$$y = \pm 3$$

$$3x + 2y + c = 0$$

$$c = -3x_0 - 2y_0$$

$$|c - 1| \text{ is min at } y_0 = 4y_0$$

$$|c - 1| \text{ is minimum at } c = 12 \text{ and } y_0 = 3$$

$$x_0 + y_0 = 3 + 3 \cdot (-2) = -3$$

ANGLE OF INTERSECTION BETWEEN TWO CURVES

16. Find the angle of intersection of the curves $y = 2 \sin^2 x$ and $y = \cos 2x$.

वक्र $y = 2 \sin^2 x$ और वक्र $y = \cos 2x$ के प्रतिच्छेद कोण का मान ज्ञात कीजिये।

Ans. $\frac{\pi}{3}$

Sol. Angle between two curves is basically the angle between their tangents at the point of intersection.

Point of intersection of the curves will be $\cos 2x = 2 \sin^2 x$

$$\Rightarrow \cot^2 x - 1 = 2$$

$$\Rightarrow \cot x = \sqrt{3}$$

$$x = \frac{\pi}{6}$$

Slope of both the curve at the point of intersection is given by

$$m_1 = -3 \sin 2x = -\sqrt{3}$$

$$m_2 = 2 \sin x \cos x = \sqrt{3}$$

$$\text{Applying the formula, } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\text{This gives } \theta = \frac{\pi}{3}$$

17. Find the angle of intersection of the curves $y^2 = 4ax$ and $x^2 = 32ay$.

वक्रों $y^2 = 4ax$ और $x^2 = 32ay$ के प्रतिच्छेद कोण का मान ज्ञात कीजिये।

Ans. $\frac{\pi}{2}, \tan^{-1}\left(\frac{3}{5}\right)$

Sol. $y^2 = 4x$ and $x^2 = 32y$ intersect at $(16, 8)$ at an angle θ

Now slope of the tangent to $y^2 = 4x$ at $(16, 8)$ is given by

$$m_1 = \left(\frac{dy}{dx}\right)(16, 8) = \left(\frac{4}{2y}\right)(16, 8) = \frac{1}{4}$$

Again slope of tangent to $x^2 = 32y$ at $(16, 8)$

$$\text{is given by } m_2 = \left(\frac{dy}{dx}\right)(16, 8) = \left(\frac{2x}{32}\right)(16, 8) = 1$$

$$\text{We know } \tan \theta = \frac{m_2 - m_1}{m_2 + m_1}$$

$$= \frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} = \frac{3}{5}$$

$$\therefore \theta = \tan^{-1}\left(\frac{3}{5}\right)$$

18. Show that the curves $\frac{x^2}{a^2 + K_1} + \frac{y^2}{b^2 + K_1} = 1$ and $\frac{x^2}{a^2 + K_2} + \frac{y^2}{b^2 + K_2} = 1$ intersect orthogonally.

सिद्ध कीजिये कि वक्र $\frac{x^2}{a^2 + K_1} + \frac{y^2}{b^2 + K_1} = 1$ तथा $\frac{x^2}{a^2 + K_2} + \frac{y^2}{b^2 + K_2} = 1$ लम्बवत् प्रतिच्छेद करते हैं।

Ans.

Sol. $\frac{x^2}{a^2 + k_1} + \frac{y^2}{b^2 + k_1} = 1$ _____(1)

$$\frac{x^2}{a^2 + k_2} + \frac{y^2}{b^2 + k_2} = 1$$
 _____(2)

$$m_1 \Rightarrow \frac{dy}{dx} = -\left(\frac{b^2 + k_1}{a^2 + k_1}\right) \frac{x}{y} \quad \text{_____ (3) from}$$

(1)

$$m_1 \Rightarrow \frac{dy}{dx} = -\left(\frac{b^2 + k_2}{a^2 + k_2}\right) \frac{x}{y} \quad \text{_____ (4) from}$$

(2)

$$m_1.m_2 \Rightarrow \frac{(b^2 + k_1)(b^2 + k_2)x^2}{(a^2 + k_1)(a^2 + k_2)y^2} \quad \text{_____ (5)}$$

from eq(1) – eq(2)

$$x^2 \left(\frac{1}{a^2 + k_1} - \frac{1}{a^2 + k_2} \right) + y^2 \left(\frac{1}{b^2 + k_1} - \frac{1}{b^2 + k_2} \right) = 0$$

$$\frac{(b^2 + k_1)(b^2 + k_2)}{(a^2 + k_1)(a^2 + k_2)} = \frac{-y^2}{x^2} \quad \text{_____ (6)}$$

Putting value of eq (6) in eq (5)

$$m_1.m_2 = -1$$

Hence the angle between two curve is $\frac{\pi}{2}$.

19. If the two curves $C_1 : x = y^2$ and $C_2 : xy = k$ cut at right angles find the value of k .

यदि दो वक्र $C_1 : x = y^2$ तथा $C_2 : xy = k$ एक दूसरे को समकोण पर प्रतिच्छेद करते हैं तो k का मान ज्ञात कीजिये।

Ans. $\pm \frac{1}{2\sqrt{2}}$

Sol. By solving the equations $x = y^2$ and $xy = k$

$$\Rightarrow y^3 = k,$$

$$\Rightarrow y = k^{\frac{1}{3}}$$

$$\Rightarrow x = k^{\frac{2}{3}}$$

$\Rightarrow$ Point of intersection of the two curves is $\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)$

Slope of tangent to the curve $x = y^2$ is $\frac{dy}{dx} = \frac{1}{2y}$

$$\Rightarrow m_1 = \left(\frac{dy}{dx}\right)_{\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)} = \frac{1}{2k^{\frac{1}{3}}}$$

Slope of tangent to the curve $xy = k$ is $\Rightarrow \frac{dy}{dx} = \frac{-k}{x^2}$

$$\Rightarrow m_2 = \left(\frac{dy}{dx}\right)_{\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)} = \frac{-k}{k^{\frac{4}{3}}} = -\frac{1}{k^{\frac{1}{3}}}$$

$\therefore$ the curves are orthogonal $\Rightarrow m_1 \cdot m_2 = -1$

$$\Rightarrow \frac{1}{2k^{\frac{1}{3}}} \cdot -\frac{1}{k^{\frac{1}{3}}} = -1$$

$$\Rightarrow k^{\frac{2}{3}} = \frac{1}{2}$$

$$\Rightarrow k^2 = \frac{1}{8}$$

SOME IMPORTANT LENGTHS (TANGENT SUBTANGENT, NORMAL SUBNORMAL)

20. Find the value of n so that the subnormal at any point on the curve $xy^n = a^{n+1}$ may be constant.

वक्र $xy^n = a^{n+1}$ में n का मान ज्ञात कीजिये यदि उस पर अधोलम्ब की लम्बाई नियत है।

Ans. $n = -2$

Sol. $xy^n = a^{n+1}$, $a \neq 0$

$$y^n + nxy^{n-1} \frac{dy}{dx} = 0$$

$$\text{i.e.} \quad \frac{dy}{dx} = -\frac{y}{nx}$$

∴ length of subnormal

$$= \left| y \cdot \frac{dy}{dx} \right| = \left| \frac{y^2}{nx} \right| = \left| \frac{y^2 \cdot y^n}{an^{n+1}} \right|$$

$$= \left| \frac{y^{n+2}}{na^{n+1}} \right| \text{ is constant only if } n = -2.$$

21. Show that in the curve $y = a \cdot \ln(x^2 - a^2)$, sum of the length of tangent & subtangent varies as the product of the coordinates of the point of contact.

सिद्ध कीजिए की वक्र $y = a \cdot \ln(x^2 - a^2)$ में स्पर्श रेखा एवं अधःस्पर्शी की लम्बाई का योग बिन्दु के निर्देशांको के गुणन के सापेक्ष समानुपाती है।

Ans.

Sol. Given, $y = a \log(x^2 - a^2)$

$$\frac{dy}{dx} = a \cdot \frac{2x}{x^2 - a^2} = \frac{2ax}{x^2 - a^2}$$

$$\therefore \text{Length of subtangent} = y \frac{dx}{dy}$$

$$= y \cdot \frac{(x^2 - a^2)}{2ax}$$

$$\text{Length of tangent} = y \sqrt{1 + \left(\frac{dx}{dy} \right)^2}$$

$$= y \sqrt{1 + \left(\frac{x^2 - a^2}{2ax} \right)^2}$$

$$= y \sqrt{\frac{4a^2x^2 + (x^2 - a^2)^2}{4a^2x^2}}$$

$$= y \cdot \frac{y}{2ax} \sqrt{(x^2 + a^2)^2} = y \frac{(x^2 + a^2)}{2ax}$$

$$\text{Hence, sum of lengths of tangents and subtangents} = y \frac{(x^2 - a^2)}{2ax} + y \frac{(x^2 - a^2)}{2ax}$$

$$= \frac{y}{2ax} (x^2 - a^2 + x^2 - a^2)$$

$$= \frac{xy}{a} = \frac{1}{a} (xy)$$

$\therefore$ Sum of lengths of t and s a(xy).

RATE MEASURE

22. A man 1.5 m tall walks away from a lamp post 4.5 m high at the rate of 4 km/hr.

(i) how fast is the farther end of the shadow moving on the pavement ?

(ii) how fast is his shadow lengthening?

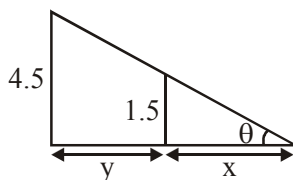
एक आदमी, जिसकी लम्बाई 1.5 m है, एक प्रकाश स्तम्भ, जिसकी लम्बाई 4.5 m है, से दूर 4 km/hr की गति से जा रहा है।

(i) किस गति से उसकी परछाई का अंतिम छोर आगे बढ़ रहा है?

(ii) किसी दर से उसकी परछाई की लम्बाई बढ़ रही है?

Ans. (i) 6 km/h ; (ii) 2 km/hr

Sol.



$$\frac{4.5}{x+y} = \frac{1.5}{x} \Rightarrow 3x = x + y \Rightarrow 2x = y$$

$$\frac{2dx}{dt} = \frac{dy}{dt}$$

$$\frac{dy}{dt} = 4 \text{ is given then } \frac{dx}{dt} = 2$$

$$(a) \quad \frac{d}{dt}(x+y) = \frac{dx}{dt} + \frac{dy}{dt} = 4 + 2 = 6 \text{ km/hr.}$$

$$(b) \quad \text{Shadow lengthening} = \frac{dx}{dt} = 2 \text{ km/hr.}$$

23. A particle moves along the curve $6y = x^3 + 2$. Find the points on the curve at which the y coordinate is changing 8 times as fast as the x coordinate.

एक कण वक्र $6y = x^3 + 2$ पर गमन करता है। यदि वक्र पर बिन्दु जिनके y निर्देशांक x निर्देशांक से 8 गुना तेजी से बदल रहा है तो वह बिन्दु ज्ञात कीजिये।

Ans. $(4, 11)$ & $(-4, -31/3)$

Sol. Let the coordinate on the curve be (x, y)

According to question

$$\frac{dy}{dx} = 8$$

$$\Rightarrow \frac{d}{dx} \left(\frac{x^3 + 2}{6} \right) = 8$$

$$\Rightarrow \frac{1}{6} \cdot (3x^2) = 8$$

$$\Rightarrow x^2 = 16 \quad \Rightarrow x = \pm 4$$

When $x = 4$

$$y = \left(\frac{x^3 + 2}{6} \right) = \left(\frac{64 + 2}{6} \right) = \frac{66}{6} = 11$$

When $x = -4$

$$y = \left(\frac{x^3 + 2}{6} \right) = \left(\frac{-64 + 2}{6} \right) = \frac{-62}{6} = -\frac{31}{3}$$

Therefore, required points are $(4, 11)$ and $\left(-4, -\frac{31}{3}\right)$

24. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of $1.5 \text{ cm}^3/\text{min}$. Find the rate at which level of water in the cone is rising, when the depth of water is 4 cm.

एक उलटे शंकु की गहराई 10 cm है तथा आधार की त्रिज्या 5 cm है, में जल $1.5 \text{ cm}^3/\text{min}$ की दर से भरा जा रहा है। जल की सतह की बढ़ने की दर ज्ञात कीजिये जब जल की गहराई 4 cm हो।

Ans. $3/8 \pi \text{ cm/min}$

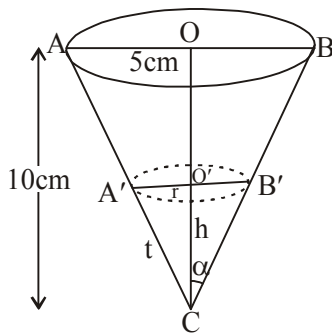
Sol. Let α be the semi-vertical angle of the cone CAB whose height CO is 10 cm and radius OB = 5 cm.

Then

$$\tan \alpha = \frac{5}{10} = \frac{1}{2}$$

$$\tan \alpha = \frac{O'B'}{CO'} = \frac{O'B'}{h}$$

$$\Rightarrow O'B' = h \tan \alpha$$



Let V be the volume of the water in the cone i.e. the volume of the cone CA'B' after time t minutes and h be the height of water. Then,

$$V = \frac{1}{3} \pi (O'B')^2 (CO')$$

$$\Rightarrow V = \frac{1}{3} \pi h^3 \tan^2 \alpha$$

$$\Rightarrow V = \frac{\pi}{12} h^3 \quad \left[\because \tan \alpha = \frac{1}{2} \right]$$

$$\therefore \frac{dv}{dt} = \frac{\pi}{12} 3h^2 \frac{dh}{dt} = \frac{\pi}{4} h^2 \frac{dh}{dt}$$

$$\Rightarrow \frac{3}{2} = \frac{\pi h^2}{4} \frac{dh}{dt} \quad \left[\because \frac{dv}{dt} = \frac{3}{2} \text{ cm}^3 / \text{minute (given)} \right]$$

$$\frac{dh}{dt} = \frac{6}{\pi h^2}$$

$$\Rightarrow \left(\frac{dh}{dt} \right)_{h=4} = \frac{6}{\pi (4)^2} = \frac{3}{8\pi} \text{ cm / min}$$

$$\therefore \text{required are} = \frac{3}{8\pi} \text{ cm/sec}.$$

25. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of 1 cu. cm/sec. Water is poured into the tank at a constant rate of C cu. cm/sec. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep.

एक पानी की टंकी जो एक शंकु के आकार की हो जिसका शीर्ष नीचे की ओर है जिसकी ऊँचाई 10 cm है तथा आधार की त्रिज्या 15 cm है। उसमें से पानी का रिसाव एक निश्चित दर 1 cu. cm/sec से हो रहा है। टंकी में एक नियत दर C cu. cm/sec से भरा जा रहा है। जब उसकी गहराई दो 2 cm है तो पानी की सतह 4 cm/sec की दर से बढ़ रही हो तो C का मान ज्ञात कीजिये।

Ans. $1 + 36\pi$ cu. cm/sec

Sol. $r = 15\text{cm}$

$$h = 10\text{cm}$$

Let r be radius of wooden-level at the instant when the water is 2cm deep

$$\therefore \frac{r'}{h'} = \frac{r}{h}$$

$$\Rightarrow \frac{r'}{h'} = \frac{15}{10}$$

$$\Rightarrow r' = \frac{3}{2} h'$$

$$\text{Volume of the right circular cone} = \frac{1}{3} \times R^2 H$$

$$= \frac{1}{3} \pi \times \frac{9}{4} K^2 h'$$

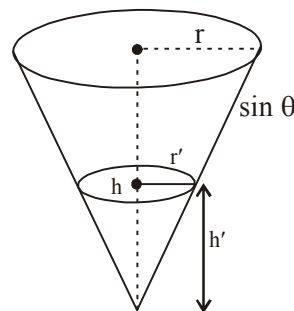
$$= \frac{3\pi}{4} K^3$$

$$\therefore \frac{dv}{dt} = \frac{3\pi}{4} \cdot 3K^2 \cdot \frac{dK}{dt}$$

$$= \frac{3\pi}{4} \times 3 \times (2)^2 \times 4 \text{ Cu.cm/sec}$$

$$= 36\pi \text{ Cu.cm/sec}$$

$$\therefore \text{value of } c = 36\pi + 1 \text{ Cu.cm/sec}$$



26. Sand is pouring from a pipe at the rate of 12 cc/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always 1/6th of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm.

एक नलिका से रेत 12 cc/sec की दर से भरी जा रही है नीचे गिरने वाली रेत जमीन पर एक शंकु का आकार इस प्रकार बना रही है कि शंकु की ऊँचाई सदैव त्रिज्या की 1/6th है। शंकु की ऊँचाई की वृद्धि दर ज्ञात कीजिये जब ऊँचाई 4 cm हो।

Ans. $1/48 \pi \text{ cm/s}$

Sol. $h = \frac{1}{6}r$, $h = 4\text{cm}$

$$\frac{dv}{dt} = 12\text{cm}^3 / \text{sec}$$

$$v = \frac{1}{3}\pi r^2 h$$

$$v = \frac{1}{3}\pi (6h)^2 h$$

$$v = 12\pi h^3$$

$$\frac{dv}{dt} = 12\pi (3h)^2 \frac{dh}{dt}$$

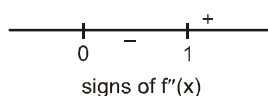
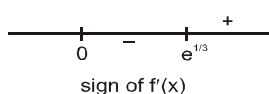
$$\frac{12}{12\pi(3 \times 16)} = \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{1}{48\pi} \text{ cm / sec}$$

27. An open Can of oil is accidentally dropped into a lake; assume the oil spreads over the surface as a circular disc of uniform thickness whose radius increases steadily at the rate of 10 cm/sec. At the moment when the radius is 1 meter, the thickness of the oil slick is decreasing at the rate of 4 mm/sec, how fast is it decreasing when the radius is 2 meters.

एक खुला हुआ तेल का केन दुर्घटनावश एक झील में गिर जाता है मान लीजिये की तेल झील की सतह पर एक वृत्ताकार चकती जिसकी एकसमान मोटाई है तथा त्रिज्या 10 cm/sec नियत दर से बढ़ रही है एक क्षण जब त्रिज्या 1 मीटर है तथा मोटाई 4 mm/sec की दर से घट रही है। मोटाई की घटने की दर क्या होगी जब त्रिज्या 2 मीटर है।

Ans. 0.05 cm/sec

Sol. $f'(x) = 16x^3(3\ln x - 1)$



$$f'(x) = 16 \cdot 3 \cdot x^2 \cdot 3 \cdot \ln x$$

$x = 1$ is point of inflection. $x = e^{1/3}$ is point of minima. In $(0, 1)$ $f(x)$ is concave downward. In $(1, \infty)$ $f(x)$ is concave upward.

MONOTONOCITY

DETERMINE MONOTONOCITY

28. Find the intervals of monotonicity for the following functions & represent your solution set on the number line.

(a) $f(x) = 2 \cdot e^{x^2-4x}$ (b) $f(x) = e^x/x$ (c) $f(x) = x^2 e^{-x}$ (d) $f(x) = 2x^2 - \ln|x|$

Also plot the graphs in each case and state their range.

निम्नलिखित फलनों के लिये एकदिष्टता के अन्तराल ज्ञात कीजिये तथा अपने हलों को संख्या रेखा पर निरूपित कीजिये।

(a) $f(x) = 2 \cdot e^{x^2-4x}$ (b) $f(x) = e^x/x$ (c) $f(x) = x^2 e^{-x}$ (d) $f(x) = 2x^2 - \ln|x|$

साथ ही प्रत्येक स्थिति के लिये फलनों के आरेख भी बनाइयें, उनका परिसर भी लिखिये।

Ans. (a) I in $(2, \infty)$ & D in $(-\infty, 2)$

(b) I in $(1, \infty)$ & D in $(-\infty, 0) \cup (0, 1)$

(c) I in $(0, 2)$ & D in $(-\infty, 0) \cup (2, \infty)$

(d) I for $x > \frac{1}{2}$ or $-\frac{1}{2} < x < 0$ & D for $x < -\frac{1}{2}$ or $0 < x < \frac{1}{2}$

Sol.

29. Find the intervals of monotonicity of the functions in $[0, 2\pi]$

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$ (b) $g(x) = 2 \sin x + \cos 2x$ in $(0 \leq x \leq 2\pi)$.

(c) $f(x) = \frac{4 \sin x - 2x - x \cos x}{2 + \cos x}$

निम्नलिखित फलनों के लिये अन्तराल $[0, 2\pi]$ में एकदिष्टता के अन्तराल ज्ञात कीजिये।

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$ (b) $g(x) = 2 \sin x + \cos 2x$ in $(0 \leq x \leq 2\pi)$.

(c) $f(x) = \frac{4 \sin x - 2x - x \cos x}{2 + \cos x}$

Ans. (a) I in $[0, 3\pi/4) \cup (7\pi/4, 2\pi]$ & D in $(3\pi/4, 7\pi/4)$

(b) I in $[0, \pi/6) \cup (\pi/2, 5\pi/6) \cup (3\pi/2, 2\pi]$ & D in $(\pi/6, \pi/2) \cup (5\pi/6, 3\pi/2)$

(c) I in $[0, \pi/2) \cup (3\pi/2, 2\pi]$ and D in $(\pi/2, 3\pi/2)$

Sol.

30. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max \{f(t) : 0 \leq t \leq x\} & , 0 \leq x \leq 1 \\ 3 - x & , 1 < x \leq 2 \end{cases}$. Discuss the conti. & differentiability of $g(x)$ in the interval $(0, 2)$.

माना $f(x) = x^3 - x^2 + x + 1$ तथा $g(x) = \begin{cases} \max \{f(t) : 0 \leq t \leq x\} & , 0 \leq x \leq 1 \\ 3 - x & , 1 < x \leq 2 \end{cases}$ $g(x)$ की अन्तराल $(0, 2)$ में सान्त्वत्य एवं

अवकलनीयता के बारे में बताइयें।

Ans. continuous but not diff. at $x = 1$

Sol. Here, $f(x) = x^3 - x^2 + x + 1$

$$\Rightarrow f'(x) = 3x^2 - 2x + 1, \text{ which is strictly}$$

Increasing in $(0, 2)$

$$\therefore g(x) = \begin{cases} f(x); & 0 \leq x \leq 1 \\ 3 - x; & 1 < x \leq 2 \end{cases}$$

[As $f(x)$ is increasing so $f(x)$ is maximum when $0 \leq t \leq x$]

$$\text{So, } g(x) = \begin{cases} x^3 - x^2 + x + 1; & 0 \leq x \leq 1 \\ 3 - x; & 1 < x \leq 2 \end{cases}$$

$$\text{Also, } g'(x) = \begin{cases} 3x^2 - 2x + 1; & 0 \leq x \leq 1 \\ -1; & 1 < x \leq 2 \end{cases}$$

Which clearly shows $g(x)$ is continuous for all $x \in [0, 2]$,

but $g(x)$ is not differentiable at $x = 1$

ALWAYS INCREASING / ALWAYS DECREASING BASED QUESTIONS

31. Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality, $1 - f(x) - f^3(x) > f(1 - 5x)$
माना $f(x) = 1 - x - x^3$, x के सभी वास्तविक मान ज्ञात कीजिये जो असमिका $1 - f(x) - f^3(x) > f(1 - 5x)$ को संतुष्ट करते हैं।

Ans. $(-2, 0) \cup (2, \infty)$

Sol. $1 - f(x) - f^3(x) > f(1 - 5x)$

$$\Rightarrow f(x) = -1 - 3x^2 < 0 \Rightarrow f \text{ is decreasing}$$

Now $f(f(x)) > f(1 - 5x)$

$\Rightarrow f(x) < 1 - 5x \quad \{ \because f \text{ is decreasing } \}$

$$1 - x - x^3 < 1 - 5x.$$

32. Let $f(x)$ be an increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. Find the range of a .

Ans. $(0, 1/3) \cup (1, 5)$

Sol. Increasing function $\Rightarrow 2a^2 + a + 1 > 3a^2 - 4a + 1$

$$\Rightarrow a^2 - 5a < 0 \quad \Rightarrow a \in (0, 5)$$

$\Rightarrow a = 1, 2, 3, 4$ are the possible integral values.

From domain conditions $2a^2 + a + 1 > 0$;

$3a^2 - 4a + 1 > 0$ we can say that first satisfies but the second one makes it $a \in \left(0, \frac{1}{3}\right) \cup (1, \infty)$.

So possible integral values are $a = 2, 3, 4$

32. एक वर्द्धमान फलन $f(x)$ अन्तराल $(0, \infty)$ में परिभाषित है। यदि $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$ है तो a का परिसर ज्ञात कीजिये।

Ans. $(0, 1/3) \cup (1, 5)$

Sol.

33. Find the set of all values of the parameter 'a' for which the function, $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$ increases for all $x \in \mathbb{R}$ and has no critical points for all $x \in \mathbb{R}$.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$, सभी x के वास्तविक मानों के लिये वर्द्धमान हो एवं कोई भी क्रांतिक बिन्दु ना हो।

Ans. $a < -(2 + \sqrt{5})$ or $a > \sqrt{5}$

Sol. $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$

$$f'(x) = 2 \cos 2x - 8(a + 1) \cos x + (4a^2 + 8a - 14)$$

$$f'(x) = 2(2\cos^2 x - 1) - 8(a + 1) \cos x + 4a^2 + 8a - 14$$

$$= 4\{\cos^2 x - 2(a + 1) \cos x\} + 4a^2 + 8a - 16$$

$$= 4\{\cos x - (a + 1)\}^2 - 20 > 0$$

$$= \{\cos x - (a + 1)\}^2 - (\sqrt{5})^2 > 0$$

$$f'(x) = \{\cos x - (a+1) - \sqrt{5}\} \{\cos x - (a+1) + \sqrt{5}\} > 0$$

$$\Rightarrow \cos x > a+1 + \sqrt{5} \text{ or } \cos x < (a+1) - \sqrt{5}$$

$$\forall x \in \mathbb{R}$$

$$a+1 + \sqrt{5} < -1 \quad \text{or} \quad (a+1) - \sqrt{5} > 1$$

$$a < -2 - \sqrt{5} \quad \text{or} \quad a > \sqrt{5}$$

$$a \in (-\infty, -2 - \sqrt{5}) \cup (\sqrt{5}, \infty)$$

34. Find all the values of the parameter 'a' for which the function; $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$ increases and has no critical points for all $x \in \mathbb{R}$.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन, $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$ सभी x के वास्तविक मानों के लिये वर्द्धमान हो एवं कोई भी क्रांतिक बिन्दु ना हो।

Ans. $(6, \infty)$

Sol. $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$

$$f'(x) = 8a - 6a \cos 6x - 7 - 5 \cos 5x > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a > \frac{7 + 5 \cos 5x}{8 - 6 \cos 6x} \quad \forall x \in \mathbb{R}$$

$\Rightarrow$ RHS assumes maximum value for $x = 0$.

$$\Rightarrow a > \frac{7+5}{8-6} \Rightarrow a > 6$$

35. Find the set of values of 'a' for which x the function $f(x) = (a^2 + a - 6)\cos 2x + (a - 2)x + \cos 1$ has no critical points.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन, $f(x) = (a^2 + a - 6)\cos 2x + (a - 2)x + \cos 1$ के कोई भी क्रांतिक बिन्दु ना हों।

Ans. $a \in \left(-\frac{7}{2}, -\frac{5}{2}\right)$

Sol. There is no critical point if $f'(x) \neq 0$: $f'(x) = -2(a^2 + a - 6)\sin 2x + (a - 2) \neq 0$

$$\Rightarrow a \neq 2$$

$$\sin 2x \neq \frac{1}{2(a+3)} \Rightarrow \begin{cases} \frac{1}{2(a+3)} < -1 \\ \frac{1}{2(a+3)} > 1 \end{cases} \Rightarrow \dots$$

36. Find the values of 'a' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ increases throughout the number line.

Ans. $[1, \infty)$

Sol. $f'(x) = \cos x - 2a \cos 2x - \cos 3x + 2a \geq 0$

$$\Rightarrow \cos x - \cos 3x + 2a(1 - \cos 2x) \geq 0$$

$$\Rightarrow 2 \sin 2x \sin x + 2a(2 \sin^2 x) \geq 0$$

$$\Rightarrow 4 \sin^2 x \cos x + 4a \sin^2 x \geq 0$$

$$\Rightarrow 4 \sin^2 x (\cos x + a) \geq 0$$

$$a \geq -\cos x$$

$$a \geq 1$$

36. 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ सम्पूर्ण संख्या रेखा पर वर्द्धमान हो।

Ans. $[1, \infty)$

Sol. Let $f(x) = \ln(1+x) - \frac{x}{1+x}$ then

$$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0$$

$$\text{for } x > 0, \text{ so } \ln(1+x) > \frac{x}{1+x} \text{ for } x > 0.$$

But the answer given in my book is $(-1, 0) \cup (0, \infty)$

37. Find the set of values of 'a' for which the function, $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is increasing at every point of its domain.

'a' के मानों का समुच्चय ज्ञात कीजिये जिनके लिये फलन $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ अपने प्रान्त के सभी बिन्दु के लिये वर्द्धमान है।

Ans. $[-7, -1) \cup [2, 3]$

Sol. $f(x) = \left(1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1}\right)x^3 + 5x + \sqrt{6}$

$$f'(x) = 3 \left(1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1}\right)x^2 + 5$$

$$f(x) \text{ is increasing} \Rightarrow f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow 1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1} \geq 0 \Rightarrow \frac{\sqrt{(b + 7)(3 - b)}}{b + 1} \leq 1$$

Case- I If $b + 1 > 0$, then $\frac{(b + 7)(3 - b)}{(b + 1)^2} \leq 1$ and $-7 \leq b \leq 3$

$$\Rightarrow b \leq -5 \text{ and } b \geq 2 \Rightarrow b \in [2, 3]$$

Case-II If $b + 1 < 0 \Rightarrow b \in [-7, -1)$

38. If $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ monotonically increases for every $x \in \mathbb{R}$ then find the range of values of 'a'.

यदि $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ प्रत्येक $x \in \mathbb{R}$ के लिये एकदिष्ट वर्द्धमान है तो 'a' के मानों का परिसर ज्ञात कीजिये।

Ans. $a \geq 0$

Sol. $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$

$$f'(x) = 2e^x + ae^{-x} + (2a + 1)$$

Let (माना) $e^x = t$; $t > 0$

$$\therefore 2t + \frac{a}{t} + (2a + 1) \Rightarrow \frac{2t^2 + (2a + 1)t + a}{t}$$

$$\Rightarrow \text{Now (पुनः)} \quad f'(x) \geq 0 \quad \therefore 2t^2 + (2a + 1)t + a \geq 0$$

Case-I (स्थिति-I) $D \leq 0$

$$(2a + 1)^2 - 8a \leq 0$$

$$\Rightarrow (2a - 1)^2 \leq 0 \Rightarrow a = \frac{1}{2}$$

Case-II (स्थिति-II)



$$\Rightarrow \text{(i)} \quad D \geq 0 \Rightarrow a \in \mathbb{R}$$

$$\text{(ii)} \quad f(0) \geq 0 \Rightarrow a \geq 0$$

$$\text{(iii)} \quad -\frac{b}{2a} < 0 \Rightarrow \frac{-(2a+1)}{4} < 0 \Rightarrow a > -\frac{1}{2} \therefore a \in [0, \infty)$$

PROVING INEQUALITIES

39. Find the set of values of x for which the inequality $\ln(1+x) > x/(1+x)$ is valid.

असमिका $\ln(1+x) > x/(1+x)$ को सन्तुष्ट करने वाले x के मानों का समुच्चय ज्ञात कीजिये।

Ans. $(-1, 0) \cup (0, \infty)$

Sol.

40. Prove that, $x^2 - 1 > 2x \ln x > 4(x-1) - 2 \ln x$ for $x > 1$.

$x > 1$ के लिये सिद्ध कीजिये $x^2 - 1 > 2x \ln x > 4(x-1) - 2 \ln x$

Ans.

Sol. $x^2 - 1 > 2x \ln x > 4(x-1) - 2 \ln x, x > 1$

(a) Consider $f(x) = x^2 - 1 - 2x \ln x$

$$f'(x) = 2 \{x - 1 - \ln x\}$$

$$f''(x) = 2 \left\{1 - \frac{1}{x}\right\}$$

$$f''(x) > 0 \quad \forall x > 1$$

$$\Rightarrow f(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow f(x) > f(1) \Rightarrow f(x) > 0 \quad \forall x > 1$$

$$\Rightarrow f(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow f(x) > f(1) \Rightarrow f(x) > 0 \quad \forall x > 1$$

(b) Consider,

$$g(x) = 2x \ln x + 2 \ln x - 4(x-1)$$

$$g'(x) = 2(1 + \ln x) + \frac{2}{x} - 4$$

$$g''(x) = \frac{2}{x} - \frac{2}{x^2}$$

$$g''(x) = \frac{2(x-1)}{x^2} > 0 \quad \forall x > 1$$

$$\Rightarrow g'(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow g'(x) > g'(1) \Rightarrow g'(x) > 0 \quad \forall x > 1$$

$$\Rightarrow g(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow g(x) > g(1) \quad \forall x > 1$$

$$\Rightarrow g(x) > 0 \quad \forall x > 1.$$

41. Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.

माना f, g दो $\mathbb{R}$ में परिभाषित अवकलनीय फलन हैं तथा $x \geq 0$ के लिये $f(0) = g(0)$ एवं $f'(x) \leq g'(x)$ है सिद्ध कीजिये $x \geq 0$ के लिये $f(x) \leq g(x)$ है।

Ans.

Sol. Let $h(x) : f(x) - g(x), x \in \mathbb{R}$

The function h is differentiable in $\mathbb{R}$ as a difference of differentiable functions, with

$h'(x) = f'(x) - g'(x), x \in \mathbb{R}$, So, according to hypothesis we obtain that :

$$h'(x) = f'(x) - g'(x) < 0 \quad \forall x \in \mathbb{R}$$

so h is (strictly) decreasing in $\mathbb{R}$, and therefore for $x > 0$ we have that

$$h(x) < h(0)$$

Thus,

$$f(x) - g(x) < f(0) - g(0) = 0 \quad \Rightarrow f(x) < g(x)$$

$$\forall x > 0$$

42. Show that exactly two real values of x satisfy the equation $x^2 = x \sin x + \cos x$.

सिद्ध कीजिये समीकरण $x^2 = x \sin x + \cos x$ को संतुष्ट करने वाले x के केवल दो वास्तविक मान हैं।

Ans.

Sol. Taking the difference $h(x) = x^2 - x \sin x - \cos x$ to show that $h'(x) > 0$ if $x > 0$ and $h'(x) < 0$ if $x < 0$ then since $h(0) = 0$ there exists exactly two sets of numbers $[a_1, b_1]$ and $[a_2, b_2]$ such that $h(a_1) > 0 > h(b_1)$ and

$h(a_2) < 0 < h(b_2)$. Therefore by the intermediate value theorem and since h is continuous there exists exactly two numbers c_1, c_2 such that $h(c_1) = h(c_2) = 0$ implying these numbers hold true in the proposed statement.

GREATEST AND LEAST VALUE OF A FUNCTION IN AN INTERVAL

43. Find the greatest & the least values of the following functions in the given interval if they exist :

(a) $f(x) = \sin^{-1} \frac{x}{\sqrt{x^2+1}} - \ln x$ in $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ (b) $f(x) = 12x^{4/3} - 6x^{1/3}, x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1$ in $[-1, 2]$

निम्नलिखित फलनों के दिये गये अन्तराल में महत्तम एवं न्यूनतम मान ज्ञात कीजिये। (यदि अस्तित्व में हो तो) :

(a) $f(x) = \sin^{-1} \frac{x}{\sqrt{x^2+1}} - \ln x$ in $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ (b) $f(x) = 12x^{4/3} - 6x^{1/3}, x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1$ in $[-1, 2]$

- Ans.** (a) $(\pi/6) + (1/2)\ln 3, (\pi/3) - (1/2)\ln 3$
 (b) Maximum at $x = 1$ and $f(-1) = 18$; Minimum at $x = 1/8$ and $f(1/8) = -9/4$
 (c) 2 & -10

Sol. (a) $f'(x) = \frac{1}{\sqrt{1 - \frac{x^2}{x^2+1}}} \left(\frac{\left(\sqrt{x^2+1}\right) \cdot 1 - \frac{x}{2\sqrt{x^2+1}} \cdot 2x}{x^2+1} \right) - \frac{1}{x}$

$$= \frac{1}{1+x^2} - \frac{1}{x} = \frac{x-1-x^2}{x(1+x^2)}$$

Let $g(x) = x - 1 - x^2$

$$g'(x) = 1 - 2x \text{ for } x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$$

$$g'(x) < 0 \Rightarrow f(x) \text{ decreasing function}$$

maximum $f(x)$ is $f\left(\frac{1}{\sqrt{3}}\right) = \sin\left(\frac{1}{2}\right) - \ln\left(\frac{1}{\sqrt{3}}\right)$

$$= \frac{\pi}{6} + \frac{1}{2} \ln 3$$

$$\text{minimum } f(x) \text{ is } = \sin\left(\frac{\sqrt{3}}{2}\right) - \ln(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2}\ln 3$$

(b) Consider, $f(x) = 12x^{4/3} - 6x^{1/3} \quad x \in [-1, 1]$

$$f'(x) = 16x^{1/3} - 2x^{-2/3}$$

$$f'(x) > 0 \Rightarrow 16x^{1/3} > 2x^{-2/3} \Rightarrow 16x > 2 \Rightarrow x > \frac{1}{8}$$

$$f'(x) < 0 \Rightarrow 16x^{1/3} < 2x^{-2/3} \Rightarrow x < \frac{1}{8}$$

$$\therefore f' \text{ increases on } \left(\frac{1}{8}, 1\right) \text{ and decreases on } \left(-1, \frac{1}{8}\right)$$

Also $f(1) = 18$

$$f\left(\frac{1}{8}\right) = \frac{9}{4}$$

$$\therefore \text{Min value} = -\frac{9}{4} \text{ and max value } u = 18$$

(c) $\therefore y'(x) = 5x^4 - 20x^3 + 15x^2$

$$= 5x^2[x^2 - 4x + 3]$$

$$= 0$$

$$x = 0, 1, 3$$

Now $y(0) = 1$ and $y(1) = 2$

3 does not belong in the given range.

Now $y(-1) = 10$

Hence, in the interval $[-1, 2]$ it attains the maximum value, $y = 2$ at $x = 1$

44. Find the minimum value of the function $f(x) = x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x .

x के सभी सम्भावित मानों के लिये फलन $f(x) = x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ का न्यूनतम मान ज्ञात कीजिये।

Ans. -10

Sol. $f(x) = x^{\frac{3}{2}} + x^{-\frac{3}{2}} - 4\left(x + \frac{1}{x}\right)$

$$f(x) = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^3 - 3\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$$

$$-4\left[\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 - 2\right]$$

Let $\sqrt{x} + \frac{1}{\sqrt{x}} = t \ (x > 0)$

Let $g(t) = t^3 - 3t - 4t^2 + 8$

$$g(t) = t^3 - 4t^2 - 3t + 8$$

$$g'(t) = 3t^2 - 8t - 3 = (t-3)(3t+1)$$

$$g'(t) = 0 \Rightarrow t = 3 \left(t \neq -\frac{1}{3} \right)$$

$$g''(t) = 6t - 8$$

$$g''(3) = 10 > 0 \Rightarrow g(3) \text{ is minimum}$$

$$g(3) = 27 - 9 - 36 + 8 = -10$$

45. If $b > a$, find the minimum value of $|x-a|^3 + |x-b|^3$, $x \in \mathbb{R}$.

यदि $b > a$, तो $|x-a|^3 + |x-b|^3$, $x \in \mathbb{R}$ का न्यूनतम मान बताइयें।

Ans. $(b-a)^3/4$

Sol. Let $a < x < b$.

Hence, since $f(x) = x^3$ is a convex function, by Jensen we obtain :

$$|(x-a)^3| + |(x-b)^3| = (x-a)^3 + (b-x)^3 \geq 2\left(\frac{x-a+b-x}{2}\right)^3 = \frac{(b-a)^3}{4}$$

Let $x \geq b$. Hence,

$$\left| (x-a)^3 \right| + (x-b)^3 = (x-a)^3 + (x-b)^3 \geq (b-a)^3 > \frac{(b-a)^3}{4}.$$

ROLLE'S THEOREM

46. Let $f(x) = 4x^3 - 3x^2 - 2x + 1$, use Rolle's theorem to prove that there exist c , $0 < c < 1$ such that $f'(c) = 0$.

माना $f(x) = 4x^3 - 3x^2 - 2x + 1$, रोले प्रमेय का प्रयोग करते हुये सिद्ध कीजिये कि c , $0 < c < 1$, अस्तित्व में है जिसके लिये $f'(c) = 0$ है।

Ans.

Sol. Let $g(x) = x^4 - x^3 - x^2 + x$. Then $g'(x) = f(x)$.

$$g(0) = g(1) = 0.$$

By Rolle's Theorem, there exist c such that $0 < c < 1$ and $g'(c) = 0$.

47. Let f and g be functions, continuous in $[a, b]$ and differentiable on $[a, b]$. If $f(a) = f(b) = 0$ then show that there is a point $c \in (a, b)$ such that $g'(c)f(c) + f'(c)g(c) = 0$.

माना f तथा g अन्तराल $[a, b]$ में सतत् तथा अन्तराल $[a, b]$ में अवकलनीय है। यदि $f(a) = f(b) = 0$ तो दर्शाइये कि एक बिन्दु $c \in (a, b)$ इस प्रकार है कि $g'(c)f(c) + f'(c)g(c) = 0$.

Ans.

Sol.

48. Assume that f is continuous on $[a, b]$, $a > 0$ and differentiable on an open interval (a, b) . Show that if

$$\frac{f(a)}{a} = \frac{f(b)}{b}, \text{ then there exist } x_0 \in (a, b) \text{ such that } x_0 f'(x_0) = f(x_0).$$

माना f अन्तराल $[a, b]$, $a > 0$ में सतत् फलन है तथा (a, b) में अवकलनीय है तो सिद्ध कीजिये यदि $\frac{f(a)}{a} = \frac{f(b)}{b}$, है तो एक $x_0 \in (a, b)$ अस्तित्व में होगा जिसके लिये $x_0 f'(x_0) = f(x_0)$ है।

Ans.

Sol. Let $g(x) = \frac{f(x)}{x}$, $x \in [a, b]$

By Rolle's theorem, $g'(x_0) = 0$

$$\Rightarrow \frac{f'(x_0)x_0 - 1.f(x_0)}{x_0^2} = 0$$

$$\Rightarrow f'(x_0) = \frac{f(x_0)}{x_0}$$

49. $f(x)$ and $g(x)$ are differentiable functions for $0 \leq x \leq 2$ such that $f(0) = 5$, $g(0) = 0$, $f(2) = 8$, $g(2) = 1$. Show that there exists a number c satisfying $0 < c < 2$ and $f'(c) = 3g'(c)$.

$f(x)$ तथा $g(x)$ अन्तराल $0 \leq x \leq 2$ में अवकलनीय फलन हैं जिनके लिये $f(0) = 5$, $g(0) = 0$, $f(2) = 8$, $g(2) = 1$ है। सिद्ध कीजिये एक संख्या c इस प्रकार अस्तित्व में है जो $0 < c < 2$ तथा $f'(c) = 3g'(c)$ को संतुष्ट करती है।

Ans.

Sol. Let $n(x) = f(x) - kg(x)$

$$n'(x) = f'(x) - kg'(x)$$

$$h'(c) = 0$$

$$h(0) = h(g)$$

$$f(0) - kg(0) = f(g) - kg(g)$$

$$S - kx_0 = 8 - k(1)$$

$$k = 3 \quad \text{H.P.}$$

50. If $p(x) = 51x^{101} - 2323x^{100} - 45x + 1035$, using Rolle's theorem, prove that at least one root of $p(x)$ lies between $(45^{1/100}, 46)$.

यदि $p(x) = 51x^{101} - 2323x^{100} - 45x + 1035$, रोले प्रमेय का प्रयोग करते हुये सिद्ध कीजिये कि $p(x)$ का कम से कम एक शून्य अन्तराल $(45^{1/100}, 46)$ में स्थित है।

Ans.

Sol. Let $g(x) = \int p(x) dx + K$

$$g(x) = \frac{x^{102}}{2} - 23x^{101} - \frac{45x^2}{2} + 1035x + K$$

$$= \frac{x^{102} - 46x^{101} - 45x^2 + 2070x}{2} + K$$

$$= \frac{x(x^{100} - 45)(x - 46)}{2} + K$$

$$g(45^{1/100}) = g(46)$$

$$\Rightarrow g'(x) = 0 \text{ has exactly one root in } (45^{1/100}, 46)$$

51. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0$, where $a < b < c < d < e$. Determine the minimum number of zeroes of $g(x) = (f'(x))^2 + f(x) \cdot f''(x)$ in the interval $[a, e]$.

दिया है $f: \mathbb{R} \rightarrow \mathbb{R}$ एक द्विक अवकलनीय फलन इस प्रकार से है कि $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0$, जहाँ $a < b < c < d < e$ तो अन्तराल $[a, e]$ में $g(x) = (f'(x))^2 + f(x) \cdot f''(x)$ के शून्यों की न्यूनतम संख्या ज्ञात करो।

Ans. 6

Sol. $g(x) = \frac{d}{dx} (f(x) \cdot f'(x))$

to get the zero of $g(x)$ we take function

$$h(x) = f(x) \cdot f'(x)$$

between any two roots of $h(x)$ there lies at least one root of $h'(x) = 0$ i.e. $g(x) = 0$

$$\text{Now } h(x) = 0 \Rightarrow f(x) = 0 \text{ or } f'(x) = 0$$

As $f(x) = 0$ has 4 minimum solutions.

and $f'(x) = 0$ minimum 3 solutions.

$\therefore h(x) = 0$ minimum 7 solutions.

$\Rightarrow h'(x) = g(x) = 0$ has minimum 6 solutions.

LMVT

52. Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = a$ and $f(b) = b$, show that there exist distinct c_1, c_2 in (a, b) such that $f'(c_1) + f'(c_2) = 2$.

माना f अन्तराल $[a, b]$ में सतत् तथा अन्तराल (a, b) में अवकलनीय है यदि $f(a) = a$ एवं $f(b) = b$ है तो सिद्ध कीजिये ऐसे दो भिन्न c_1, c_2 , अन्तराल (a, b) में, अस्तित्व में हैं जिनके लिये $f'(c_1) + f'(c_2) = 2$ है।

Ans.

Sol. By LMVT in $\left[a, \frac{a+b}{2}\right]$

$$\frac{f\left(\frac{a+b}{2}\right) - f(a)}{\frac{a+b}{2} - a} = f'(c_1), c_1 \in \left(a, \frac{a+b}{2}\right) \dots (i)$$

By LMVT in $\left[\frac{a+b}{2}, b\right]$

$$\frac{f(b) - f\left(\frac{a+b}{2}\right)}{b - \frac{a+b}{2}} = f'(c_2), c_2 \in \left(\frac{a+b}{2}, b\right) \dots (ii)$$

$$(i) + (ii) \Rightarrow f(c_1) + f(c_2) = 2$$

53. For what value of a , m and b does the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases}$ satisfy the hypothesis of the mean

value theorem for the interval $[0, 2]$.

$$a, m \text{ तथा } b \text{ के किन मानों के लिये फलन } f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases} \text{ माध्य मान प्रमेय को अन्तराल } [0, 2] \text{ में संतुष्ट करता}$$

है।

Ans. $a = 3$, $b = 4$ and $m = 1$

Sol. Using Lagrange mean value theorem

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Hence $f(x)$ is continuous

$$f(0) = 3 = a$$

$$f(1) = 2 = m + b$$

Hence $b = 4$

$f(x)$ is differentiable at 1

$$\text{So } f'(1) = -2 = m$$

54. Let $a > 0$ and f be continuous in $[-a, a]$. Suppose that $f'(x)$ exists and $f'(x) \leq 1$ for all $x \in (-a, a)$. If $f(a) = a$ and $f(-a) = -a$, show that $f(0) = 0$.

माना $a > 0$ है तथा अन्तराल $[-a, a]$ में फलन f सतत् है। माना $f'(x)$ अस्तित्व में है एवं $f'(x) \leq 1 \quad \forall \quad x \in (-a, a)$. यदि $f(a) = a$ तथा $f(-a) = -a$ तो सिद्ध कीजिये $f(0) = 0$ है।

Ans.

Sol. As f is continuous in $[-a, a]$ and differential in $(-a, a)$

So using Lagrange's mean value theorem we get some $c \in (-a, a)$ such that $f'(c) = \frac{f(-a) - f(a)}{-a - a} = 1$

And $f(c) = c + d$ from $f(a) = a$ we get $d = 0$

Hence $f(c) = c$

$f(0) = 0$

55. Let f be a differentiable function for all x and that $|f'(x)| \leq 2$ for all x . If $f(1) = 2$ and $f(4) = 8$ then compute the value of $f^2(2) + f^2(3)$.

माना f सभी x के लिये अवकलनीय फलन है तथा सभी x के लिये $|f'(x)| \leq 2$ है। यदि $f(1) = 2$ तथा $f(4) = 8$ है तो $f^2(2) + f^2(3)$ का मान ज्ञात करें।

Ans. 52

Sol. Since, f is differentiable for all x .

$\Rightarrow f$ is differentiable in $(1, 4)$

Also, f is continuous on $[1, 4]$

So, by Lagrange mean value theorem in $[1, 4]$, there exists $c \in (0, 4)$ such that $f'(c) = \frac{f(4) - f(1)}{3}$

$f'(c) = 2$

$\Rightarrow f'(x) = 2$

$\Rightarrow f(x) = 2x + k$

$f(1) = 2 + k$

$k = 0$

$\Rightarrow f(x) = 2x$

MAXIMA & MINIMA

MAXIMA AND MINIMA AT A POINT

56. Find the greatest & least value for the function :

(a) $y = x + \sin 2x, 0 \leq x \leq 2\pi$

(b) $y = 2 \cos 2x - \cos 4x, 0 \leq x \leq \pi$

फलन का महत्तम एवं न्यूनतम मान बताइये :

(a) $y = x + \sin 2x, 0 \leq x \leq 2\pi$

(b) $y = 2 \cos 2x - \cos 4x, 0 \leq x \leq \pi$

Ans. (a) Max at $x = 2\pi$, Max value = 2π , Min. at $x = 0$, Min value = 0

(b) Max at $x = \pi/6$ & also at $x = 5\pi/6$ and Max value = $3/2$, Min at $x = \pi/2$, Min value = -3

Sol.

57. Let $P(x)$ be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4$. If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set $A = \{x \mid x^2 + 6 \leq 5x\}$ then find $\frac{m}{M}$.

माना $P(x)$ एक 5 घातीय बहुपद है जिसके चरम बिन्दु $x = -1, 1$ है तथा $\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4$ है। यदि M तथा m फलन

$y = P'(x)$ के समुच्चय $A = \{x \mid x^2 + 6 \leq 5x\}$ में अधिकतम तथा न्यूनतम मान है तो $\frac{m}{M}$ का मान ज्ञात कीजिये।

Ans. 6

Sol. $P'(x) = -18x^4 + 18x^2 = 18(x^2 - x^4)$

and $P''(x) = 18(2x - 4x^3) = 36 \times (1 - 2x^2)$

Also, $A = \{x : x^2 + 6 \leq 5x\}$

$\Rightarrow x \in [2, 3]$

Clearly, $P''(x) < 0 \forall x \in [2, 3]$

So, $y = P'(x)$ is decreasing function in $[2, 3]$

$\therefore M = P'_{\max}(x = 2) = 18(4 - 16) = -18 \times 12$

and $m = P'_{\min}(x = 3) = 18(9 - 81) = -18 \times 72$

Hence, $\frac{m}{M} = \frac{-18 \times 72}{-18 \times 12} = 6$

58. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2 + 3}$, find the equations of the tangent lines of minimum and maximum slope.

वक्र $y = \frac{6}{x^2 + 3}$ कि सभी स्पर्श रेखाओं के लिये अधिकतम एवं न्यूनतम प्रवणता की स्पर्श रेखाओं का समीकरण बताइये।

Ans. $3x + 4y - 9 = 0$; $3x - 4y + 9 = 0$

Sol. The given function is $f(x) = y = \frac{6}{x^2 + 3}$.

$$\Rightarrow f'(x) = -\frac{12x}{(x^2 + 3)^2}$$

$$\text{Let } g(x) = f'(x) = -\left(\frac{12x}{(x^2 + 3)^2}\right)$$

When the tangent line has the minimum or maximum slope the first derivative of the function representing the tangent lines (i.e. $g(x)$) should be equal to zero.

$$\Rightarrow g'(x) = -\left(\frac{12(x^2 + 3)^2 - 48x^2(x^2 + 3)}{(x^2 + 3)^4}\right) = 0.$$

$$\Rightarrow 12(x^2 + 3)^2 - 48x^2(x^2 + 3) = 0$$

$$\Rightarrow 36(x^2 + 3)(1 - x^2) = 0$$

Assuming that the domain of the function is the set of real numbers, we get, $x = \pm 1$.

$$\text{It can be seen that } g(-1) = -\left(\frac{-12}{(1+3)^2}\right) = \frac{3}{4}$$

and

$\Rightarrow$ The value of the derivative of the function $f(x)$ is maximum at $x = -1$ and minimum at $x = 1$

$$\text{when } x = -1, \quad y = \frac{6}{1+3} = \frac{3}{2}$$

$$\text{when } x = 1, \quad y = \frac{6}{1+3} = \frac{3}{2}$$

The tangent the having the maximum slope has a slope of $\frac{3}{4}$ and passes through point $\left(-1, \frac{3}{2}\right)$

$\Rightarrow$ The equation of the tangent line having the maximum slope is $\frac{y - \frac{3}{2}}{x + 1} = \frac{3}{4}$.

$$\Rightarrow 3x - 4y + 9 = 0$$

The tangent line having the minimum slope has a slope of $-\frac{3}{4}$ and passes through point $\left(1, \frac{3}{2}\right)$.

$$\Rightarrow 3x + 4y - 9 = 0$$

59. If $y = \frac{ax + b}{(x-1)(x-4)}$ has a turning value at $(2, -1)$ find a & b and show that the turning value is a maximum.

यदि $y = \frac{ax + b}{(x-1)(x-4)}$ का चरम बिन्दु $(2, -1)$ है तो a तथा b का मान ज्ञात कीजिये सिद्ध कीजिये कि चरम बिन्दु उच्चिष्ठ है।

Ans. $a = 1, b = 0$

Sol. $(2, -1) \Rightarrow -1 = \frac{2a + b}{(+1)(-2)}$

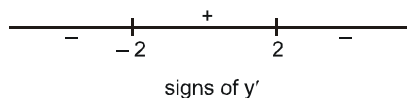
$$2a + b = 2$$

$$y' = \frac{a(x-1)(x-4) - (ax+b)(2x-5)}{(x-1)^2(x-4)^2}$$

$$y' = 0 \text{ at } x = 2 \quad \Rightarrow \quad b = 0 \quad \Rightarrow \quad a = 1$$

$$y = \frac{x}{(x-1)(x-4)}$$

$$y' = \frac{(2+x)(2-x)}{(x-1)^2(x-4)^2}$$



At $x = 2$, y' changes sign from positive to negative

$\Rightarrow x = 2$ is point of maxima.

60. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has a local minimum at $x = 1$. If $f(-1) = 10$ and $f(3) = -22$, then find the distance between its two horizontal tangents.

माना $f(x)$ तीन घात का बहुपद है जिसका स्थानीय उच्चिष्ठ $x = -1$ पर है एवं $f'(x)$ का स्थानिय निम्ननिष्ठ $x = 1$ पर है। यदि $f(-1) = 10$ एवं $f(3) = -22$ तो फलन की दो क्षैतिज स्पर्श रेखाओं के मध्य दूरी ज्ञात कीजिये।

Ans. 32

Sol. $x = 1 \Rightarrow 3 = a + b$

$$\left. \frac{d^2y}{dx^2} \right|_{x=1} = 0 \Rightarrow 3a + b = 0$$

$$a = -\frac{3}{2}, \quad b = \frac{9}{2}$$

61. Consider the function $f(x) = \begin{cases} \sqrt{x} \ln x & \text{when } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$:

- Find whether f is continuous at $x = 0$ or not.
- Find the minima and maxima if they exist.
- Does $f'(0)$ exist? Find $\lim_{x \rightarrow 0} f'(x)$.
- Find the inflection points of the graph of $y = f(x)$.

$$\text{माना फलन } f(x) = \begin{cases} \sqrt{x} \ln x & \text{when } x > 0 \\ 0 & \text{for } x = 0 \end{cases} :$$

- ज्ञात कीजिये कि f बिन्दु $x = 0$ पर सतत् है अथवा नहीं

- (b) उच्चिष्ठ एवं निम्निष्ठ मान ज्ञात कीजिये यदि अस्तित्व में हो तो
 (c) क्या $f'(0)$ अस्तित्व में है ? $\lim_{x \rightarrow 0} f'(x)$ ज्ञात कीजिये।
 (d) आरेख $y = f(x)$ के नति परिवर्तन बिन्दु ज्ञात कीजिये।

Ans. (a) f is continuous at $x = 0$; (b) $x = e^{-2}, 0$; (c) does not exist, does not exist ; (d) pt. of inflection $x = 1$

Sol. $f(x) = \sqrt{x} \ln x$

$$f'(x) = \frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}}$$

$$f''(x) = -\frac{1}{2x\sqrt{x}} + \frac{1}{2} \frac{\frac{1}{\sqrt{x}} + \frac{\ln x}{2x\sqrt{x}}}{x}$$

For point of inflection $f''(x) = 0 \Rightarrow x = 1$

Since $x = k \rightarrow k = 1$

62. Let $p(x)$ be a polynomial of degree 4 having extremum at $x = 1, 2$ and $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2$. Then the value of $p(2)$ is. [JEE-2009]

माना $p(x)$ एक चार घात का बहुपद है जिसका चरम बिन्दु $x = 1, 2$ पर है तथा $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2$ है। तो $p(2)$ का मान है। [JEE-2009]

Ans. 0

Sol. $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{p(x)}{x^2} = 1$$

Let $p(x) = ax^4 + bx^3 + cx^2$

$$\therefore \lim_{x \rightarrow 0} \frac{p(x)}{x^2} = 1 \Rightarrow c = 1$$

$p(x) = ax^4 + bx^3 + x^2$

Now, $p'(x) = 4ax^3 + 3bx^2 + 2x$

$$\because p'(1) = 0, p'(2) = 0$$

$$\Rightarrow 4a + 3b + 2 = 0$$

$$32a + 12b + 4 = 0 \Rightarrow a = \frac{1}{4}, b = -1$$

$$\Rightarrow p(x) = \frac{1}{4}x^4 - x^3 + x^2 \Rightarrow p(2) = 4 - 8 + 4 = 0$$

63. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x \mid x^2 + 20 \leq 9x\}$ is.

[JEE-2009]

फलन $f(x) = 2x^3 - 15x^2 + 36x - 48$ का समुच्चय $A = \{x \mid x^2 + 20 \leq 9x\}$ में अधिकतम मान है।

[JEE-2009]

Ans. 7

Sol. $f(x) = 2x^3 - 15x^2 + 36x - 48$

$$\text{Set } A = \{x \mid x^2 + 20 \leq 9x\}$$

$$\because x^2 - 9x + 20 \leq 0$$

$$(x-5)(x-4) \leq 0$$

$$\Rightarrow x \in [4, 5]$$

$$\text{Now, } f'(x) = 6x^2 - 30x + 36 = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$x = 2, 3 \text{ and } f(x) \uparrow \text{ in } x \in (-\infty, 2) \cup (3, \infty)$$

$\Rightarrow$ In the set A, $f(x)$ is increasing

$$\Rightarrow f(x)_{\max} = f(5)$$

$$= 2.125 - 15.25 + 36.5 - 48$$

$$= 7$$

64. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in \mathbb{R}$.

If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is.

[JEE-2010]

माना $\mathbb{R}$ (सभी वास्तविक संख्याओं का समुच्चय) में परिभाषित फलन इस प्रकार है कि

$f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, सभी $x \in \mathbb{R}$ के लिये यदि g फलन $\mathbb{R}$ परिभाषित है जिसके मान अन्तराल $(0, \infty)$ में स्थित है इस प्रकार है कि $f(x) = \ln(g(x))$, सभी $x \in \mathbb{R}$ के लिये, तो $\mathbb{R}$ में स्थित उन बिन्दुओं की संख्या जिसके लिये g का एक स्थानीय उच्चिष्ठ हो।

[JEE-2010]

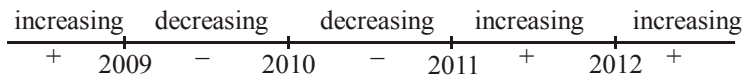
Ans. (b) 1

Sol. $f(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4 \forall x \in \mathbb{R}$

$$f(x) = \ln(g(x)) \forall x \in \mathbb{R}$$

$$g(x) = e^{f(x)}$$

$$g'(x) = 0 \Rightarrow e^{f(x)} \cdot f'(x) = 0 \Rightarrow f'(x) = 0$$



local maximum at $x = 2009$, hence only 1 point.

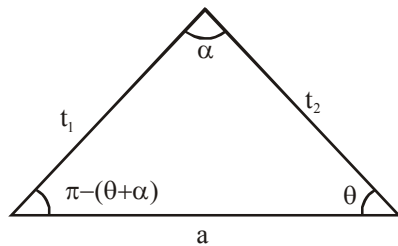
GEOMETRY BASED QUESTIONS ON MAXIMA AND MINIMA

65. Find the maximum perimeter of a triangle on a given base 'a' and having the given vertical angle α .

दिये गए आधार 'a' तथा दिये गये शीर्ष कोण α के लिये त्रिभुज का अधिकतम परिमाण ज्ञात कीजिये।

Ans. $P_{\max} = a \left(1 + \csc \frac{\alpha}{2} \right)$

Sol.



$$\frac{a}{\sin \alpha} = \frac{t_1}{\sin \theta} = \frac{t_2}{\sin(\theta + \alpha)}$$

$$t_1 = \frac{a}{\sin \alpha} \sin \theta, \quad t_2 = \frac{a}{\sin \alpha} \sin(\theta + \alpha)$$

to maximize perimeter we maximize $t_1 + t_2$

$$t_1 + t_2 = f(\theta) = 2 \frac{a}{\sin \alpha} \{ \sin(\theta) + \sin(\theta + \alpha) \}$$

$$= \frac{a}{\sin \alpha} \left\{ \sin \left(\theta + \frac{\alpha}{2} \right) \cos \frac{\alpha}{2} \right\}$$

$$\text{for maximum } \sin \left(\theta + \frac{\alpha}{2} \right) = 1$$

66. The length of three sides of a trapezium are equal, each being 10 cms. Find the maximum area of such a trapezium.

एक समलम्ब चतुर्भुज की तीन भुजाये समान है एवं 10 से.मी. के बराबर है। चतुर्भुज का अधिकतम क्षेत्रफल ज्ञात कीजिये।

Ans. $75\sqrt{3}$ sq. units

Sol. Consider the trapezium ABCD, where AD // BC, AB = CD = AD = 10 cm

Let BC be x

Draw AM $\perp$ BC and DN $\perp$ BC

$\therefore$ MN = 10 cm

Let BC = x cm

$$\text{Now, } CN = \left(\frac{x-10}{2} \right) \text{ cm} = \left(\frac{x}{2} - 5 \right) \text{ cm}$$

Using Pythagoras theorem in $\triangle DNC$,

$$DN = \sqrt{CD^2 - CN^2} = \sqrt{(10)^2 - \left(\frac{x}{2} - 5 \right)^2} \text{ cm}$$

$$\text{Thus, Area of trapezium ABCD} = \frac{1}{2}(BC + AD) \times DN = \frac{1}{2}(x + 10) \sqrt{(10)^2 - \left(\frac{x}{2} - 5 \right)^2} \text{ cm}^2$$

$$\text{Let } A = \frac{1}{2}(x + 10) \sqrt{(10)^2 - \left(\frac{x}{2} - 5 \right)^2} \text{ cm}^2$$

$$\Rightarrow A^2 = \frac{1}{4}(x + 10)^2 \left[(10)^2 - \left(\frac{x}{2} - 5 \right)^2 \right] \text{ cm}^2$$

$$\Rightarrow A^2 = \frac{1}{4}(x + 10)^2 \left[10 + \left(\frac{x}{2} - 5 \right) \right] \left[10 - \left(\frac{x}{2} - 5 \right) \right]$$

$$\Rightarrow A = \frac{1}{4}(x + 10)^2 \left(\frac{x}{2} + 5 \right) \left(15 - \frac{x}{2} \right)$$

$$\Rightarrow A^2 = \frac{1}{16}(x + 10)^3 (30 - x)$$

$$\Rightarrow \text{Let } P = 16A^2 = (x + 10)^3 (30 - x)$$

$$\text{More, area is more } \frac{dp}{dx} = (x + 10)^3 (-1) + 3(x + 10)^2 (30 - x)$$

$$= (x + 10)^2 [-(x + 10) + 3(30 - x)]$$

$$= (x + 10)^2 (80 - 4x)$$

$$= 4(x + 10)^2 (20 - x)$$

$$\frac{d^2P}{dx^2} = 4 \left[2(x + 10)(20 - x) + (x + 10)^2 (-1) \right]$$

$$= 4(x + 10) [2(20 - x) - (x + 10)]$$

$$= 4(x + 10)(30 - 3x)$$

$$= 12(x + 10)(10 - x)$$

$$\frac{dP}{dx} = 0 \text{ Implies that } x + 10 = 0 \text{ or } 20 - x = 0$$

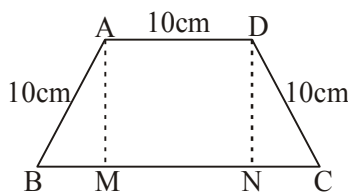
$$\Rightarrow x = -10 \text{ or } x = 20$$

x cannot be negative

$$\therefore x = 20$$

$$\text{Now, for } x = 20, \frac{d^2P}{dx^2} = 12 \times 30 \times (-10) = -3600 < 0$$

$$\begin{aligned} \text{Thus, } P \text{ is maximum area of the trapezium} &= \frac{1}{4} \sqrt{(20 + 10)^3 (30 - 20)} \\ &= \frac{1}{4} \times \sqrt{(30)^3 \times 10} \text{ cm}^2 \\ &= \frac{1}{4} \times 30 \times 10 \sqrt{3} \text{ cm}^2 = 75\sqrt{3} \text{ cm}^2 \end{aligned}$$



67. The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of length ' $2r$ ' and width ' s '. If the surface area A of the pool is fixed, for what value of ' r ' and ' s ' the perimeter ' P ' of the pool is minimum.

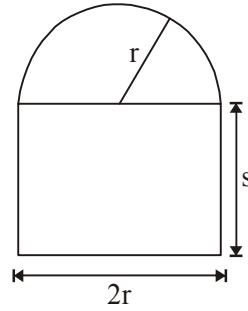
एक तरणताल का तलीय दर्शन में एक अर्धवृत्त (r त्रिज्या) के साथ एक ' $2r$ ' लम्बाई एवं ' s ' चौड़ाई का आयत जुड़ा हुआ है। यदि ताल का पृष्ठीय क्षेत्रफल A नियत है, तो ' r ' तथा ' s ' के किन मानों के लिये परिमाप ' P ' अधिकतम होगा।

Ans. $r = \sqrt{\frac{2A}{\pi+4}}, s = \sqrt{\frac{2A}{\pi+4}}$

Sol. $A = \frac{\pi r^2}{2} + rs$ $P = 2s + 2r + \pi r$

$$P = \left(\frac{\pi r}{2} + 2s \right) + 2r + \frac{\pi r}{2} \quad P = \frac{A}{r} + 2r + \frac{\pi r}{2}$$

Now $\frac{dP}{dr} = 0 \Rightarrow r = \sqrt{\frac{2A}{\pi+4}} \Rightarrow s = \sqrt{\frac{2A}{\pi+4}}$

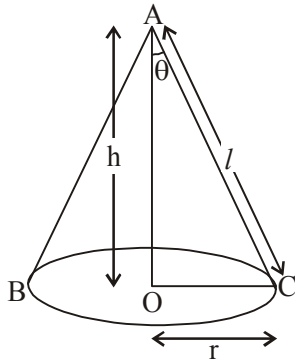


68. For a given curved surface of a right circular cone when the volume is maximum, prove that the semi vertical angle is $\sin^{-1} \frac{1}{\sqrt{3}}$.

दिये गये पृष्ठीय क्षेत्रफल के लिये एक सम वृत्तीय शंकु का आयतन जब अधिकतम हो तो सिद्ध कीजिये कि अर्द्ध शीर्ष कोण $\sin^{-1} \frac{1}{\sqrt{3}}$ होगा।

Ans.

Sol.



With usual notation, given that total surface area $S = \pi r l + \pi r^2$

$$\Rightarrow S = \pi r \sqrt{r^2 + h^2} + \pi r^2 \quad \left(\because l = \sqrt{r^2 + h^2} \right)$$

$$\Rightarrow \frac{S}{\pi r} - r = \sqrt{r^2 + h^2} \Rightarrow \frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi} = h^2$$

$$\Rightarrow h = \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}} \quad \left(\because \frac{S^2}{\pi^2 r^2} > \frac{2S}{\pi} \right) \quad \text{—————(1)}$$

$$\text{and volume } V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}}$$

$$\Rightarrow V = \frac{r}{3} \sqrt{S^2 - 2S\pi r^2}, r^2 < \frac{S}{2\pi} \text{ i.e., } 0 < r < \sqrt{\frac{S}{2\pi}}$$

Since, V is maximum, then V^2 is maximum

$$\text{Now, } V^2 = \frac{S^2 r^2}{9} - \frac{2S\pi r^4}{9}, 0 < r < \sqrt{\frac{S}{2\pi}}$$

$$\therefore \frac{d}{dr}(V^2) = \frac{2S^2}{9} - \frac{8S\pi r^3}{9}$$

$$\text{and } \frac{d}{dr^2}(V^2) = \frac{2S^2}{9} - \frac{24S\pi r^2}{9}$$

$$\text{For maxima put } \frac{dV}{dr} = 0$$

$$\Rightarrow \frac{2rS^2}{9} - \frac{8S\pi r^3}{9} = 0 \Rightarrow r^2 = \frac{S}{4\pi}$$

$$\Rightarrow \frac{d^2(V^2)}{dr^2} < 0 \text{ for } r = \sqrt{\frac{S}{4\pi}}$$

$$\text{From equ. (i) } h = \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}} = \sqrt{\frac{S^2(4\pi)}{\pi^2 S} - \frac{2S}{\pi}} = \sqrt{\frac{2S}{\pi}}$$

If θ is semi-vertical angle of the cone when the volume is maximum, then in right triangle AOC,

$$\sin \theta = \frac{r}{r^2 + h^2} = \frac{\sqrt{\frac{S}{4\pi}}}{\sqrt{\frac{S}{4\pi} + \frac{S\pi}{2}}} = \frac{1}{\sqrt{1+8}} \text{ i.e. } \theta = \sin^{-1}\left(\frac{1}{3}\right)$$

69. By the post office regulations, the combined length & girth of a parcel must not exceed 3 metre. Find the volume of the biggest cylindrical (right circular) packet that can be sent by the parcel post.

पोस्ट ऑफिस के नियमों के अनुसार एक पार्सल की लम्बाई एवं घेरा का योग दोनों 3 मीटर से अधिक नहीं होनी चाहिये समवृत्ताकार बेलन का अधिकतम आयतन ज्ञात कीजिये जिसे पार्सल पोस्ट द्वारा भेजा जा सकता हो।

Ans. $1/\pi$ cu m

Sol. Let the dimension of the box with square base be x, x, h

Volume of the rectangular box $V = x^2h$

$$x^2h = 1000$$

$$\text{Area of top} = x^2$$

$$\text{Area of base} = x^2$$

$$\text{Area of faces} = 4xh$$

$$E = 15x^2 + 25x^2 + 20(4xh) + 300$$

$$E = 40x^2 + 80x \left(\frac{1000}{x^2} \right) + 300$$

$$= 40x^2 + \frac{80 \times 1000}{x} + 300$$

$$\text{For maxima or minima, } \frac{dE}{dx} = 0$$

$$\therefore 80x - \frac{80 \times 1000}{x^2} = 0$$

$$\therefore x^3 = 1000$$

$$x = 10$$

$$\frac{d^2E}{dx^2} = 80 + \frac{2 \times 80 \times 1000}{x^3} = +\text{ive}$$

$$\therefore \text{Min. when } x = 10$$

$$h = \frac{1000}{x^2} = 10$$

Hence the box should be a cube of edge 10 feet.

70. A running track of 440 ft. is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, find the length of its sides. (Use : $\pi \approx 22/7$.)

एक फुटबल के मैदान के चारों ओर एक 440 ft. लम्बाई की धावक पट्टिका बनाई गई है। फुटबल का मैदान आयताकार है जिसके छोर अर्धवृत्ताकार हैं। यदि आयताकार भाग का क्षेत्रफल अधिकतम कर दिया जाये तो इसके लम्बाईयों क्या होंगी। (Use : $\pi \approx 22/7$.)

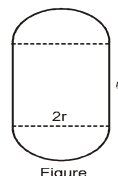
Ans. 110', 70'

Sol. $2\ell + 2\pi r = 440$

$$A = \ell \cdot 2r = -2\pi r^2 + 440r$$

$$\frac{dA}{dr} = -4\pi r + 440 = 0$$

$$\text{at } r = \frac{110}{\pi}$$



71. A closed rectangular box with a square base is to be made to contain 1000 cubic feet. The cost of the material per square foot for the bottom is 15 paise, for the top 25 paise and for the sides 20 paise. The labour charges for making the box are Rs. 3/-. Find the dimensions of the box when the cost is minimum.

एक बंद आयताकार डिब्बे का आधार वर्ग है एवं आयतन 1000 cubic feet है। आधार को बनाने में काम में लिये गये माल की प्रति स्वायर फीट की दर 15 पैसे है, छत के माल की 25 पैसे है तथा दिवारों के लिये 20 पैसे है। मजूदरी का चार्ज 3 रु. है। न्यूनतम लागत के बक्से की विमाये बताइयें।

Ans. side 10', height 10'

Sol.

72. Find the area of the largest rectangle with lower base on the x-axis & upper vertices on the curve $y = 12 - x^2$.

ऐसे आयत का अधिकतम क्षेत्रफल ज्ञात कीजिये जिसका आधार x-अक्ष पर एवं दोनों शीर्ष वक्र $y = 12 - x^2$ पर स्थित हैं।

Ans. 32 sq. units

Sol. Let A be area of rectangle.

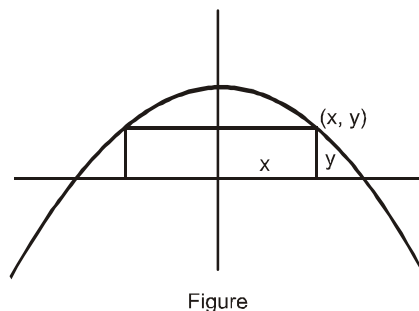
$$A = (2x)(y)$$

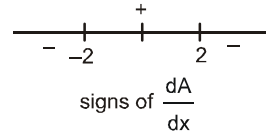
$$= 2x(12 - x^2)$$

$$\frac{dA}{dx} = 6(2 - x)(2 - x)$$

At $x = 2$, A has largest value.

Largest A = 32



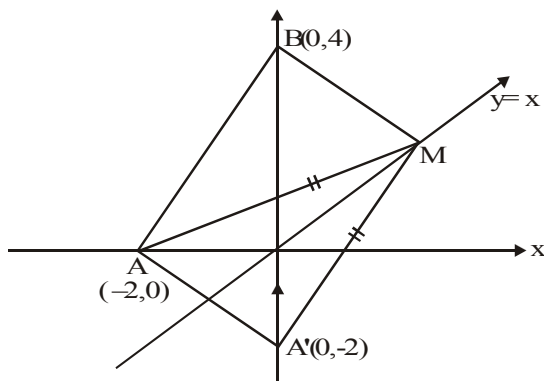


73. Given two points $A(-2, 0)$ & $B(0, 4)$ and a line $y = x$. Find the co-ordinates of a point M on this line so that the perimeter of the ΔAMB is least.

दिया गया है दो बिन्दु $A(-2, 0)$ और $B(0, 4)$ तथा एक रेखा $y = x$ है। रेखा पर स्थित बिन्दु M के निर्देशांक बताइये जब ΔAMB का परिमाण न्यूनतम हो।

Ans. $(0, 0)$

Sol.



To minimize the perimeter.

$AM + MB$ is to be minimized.

i.e. $A'M + MB$ is to be minimized.

[where A' is image of A in $y = x$.]

Obviously $A'M + MB$ is minimized.

when A', M and B are collinear.

i.e. M coincide with origin.

$\therefore M \equiv (0, 0)$

POINT OF INFLECTION & CONCAVITY OF GRAPH

74. Consider the function $y = f(x) = \ln(1 + \sin x)$ with $-2\pi \leq x \leq 2\pi$. Find :

- the zeroes of $f(x)$
- inflection points if any on the graph
- local maxima and minima of $f(x)$
- asymptotes of the graph

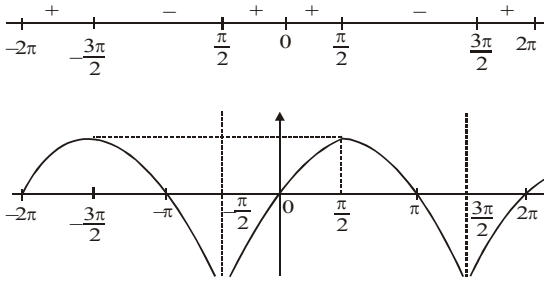
- sketch the graph of $f(x)$ and compute the value of the definite integral $\int_{-\pi/2}^{\pi/2} f(x) dx$.

माना कि फलन $y = f(x) = \ln(1 + \sin x)$ में $-2\pi \leq x \leq 2\pi$ है। ज्ञात कीजिये :

- (a) $f(x)$ के शून्य
- (b) आरेख पर स्थित नति परिवर्तन बिन्दु यदि होतो
- (c) $f(x)$ का स्थानीय उच्चिष्ठ एवं निम्निष्ठ बिन्दु
- (d) वक्र की अनन्त स्पर्शी रेखायें
- (e) $f(x)$ का आरेख बनाइये साथ ही निश्चित समाकलन $\int_{-\pi/2}^{\pi/2} f(x) dx$ की गणना कीजिये।

Ans. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$, (b) no inflection point, (c) maxima at $x = \pi/2$ and $-3\pi/2$ and no minima, (d) $x = 3\pi/2$ and $x = -\pi/2$, (e) $-\pi \ln 2$

Sol. $f(x) = \frac{\cos x}{1 + \sin x}$



75. Prove that $\sin x + 2x \geq \frac{3x \cdot (x+1)}{\pi} \quad \forall x \in \left[0, \frac{\pi}{2}\right]$. (Justify the inequality, if any used). [JEE 2004]

सिद्ध कीजिये $\sin x + 2x \geq \frac{3x \cdot (x+1)}{\pi} \quad \forall x \in \left[0, \frac{\pi}{2}\right]$. (असमिका के लिये न्यायसंगत तर्क दीजिये, यदि कही उपयोग हो तो).

Ans.

Sol. Let $f(x) = \sin x + 2x$ & $g(x) = \frac{3x^2 + 3x}{\pi}$

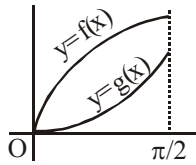
$$f'(x) = \cos x + 2 \quad g'(x) = \frac{6x + 3}{\pi}$$

$$f''(x) = -\sin x \quad g''(x) = \frac{6}{\pi}$$

$$\Rightarrow f \text{ is increasing \& concave down in } \left[0, \frac{\pi}{2}\right]$$

and 'g' is increasing & concave up in $\left[0, \frac{\pi}{2}\right]$

$$\& f\left(\frac{\pi}{2}\right) > g\left(\frac{\pi}{2}\right).$$



from the graph $f(x) \geq g(x) \forall x \in \left[0, \frac{\pi}{2}\right]$

76. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is.

[JEE-2011]

समीकरण $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ के भिन्न वास्तविक मूलों की संख्या होगी।

[JEE-2011]

Ans. 2

Sol. Let $f(x) = x^4 - 4x^3 + 12x^2 + x - 1$

$$f'(x) = 4x^3 - 12x^2 + 24x$$

$$f''(x) = 12x^2 - 24x + 24$$

$$= 12(x^2 - 2x + 2) > 0$$

$\Rightarrow f'(x)$ is strictly increasing function

$\therefore f'(x)$ is cubic polynomial

hence number of roots of $f'(x) = 0$ is 1

$\Rightarrow$ Number of maximum roots of $f(x) = 0$ are 2

Now $f(0) = -1$, $f(1) = 9$, $f(-1) = 15$

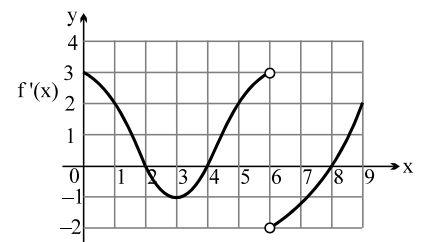
$\Rightarrow f(x)$ has exactly 2 distinct real roots.

NATURE OF $F(X)$ FROM GRAPH OF $F'(X)$

77. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$. If:

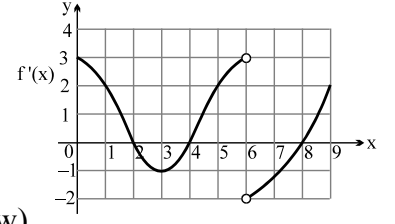
- f is monotonic increasing in the interval $[a, b) \cup (c, d) \cup (e, f]$ and decreasing in $(p, q) \cup (r, s)$.
- f has a local minima at $x = x_1$ and $x = x_2$.
- f is concave up in $(l, m) \cup (n, t]$
- f has inflection point at $x = k$
- number of critical points of $y = f(x)$ is 'w'.

Find the value of $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$.



सतत् फलन f के अवकलज f' का आरेख बनाया गया है जहाँ $f(0) = 0$ है। यदि :

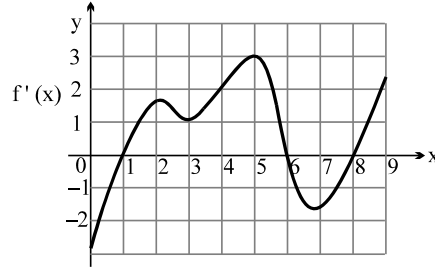
- f अन्तराल $[a, b) \cup (c, d) \cup (e, f]$ में एकदिष्ट वर्द्धमान है एवं अन्तराल $(p, q) \cup (r, s)$ में एकदिष्ट ह्रासमान है।
 - $x = x_1$ एवं $x = x_2$ पर f का स्थानीय निम्निष्ठ है।
 - अन्तराल $(l, m) \cup (n, t]$ में f की नति उपर है (Concave up)
 - f का नति परिवर्तन बिन्दु $x = k$ पर है
 - $y = f(x)$ के चरम बिन्दुओं की संख्या 'w' है।
- $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$
का मान ज्ञात कीजिये।



Ans. 74

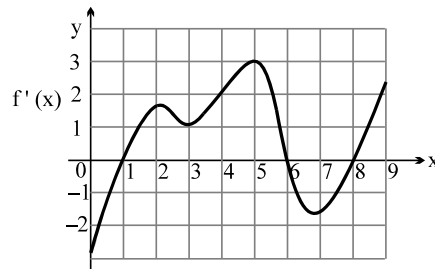
Sol.

78. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$:



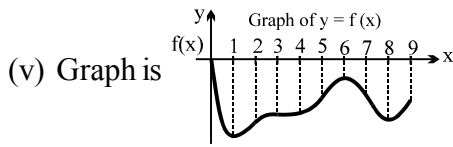
- On what intervals is f increasing or decreasing?
- At what values of x does f have a local maximum or minimum?
- On what intervals is f concave upward or downward?
- State the x -coordinate(s) of the point(s) of inflection.
- Assuming that $f(0) = 0$, sketch a graph of f .

सतत् फलन f के अवकलज f' का आरेख जहाँ $f(0) = 0$ है, दिया गया है :



- किन अन्तरालों में f वर्द्धमान या ह्रासमान है ?
- x के किन मानों के लिये f का स्थानीय उच्चिष्ठ या स्थानीय निम्निष्ठ प्राप्त होता है ?
- किन अन्तरालों में f की नति उपर (concave upward) या नति नीचे (downward) है?
- नति परिवर्तन बिन्दुओं के x -निर्देशांक बताइयें।
- $f(0) = 0$ मानते हुये f का आरेख बनाइये।

Ans. (i) I in $(1, 6) \cup (8, 9)$ and D in $[0, 1) \cup (6, 8)$; (ii). Min. at $x = 1$ and $x = 8$; L.Max. $x = 6$
 (iii) CU in $(0, 2) \cup (3, 5) \cup (7, 9)$ and CD in $(2, 3) \cup (5, 7)$; (iv) $x = 2, 3, 5, 7$



Sol.

ANALYSIS OF CUBIC EQUATION

79. The value of 'a' for which $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

'a' वह मान जिनके लिये $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ के उच्चिष्ठ मान के धनात्मक बिन्दु विद्यमान है, अन्तराल $(a_1, a_2) \cup (a_3, a_4)$ में स्थित है। $a_2 + 11a_3 + 70a_4$ का मान ज्ञात कीजिये।

Ans. 320

Sol. $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$

$$f'(x) = 3x^2 + 6(a-7)x + 3(a^2-9)$$

$$D > 0, f'(0) > 0$$

$$D > 0 \Rightarrow 36(a-7)^2 - 36(a^2-9) > 0$$

$$\Rightarrow a < \frac{29}{7}$$

$$f'(0) = 3(a^2-9) > 0$$

$$\Rightarrow (a-3)(a+3) > 0$$

$$\Rightarrow a < -3 \text{ or } a > 3$$

$$\text{Hence } a \in (-\infty, -3) \cup \left(3, \frac{29}{7}\right)$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 80 to 82

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x -intercept and b be the y -intercept of a tangent to $y = f(x)$, then

माना $f(x) = \frac{1}{1+x^2}$. माना $y = f(x)$ पर स्पर्श रेखा का ढाल m , x -अन्तःखण्ड a और y -अन्तःखण्ड b है, तो

80. Abscissa of the point of contact of the tangent for which m is greatest, is :

स्पर्श रेखा के स्पर्शी बिन्दु का भुज जिसके लिए m महत्तम है :

- (A) $\frac{1}{\sqrt{3}}$ (B) 1 (C) -1 (D) $-\frac{1}{\sqrt{3}}$

Ans. D

Sol. $f'(x) = -\frac{2x}{(1+x^2)^2}$

$$f''(x) = \frac{-2+6x^2}{(1+x^2)^3}$$

$$f''(x) = 0 \text{ if } x = \pm \frac{1}{\sqrt{3}}$$

Hence, $f'(x)$ is greatest at $x = -\frac{1}{\sqrt{3}}$.

81. Greatest value of b is :

b का अधिकतम मान है :

- (A) $\frac{9}{8}$ (B) $\frac{3}{8}$ (C) $\frac{1}{8}$ (D) $\frac{5}{8}$

Ans. A

Sol.

82. Abscissa of the point of contact of tangent for which $\frac{1}{a}$ is greatest, is :

स्पर्श रेखा के स्पर्शी बिन्दु का भुज जिसके लिए $\frac{1}{a}$ महत्तम है :

- (A) $\frac{1}{\sqrt{3}}$ (B) 1 (C) -1 (D) $-\frac{1}{\sqrt{3}}$

Ans. A

Sol.

Paragraph for Question nos. 83 to 85

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in \mathbb{R}$.

माना घन फलन $f(x) = 8x^3 + 4ax^2 + 2bx + a$ जहाँ $a, b \in \mathbb{R}$.

83. For $a = 1$ if $y = f(x)$ is strictly increasing $\forall x \in \mathbb{R}$ then maximum range of values of b is :

यदि $a = 1$ के लिये $y = f(x)$ एकदिष्ट वर्धमान है $\forall x \in \mathbb{R}$ तो b के मानों का अधिकतम परास है :

- (A) $\left(-\infty, \frac{1}{3}\right]$ (B) $\left(\frac{1}{3}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $(-\infty, \infty)$

Ans. C

Sol. $a = 1$

$$f(x) = 8x^3 + 4x^2 + 2bx + 1$$

$$f'(x) = 24x^2 + 8x + 2b = 2(12x^2 + 4x + b)$$

for increasing function, $f'(x) \geq 0 \quad \forall x \in \mathbb{R}$

$$\therefore D \leq 0 \Rightarrow 16 - 48b \leq 0 \Rightarrow b \geq \frac{1}{3} \Rightarrow \text{(C)}$$

84. For $b = 1$, if $y = f(x)$ is non monotonic then the sum of all the integral values of $a \in [1, 100]$, is :

यदि $b = 1$ के लिये, $y = f(x)$ एकदिष्ट नहीं है तब $a \in [1, 100]$ में सभी पूर्णांक मानों का योग होगा :

- (A) 4950 (B) 5049 (C) 5050 (D) 5047

Ans. B

Sol. if $b = 1$

$$f(x) = 8x^3 + 4ax^2 + 2x + a$$

$$f'(x) = 24x^2 + 8ax + 2 \quad \text{or} \quad 2(12x^2 + 4ax + 1)$$

for non monotonic $f'(x) = 0$ must have distinct roots

$$\text{hence } D > 0 \text{ i.e. } 16a^2 - 48 > 0 \Rightarrow a^2 > 3; \therefore a > \sqrt{3} \text{ or } a < -\sqrt{3}$$

$$\therefore a \in 2, 3, 4, \dots$$

$$\text{sum} = 5050 - 1 = 5049$$

85. If the sum of the base 2 logarithms of the roots of the cubic $f(x) = 0$ is 5 then the value of 'a' is :

$f(x) = 0$ के मूलों के आधार 2 पर लघुगणक का योग 5 है तो 'a' बराबर :

- (A) -64 (B) -8 (C) -128 (D) -256

Ans. D

Sol. If x_1, x_2 and x_3 are the roots then $\log_2 x_1 + \log_2 x_2 + \log_2 x_3 = 5$

$$\log_2(x_1 x_2 x_3) = 5$$

$$x_1 x_2 x_3 = 32$$

$$-\frac{a}{8} = 32 \Rightarrow a = -256$$

Paragraph for Question nos. 86 to 88

If a continuous function f defined on the real line R , assumes positive and negative values in R then the equation $f(x) = 0$ has a root in R . For example, if it is known that a continuous function f on R is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in R .

Consider $f(x) = ke^x - x$ for all real x where k is a real constant.

[JEE 2007]

यदि संख्या रेखा R पर परिभाषित f एक सतत फलन है, जो R में ऋणात्मक एवं धनात्मक मान लेता है, तो समीकरण $f(x) = 0$ के R में एक मूल होता है। उदाहरण के लिये यदि f , R में किसी बिन्दु पर धनात्मक है तथा उसका न्यूनतम मान शून्य है तो समीकरण $f(x) = 0$ का R में एक हल होगा।

मान लिये $f(x) = ke^x - x$ सभी वास्तविक x के लिये है, जहाँ k एक वास्तविक नियतांक है।

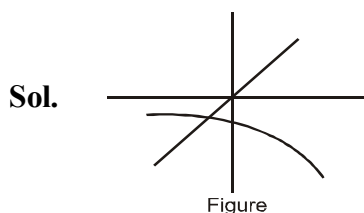
86. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at :

- (A) no point (B) one point (C) two points (D) more than two points

रेखा $y = x$ वक्र $y = ke^x$ को $k \leq 0$ के लिये मिलती है :

- (A) किसी बिन्दु पर नहीं (B) एक बिन्दु पर (C) दो बिन्दुओं पर (D) दो से अधिक बिन्दुओं पर

Ans. B



87. The positive value of k for which $ke^x - x = 0$ has only one root is :

k का धनात्मक मान जिसके लिये $ke^x - x = 0$ का केवल एक मूल है :

- (A) $1/e$ (B) 1 (C) e (D) $\log_e 2$

Ans. A

Sol. Consider $y = ke^x$ and $y = x$

Let (α, ke^α) be a point on $y = ke^x$

if it lies on $y = x$ also then $\alpha = ke^\alpha$

$$\frac{dy}{dx} = ke^x$$

$$\therefore \left. \frac{dy}{dx} \right|_{x=\alpha} = ke^\alpha = \alpha = 1 \quad \{ \because y = x \text{ is tangent to } y = ke^x \text{ at one point} \}$$

$$\therefore 1 = ke \quad \text{i.e. } k = 1/e$$

88. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is :

$k > 0$ के लिये k मानों का समुच्चय जिसके लिये $ke^x - x = 0$ के दो भिन्न मूल हैं :

- (A) $(0, 1/e)$ (B) $(1/e, 1)$ (C) $(1/e, \infty)$ (D) $(0, 1)$

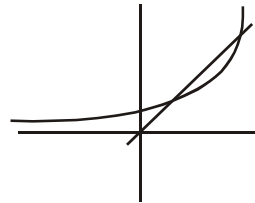
Ans. A

Sol. Consider $y = ke^x$ and $y = x$

from above question $e^x = \frac{x}{k}$ if we decrease the value of k from $\frac{1}{e}$, then

slope of $y = \frac{x}{k}$ increases

$\therefore y = e^x$ and $y = \frac{x}{k}$ intersect at two distinct points



$$\therefore k \in \left(0, \frac{1}{e}\right)$$

Paragraph for Question nos. 89 to 91

Consider the function $f(x) = \frac{x^2}{x^2 - 1}$

माना फलन $f(x) = \frac{x^2}{x^2 - 1}$

89. The interval in which f is increasing is :

वह अन्तराल जहाँ f वर्धमान है, है :

- (A) $(-1, 1)$ (B) $(-\infty, -1) \cup (-1, 0)$
(C) $(-\infty, \infty) - \{-1, 1\}$ (C) $(0, 1) \cup (1, \infty)$

Ans. B

Sol. $y = \frac{x^2}{x^2 - 1}$; not defined at $x = \pm 1$ [13th, 04-11-2007] [12th, quiz]

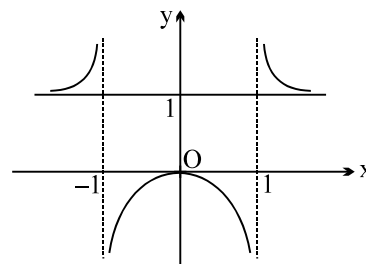
$$= 1 + \frac{1}{x^2 - 1}; \quad y' = -\frac{2x}{(x^2 - 1)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow x = 0 \text{ (point of maxima)}$$

$$\text{as } x \rightarrow 1^+, y \rightarrow \infty; \quad x \rightarrow 1^-, y \rightarrow -\infty$$

$$\text{||ly } x \rightarrow -1^+, y \rightarrow -\infty; \quad x \rightarrow -1^-, y \rightarrow \infty$$

The graph of $y = \frac{x^2}{x^2 - 1}$ is as shown



verify all alternatives from the graph.

90. If f is defined from $\mathbb{R} - \{-1, 1\} \rightarrow \mathbb{R}$ then f is :

- (A) injective but not surjective (B) surjective but not injective
(C) injective as well as surjective (D) neither injective nor surjective.

यदि $f, \mathbb{R} - \{-1, 1\} \rightarrow \mathbb{R}$ पर परिभाषित है, तब

- (A) f एकैकी है लेकिन आच्छादक नहीं (B) आच्छादक है लेकिन एकैकी नहीं
(C) एकैकी और आच्छादक दोनों (D) न तो एकैकी ना ही आच्छादक

Ans. D

Sol.

91. f has :

- (A) local maxima but no local minima (B) local minima but no local maxima
(C) both local maxima and local minima (D) neither local maxima nor local minima

f में :

- (A) स्थानीय उच्चिष्ठ है लेकिन स्थानीय निम्निष्ठ नहीं (B) स्थानीय निम्निष्ठ है लेकिन स्थानीय उच्चिष्ठ नहीं
(C) दोनों स्थानीय निम्निष्ठ और उच्चिष्ठ रखता है (D) न तो स्थानीय निम्निष्ठ है ना ही स्थानीय उच्चिष्ठ

Ans. A

Sol.

Paragraph for Question nos. 92 to 94

Consider a rational function $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4}$ and a quadratic function $g(x) = (1 + m)x^2 - 2(1 + 3m)x - 2(1 + m)$ where m is a parameter.

माना परिमेय फलन $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4}$ और द्विघात फलन $g(x) = (1 + m)x^2 - 2(1 + 3m)x - 2(1 + m)$ जहाँ m प्राचाल है।

92. Which one of the following statement does not hold good for the rational function $f(x)$?

- (A) It is a continuous function (B) It has only one asymptote
(C) It has exactly one maxima and one minima (D) $f(x)$ is monotonic in $(0, \infty)$

परिमेय फलन $f(x)$ के लिये निम्न में से असत्य है?

- (A) यह सतत फलन है (B) $f(x)$ की केवल एक अनन्त स्पर्शी है
(C) $f(x)$ का केवल एक उच्चिष्ठ और एक ही निम्निष्ठ है (D) $f(x)$, $(0, \infty)$ में एकदिष्ट है

Ans. D

Sol. $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4} = 1 - \frac{8x}{x^2 + 2x + 4}$

$$f'(x) = -8 \left[\frac{(x^2 + 2x + 4) - x(2x + 2)}{(x^2 + 2x + 4)^2} \right] = -8 \left[\frac{-x^2 + 4}{(x^2 + 2x + 4)^2} \right] = \frac{8(x^2 - 4)}{(x^2 + 2x + 4)^2}$$

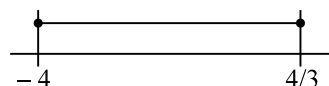
$$f'(x) = 0 \Rightarrow x = 2 \text{ or } -2$$

$$f(2) = \frac{4 - 12 + 4}{4 + 4 + 4} = \frac{-4}{12} = \frac{-1}{3}$$

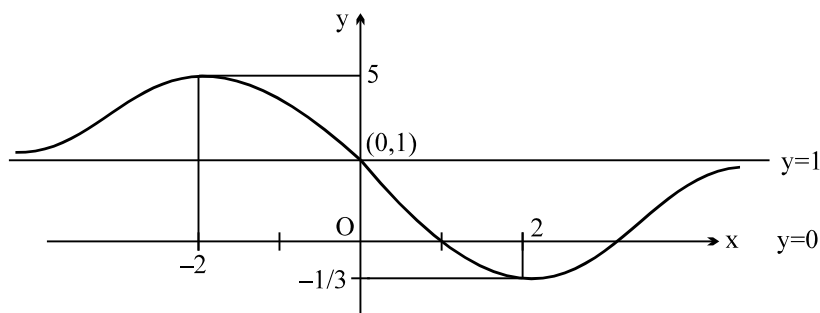
$$f(-2) = \frac{4 + 12 + 4}{4 - 4 + 4} = 5$$

$$\text{Hence range of } f(x) \text{ is } \left[-\frac{1}{3}, 5 \right]$$

the graph of $y = f(x)$ is as shown



$$\text{hence } -\frac{1}{3} \leq f(x) \leq 5$$



(i) obviously f is continuous and has $y=1$ as its asymptote with one maxima and one minima and f is non monotonic in $(0, \infty) \Rightarrow$ (D) is incorrect

(ii) $\cot(\cot^{-1} f(x)) = k$

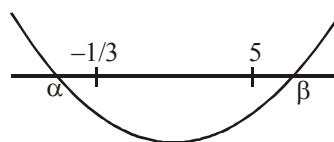
$f(x) = k$

$y = k$ and $y = f(x)$

hence only $y = 0$ cuts the graph at two distinct points $\Rightarrow$ (A)

(iii) now $g(x) = 0$

$$x^2 - \frac{2(1+3m)}{1+m}x - \frac{2(1+m)}{1+m} = 0$$



or $g(x) = x^2 - \frac{2(1+3m)}{1+m}x - 2 = 0$

as range of $f(x)$ is $\left[-\frac{1}{3}, 5\right]$

so one root of $g(x)$ is less than $\left(-\frac{1}{3}\right)$ and other is greater than 5

$$g\left(-\frac{1}{3}\right) < 0 \quad \text{and} \quad g(5) < 0$$

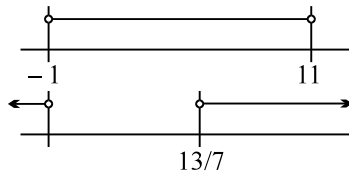
now $g\left(-\frac{1}{3}\right) = \frac{1}{9} + \frac{2}{3} \frac{(1+3m)}{1+m} - 2 < 0 = 1 + \frac{6(1+3m)}{1+m} - 18 < 0 = \frac{6(1+3m)}{1+m} - 17 < 0$

$$= \frac{6+18m-17-17m}{1+m} < 0 = \frac{m-11}{1+m} < 0$$

$g(5) < 0$

$$25 - \frac{10(1+3m)}{1+m} - 2 < 0; \quad 23 - \frac{10(1+3m)}{1+m} < 0; \quad \frac{23+23m-10-30m}{1+m} < 0$$

$$\frac{13-7m}{1+m} < 0; \quad \frac{7m-13}{m+1} > 0$$



$$m \in \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

93. If $\cot(\cot^{-1} f(x)) = k$, has exactly two distinct real solutions then the integral value of 'k' can be :

यदि $\cot(\cot^{-1} f(x)) = k$, के दो वास्तविक और भिन्न-भिन्न हल है तब 'k' का पूर्णांक मान होगा :

- (A) 0 (B) -1 (C) 1 (D) 5

Ans. A

Sol.

94. If the range of the function $f(x)$ lies between the roots of $g(x) = 0$ then the number of integral values of m equals :

यदि $f(x)$ का परिसर $g(x) = 0$ के मूलों के मध्य विद्यमान है तब m के पूर्णांक मानों की संख्या है :

- (A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol.

Paragraph for Question nos.95 to 97

Consider the function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

माना फलन $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ परिभाषित है। $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

95. Which of the following is true?

निम्न में से कौन सत्य है?

- (A) $(2+a)^2 f''(1) + (2-a)^2 f''(-1) = 0$ (B) $(2-a)^2 f''(1) - (2+a)^2 f''(-1) = 0$
 (C) $f'(1) f'(-1) = (2-a)^2$ (D) $f'(1) f'(-1) = -(2+a)^2$

Ans. A

Sol. $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$

$$f(x) = \frac{2a(x^2 - 1)}{(x^2 + ax + 1)^2} \text{ and } f'(x) = \frac{4a(-x^3 + 3x + a)}{(x^2 + ax + 1)^3}$$

$$f'(1) = \frac{4a}{(a+2)^2} \text{ and } f'(-1) = \frac{-4a}{(a-2)^2}$$

$$\therefore (a+2)^2 f'(1) + (2-a)^2 f'(-1) = 0$$

96. Which of the following is true?

- (A) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$
 (B) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$
 (C) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$.
 (D) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$.

निम्न में से कौन सत्य है ?

- (A) $f(x)$ अन्तराल $(-1, 1)$ में घटसमान है तथा स्थानीय निम्निष्ठ $x = 1$ पर है
 (B) $f(x)$ अन्तराल $(-1, 1)$ में वर्द्धमान है तथा स्थानीय उच्चिष्ठ $x = 1$ पर है
 (C) $f(x)$ अन्तराल $(-1, 1)$ में वर्द्धमान है लेकिन $x = 1$ कोई स्थानीय उच्चिष्ठ अथवा निम्निष्ठ नहीं है
 (D) $f(x)$ अन्तराल $(-1, 1)$ में घटसमान है लेकिन $x = 1$ पर कोई स्थानीय उच्चिष्ठ अथवा निम्निष्ठ नहीं है

Ans. A

Sol. As when $x \in (-1, 1)$, $f(x) < 0$

so $f(x)$ is decreasing on $(-1, 1)$ at $x = 1$

$$f'(1) = \frac{4a}{(a+2)^2} > 0 \quad \text{so local minima at } x = 1.$$

97. Let $g(x) = \int_0^x \frac{f'(t)}{1+t^2} dt$

Which of the following is true?

[JEE-2008]

- (A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$
 (B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
 (C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
 (D) $g'(x)$ does not change sign on $(-\infty, \infty)$

माना $g(x) = \int_0^x \frac{f'(t)}{1+t^2} dt$ निम्न में से कौन सत्य है?

[JEE-2008]

- (A) $g'(x)$, $(-\infty, 0)$ में धनात्मक है तथा $(0, \infty)$ में ऋणात्मक है
 (B) $g'(x)$, $(-\infty, 0)$ में ऋणात्मक है तथा $(0, \infty)$ में धनात्मक है

- (C) $g'(x)$ दोनो अन्तरालो $(-\infty, 0)$ तथा $(0, \infty)$ में अपने चिन्ह बदलता है
 (D) $g'(x)$ कही भी $(-\infty, \infty)$ में अपना चिन्ह नहीं बदलता है

Ans. B

Sol. $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$

$$g'(x) = \frac{f'(e^x)}{1+e^{2x}} e^x = \frac{2a(e^{2x}-1)e^x}{(e^{2x}+ae^x+1)^2(1+e^{2x})}$$

$$g'(x) > 0 \quad \text{when } x > 0$$

$$g'(x) < 0 \quad \text{when } x < 0$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

98. The local maximum value of $x(1-x)^2$, $0 \leq x \leq 2$ is :

$x(1-x)^2$, $0 \leq x \leq 2$ का स्थानीय उच्चिष्ठ मान है :

- (A) 2 (B) 4/27 (C) 5 (D) 0

Ans. AB

Sol. Let $f(x) = x(1-x)^2$

$$f'(x) = 1(1-x)^2 + x \times 2(1-x) \times -1$$

$$= (1-x)^2 - 2x(1-x)$$

$$f''(x) = -2(1-x) - 2[1-x+x \times -1]$$

$$= -2 + 2x - 2[1-x-x]$$

$$= -2 + 2x - 2 + 4x$$

$$= 6x - 4$$

For critical points, $F'(x) = 0$

$$\Rightarrow (1-x)^2 - 2x(1-x) = 0$$

$$\Rightarrow (1-x)(1-x-2x) = 0$$

$$\Rightarrow (1-x)(1-3x) = 0$$

$$\Rightarrow (x-1)(3x-1) = 0$$

$$x = 1, \frac{1}{3}$$

$$f'(1) = 6(1) - 4 = 6 - 4 = 2 > 0 \text{ minimum}$$

$$f''\left(\frac{1}{3}\right) = 6 \times \frac{1}{3} - 4 = 2 - 4 = -2 < 0 \text{ minimum}$$

$$f\left(\frac{1}{3}\right) = \frac{1}{3} \left(1 - \frac{1}{3}\right)^2 = \frac{1}{3} \times \frac{4}{9} = \frac{4}{27}$$

99. The function $f(x) = \int_0^x \sqrt{1-t^4} dt$ is such that :

(A) it is defined on the interval $[-1, 1]$

(B) it is an increasing function

(C) it is an odd function

(D) the point $(0, 0)$ is the point of inflection

फलन $f(x) = \int_0^x \sqrt{1-t^4} dt$ के लिये :

(A) $[-1, 1]$ में परिभाषित है

(B) वर्धमान फलन है

(C) एक विषम फलन है

(D) $(0, 0)$ फलन का नति परिवर्तन बिन्दु है

Ans. ABCD

Sol. $f'(x) = \sqrt{1-x^4} > 0$ in $(-1, 1) \Rightarrow f$ is $\uparrow$

$$\begin{aligned} \text{Now } f(x) + f(-x) &= \int_0^x \sqrt{1-t^4} dt + \int_0^{-x} \sqrt{1-t^4} dt \Rightarrow \int_0^x \sqrt{1-t^4} dt + \left(- \int_0^y \sqrt{1-y^4} dy \right) (t = -y) \\ &= 0 \Rightarrow f(x) \text{ is odd} \end{aligned}$$

again $f''(x) = \frac{-4x^3}{2\sqrt{1-x^4}}$ which vanishes at $x = 0$ and changes sign $\Rightarrow (0, 0)$ is inflection since f is well defined in $[-1, 1] \Rightarrow A, B, C, D$

100. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is/are correct :

(A) $f(x)$ has maxima but no minima

(B) $f(x)$ increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$

(C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$

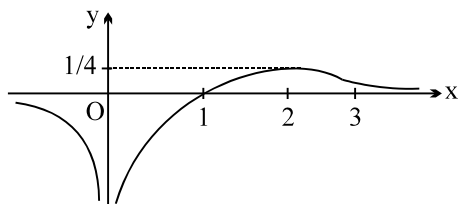
(D) $x = 3$ is the point of inflection

माना $f(x) = \frac{x-1}{x^2}$ तो निम्न में से सत्य है :

- (A) $f(x)$ का उच्चिष्ठ है लेकिन निम्निष्ठ नहीं
 (B) $f(x)$; $(0, 2)$ में वर्धमान है और $(-\infty, 0) \cup (2, \infty)$ में
 (C) $f(x)$; $(-\infty, 0) \cup (0, 3)$ में नीचे की ओर अवतल है
 (D) $x = 3$ नति परिवर्तन बिन्दु है

Ans. ABCD

Sol. $f'(x) = \frac{2-x}{x^3}$ and $f''(x) = \frac{x-3}{x^4}$. Now interpret



101. Which of the following functions fail to satisfy the condition of Rolle's theorem on the interval $[-1, 1]$, where $[x]$ denotes the greatest integer less or equal to x and $\{x\}$ denotes the fractional part of x respectively :

निम्न में से कौनसे फलन अन्तराल $[-1, 1]$ में, रोले प्रमेय की शर्तों को संतुष्ट नहीं करते हैं, जहाँ $[x]$ महत्तम पूर्णांक फलन है एवं $\{x\}$ भिन्नात्मक भाग फलन है :

- (A) $f(x) = |x| [x]$ (B) $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
 (C) $f(x) = \{x\} + \{-x\}$ (D) $f(x) = |x| - |\sin x|$

Ans. ABC

Sol. (A) $f(x) = \begin{cases} 1, & x = 1 \\ 0, & 0 \leq x < 1 \\ x, & -1 \leq x < 0 \end{cases} \Rightarrow$ not differentiable at $x = 0$ in $(-1, 1)$

(B) $f(0) = 0$ and $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \Rightarrow$ not continuous at $x = 0$

(C) $f(x) = \begin{cases} 1, & x \notin I \\ 0, & x \in I \end{cases} \Rightarrow$ not continuous at $x = 0$ [12th, 29-08-2010, P-2]

(D) $f(x) = \begin{cases} x - \sin x & 0 \leq x \leq 1 \\ -x + \sin x & -1 \leq x < 0 \end{cases} \Rightarrow$ continuous & differentiable at $x = 0$

102. Which of the following is/are correct?

- (A) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $\tan x = 1$
 (B) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $\tan x = -1$
 (C) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $e^x \sin x = 1$
 (D) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $e^x \cos x = 1$

निम्न में से कौन-कौन से कथन सत्य है?

- (A) $e^x \cos x = 1$ के किन्ही दो मूलों के मध्य, $\tan x = 1$ का कम से कम एक मूल विद्यमान होगा
 (B) $e^x \sin x = 1$ के किन्ही दो मूलों के मध्य, $\tan x = -1$ का कम से कम एक मूल विद्यमान होगा
 (C) $e^x \cos x = 1$ के किन्ही दो मूलों के मध्य, $e^x \sin x = 1$ का कम से कम एक मूल विद्यमान होगा
 (D) $e^x \sin x = 1$ के किन्ही दो मूलों के मध्य, $e^x \cos x = 1$ का कम से कम एक मूल विद्यमान होगा

Ans. ABC

- Sol.** (A) Let $f(x) = e^x \cos x - 1$
 $\Rightarrow f'(x) = e^x(\cos x - \sin x) = 0$
 $\Rightarrow \tan x = 1$, between two roots of $f(x)$. (Rolle's theorem)
 (B) Let $g(x) = e^x \sin x - 1$,
 $g'(x) = e^x(\sin x + \cos x) = 0$
 $\Rightarrow \tan x = -1$, between two roots of $g(x)$.
 (C) Let $h(x) = e^{-x} - \cos x$,
 $h'(x) = -e^{-x} + \sin x = 0$
 $\Rightarrow e^{-x} = \sin x$, between two roots of $h(x)$

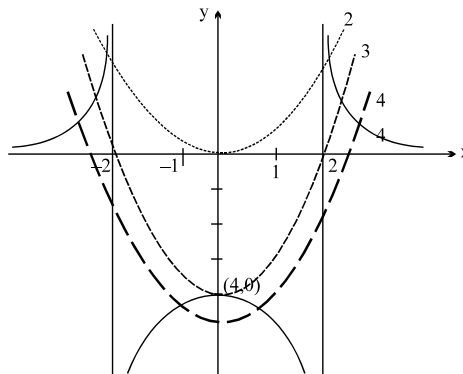
103. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. Number of possible distinct intersection(s) of these graph is :

परवलय $y = ax^2 + bx + c$; $a > 0$ और $f(x) = \frac{1}{x^2 - 4}$ के सम्भव प्रतिच्छेदन बिन्दुओं की संख्या :

- (A) 0 (B) 2 (C) 3 (D) 4

Ans. BCD

Sol. Figure is self explanatory



104. If $f(x) = \begin{cases} 3x^2 + 12x - 1 & , -1 \leq x \leq 2 \\ 37 - x & , 2 < x \leq 3 \end{cases}$ then :

(A) $f(x)$ is increasing on $[-1, 2]$

(B) $f(x)$ is continuous on $[-1, 3]$

(C) $f'(2)$ does not exist

(D) $f(x)$ has the maximum value at $x = 2$

यदि $f(x) = \begin{cases} 3x^2 + 12x - 1 & , -1 \leq x \leq 2 \\ 37 - x & , 2 < x \leq 3 \end{cases}$ तो :

(A) $f(x)$; $[-1, 2]$ में वर्धमान है

(B) $f(x)$; $[-1, 3]$ में सतत् है

(C) $f'(2)$ विद्यमान नहीं है

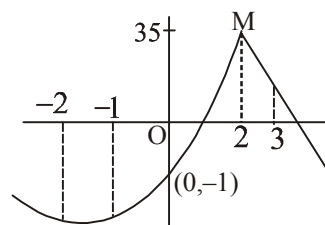
(D) $f(x)$ का $x = 2$ पर अधिकतम मान प्राप्त होगा

Ans. ABCD

Sol. Note that $f(x)$ is continuous at $x = 2$

and f is decreasing for $(2, 3)$ and increasing for $[-1, 2]$.

At $x = 2$ f has a maxima



105. Assume that inverse of the function f is denoted by g . Then which of the following statement hold good?

(A) If f is increasing then g is also increasing

(B) If f is decreasing then g is increasing

(C) The function f is injective

(D) The function g is onto

माना f का प्रतिलोम फलन g है तो निम्न में से सत्य कथन होंगे ?

(A) यदि f वर्धमान है तब g भी वर्धमान होगा

(B) यदि f ह्रसमान है तब g वर्धमान होगा

(C) फलन f एकैकी होगा

(D) फलन g आच्छादक होगा

Ans. ACD

Sol. If f and g are inverse then $(f \circ g)(x) = x$

$$f'[g(x)] g'(x) = 1$$

if f is increasing $\Rightarrow f' > 0 \Rightarrow$ sign of g' is also +ve $\Rightarrow$ (A) is correct

If f is decreasing $\Rightarrow f' < 0 \Rightarrow$ sign of g' is -ve $\Rightarrow$ (B) is false

since f has an inverse $\Rightarrow f$ is bijective $\Rightarrow f$ is injective $\Rightarrow$ (C) is correct

inverse of a bijective mapping is bijective

$\Rightarrow g$ is also bijective $\Rightarrow g$ is onto $\Rightarrow$ (D) is correct

106. If $f(x)$ is continuous and derivable $\forall x \in \mathbb{R}$ and $f'(c) = 0$ for exactly 2 real values of ' c ' then the number of real and distinct values of ' d ' for which $f(d) = 0$ can be :

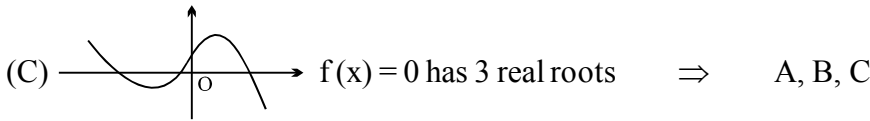
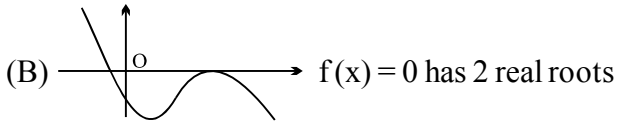
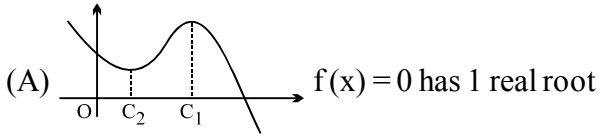
यदि $f(x)$, $\forall x \in \mathbb{R}$ में सतत् व अवकलनीय है और $f'(c) = 0$, ' c ' के दो मानों के लिये तो ' d ' के वास्तविक अलग-अलग मानों की संख्या जिसके लिये $f(d) = 0$:

(A) 1

(B) 2

(C) 3

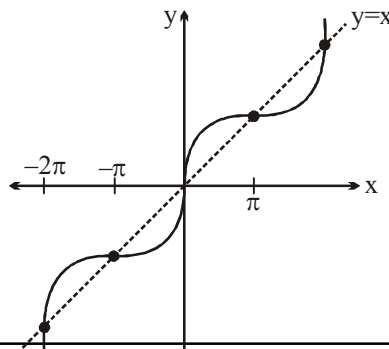
(D) 4

Ans. ABC**Sol.** If $f'(x) = 0$ has n real roots $\Rightarrow f(x) = 0$ has atmost $(n+1)$ roots107. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x + \sin x$. Which of the following is/are the correct statement(s)?

- (A) The function is strictly increasing at every point on $\mathbb{R}$ except at 'x' equal to an odd integral multiple of π where the derivative of $f(x)$ is zero and where the function f is not strictly increasing.
 (B) The function is bounded in every bounded interval but unbounded on whole real line.
 (C) The graph of the function $y = f(x)$ lies in the first and third quadrants only.
 (D) The graph of the function $y = f(x)$ cuts the line $y = x$ at infinitely many points.

 $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + \sin x$ तो निम्न में से सत्य है :

- (A) $x = (2n+1)\pi$; $n \in \mathbb{I}$ को छोड़कर, जहाँ $f'(x) = 0$, तथा एक दिष्ट वर्धमान नहीं, $f(x)$ सभी $x \in \mathbb{R}$ के लिये एकदिष्ट वर्धमान है
 (B) $f(x)$ सभी परिबद्ध अन्तरालों में परिबद्ध है लेकिन $x \in \mathbb{R}$ पर अपरिबद्ध
 (C) $y = f(x)$ का ग्राफ केवल प्रथम और तृतीय चतुर्थांश में है
 (D) $y = f(x)$ का ग्राफ $y = x$ को अनन्त बिन्दुओं पर प्रतिच्छेद करता है

Ans. BCD**Sol.** We have $f'(x) = 1 + \cos x \Rightarrow f$ is strictly increasing and has inflection point at $x = (2n+1)\pi$ Also there is no x for which $f(x)$ is not increasing. $\Rightarrow$ (A) is not correctAs f is continuous $\forall x \in \mathbb{R}$ hence bounded in every closed interval.

As f is odd hence symmetric w.r.t origin and its graphs lies in 1st and 3rd quadrant and $y = x$ cut the graph at infinitely many points.

$\Rightarrow$ B, C, D are correct

108. Equation of a line which is tangent to both the curves $y = x^2 + 1$ and $y = -x^2$ is :

उस सरल रेखा का समीकरण जो वक्रों $y = x^2 + 1$ और $y = -x^2$ दोनों की स्पर्श रेखा है :

(A) $y = \sqrt{2}x + \frac{1}{2}$ (B) $y = \sqrt{2}x - \frac{1}{2}$ (C) $y = -\sqrt{2}x + \frac{1}{2}$ (D) $y = -\sqrt{2}x - \frac{1}{2}$

Ans. AC

Sol. Let the tangent line be $y = ax + b$

The equation for its intersection with the upper parabola is

$$x^2 + 1 = ax + b$$

$$x^2 - ax + (1 - b) = 0$$

This has a double root when $a^2 - 4(1 - b) = 0$ or $a^2 + 4b = 4$

For the lower parabola

$$ax + b = -x^2$$

$$x^2 + ax + b = 0$$

This has a double root when $a^2 - 4b = 0$

subtract these two equations to get $8b = 4$ or $b = 1/4$

add them to get $2a^2 = 4$ or $a = \pm \sqrt{2}$

The tangent lines are $y = \sqrt{2}x + \frac{1}{2}$ and $y = -\sqrt{2}x + \frac{1}{2}$

Alternatively: Let the common tangent is

$$y = mx + c$$

Solving with first curve we get

$$x^2 + 1 = mx + c$$

$$x^2 - mx + 1 - c = 0$$

$$D = 0$$

$$\boxed{m^2 = 4(1 - c)} \quad \dots\dots\dots(A)$$

Solving with second curve

$$mx + c = -x^2 \Rightarrow x^2 + mx + c = 0$$

$$D = 0$$

$$\boxed{m^2 = 4c} \quad \dots\dots\dots(B)$$

From A and B

$$8c = 4, c = \frac{1}{2}$$

$$m = \pm\sqrt{2}$$

The tangent lines are $y = \sqrt{2}x + \frac{1}{2}$ and $y = -\sqrt{2}x + \frac{1}{2}$

109. Which of the following pair(s) of curves touch each other?

(A) $y^2 = 4x$ and $x^2 + y^2 = 6x - 1$

(B) $xy = 4$ and $x^2 + y^2 = 8$

(C) $y = 2x - 3$ and $y = x^3 - x + 1$

(D) $y = 6 + x - x^2$ and $y = \frac{x+2}{x-2}$

निम्न में से कौन-कौन से वक्र एक दूसरे को स्पर्श करते हैं?

(A) $y^2 = 4x$ तथा $x^2 + y^2 = 6x - 1$

(B) $xy = 4$ तथा $x^2 + y^2 = 8$

(C) $y = 2x - 3$ तथा $y = x^3 - x + 1$

(D) $y = 6 + x - x^2$ तथा $y = \frac{x+2}{x-2}$

Ans. AB

Sol. (A) $y^2 = 4x$; $x^2 + y^2 = 6x - 1$

solving, $x^2 + 4x = 6x - 1$

$x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$

touch at $(1, 2)$ and $(1, -2) \Rightarrow$ (A) is correct

(B) $xy = 4$; $x^2 + y^2 = 8$

$x^2 + y^2 = 16 \Rightarrow x^2(8 - x^2) = 16$; put $x^2 = t > 0$

$t^2 - 8t + 16 = 0 \Rightarrow (t - 4)^2 = 0$

$t = 4 \Rightarrow x^2 = 4 \Rightarrow x = 2$ or $-2 \Rightarrow (2, 2)$ or $(-2, -2)$

(C) solving the points of intersection are $(1, 1)$; $(-2, -5)$; also $\frac{dy}{dx}$ is same at $(1, 1)$.

Hence they touch at $(1, 1)$

110. The co-ordinates of a point on the parabola $2y = x^2$ which is nearest to the point $(0, 3)$ is :

परवलय $2y = x^2$ पर स्थित बिन्दु के निर्देशांक जो कि बिन्दु $(0, 3)$ से न्यूनतम दूरी पर है :

(A) $(2, 2)$

(B) $(-\sqrt{2}, 1)$

(C) $(\sqrt{2}, 1)$

(D) $(-2, 2)$

Ans. AD

Sol. Let (a, b) is the point on the parabola from which its distance from $(0, 3)$ is minimum

$\Rightarrow a^2 = 2b$

Assume d is the distance

$d^2 = a^2 + (b - 3)^2 = 2b + (b - 3)^2$

For minimum value of $d \frac{d^2}{db^2} = 0 = 2 + 2(b-3)$

$$\Rightarrow b = 2$$

$$\text{and } a^2 = 2b = 4 \quad a = \pm 2$$

Hence require points on the parabola are (2, -2) and (2, 2)

111. In the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$, at point (2, -1) :

(A) length of subtangent is $7/6$ (B) slope of tangent = $6/7$

(C) length of tangent = $\sqrt{(85)}/6$ (D) None of these

वक्र $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ के बिन्दु (2, -1) पर :

(A) अद्यः स्पर्शी की लम्बाई $7/6$ हैं (B) स्पर्श रेखा की प्रवणता = $6/7$

(C) स्पर्श रेखा की लम्बाई = $\sqrt{(85)}/6$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$

$$\frac{dx}{dt} = 2t + 3, \quad \frac{dy}{dt} = 4t - 2$$

$$\Rightarrow \frac{dy}{dx} = \left(\frac{dy}{dt} \right) \cdot \left(\frac{dt}{dx} \right) = \left(\frac{4t - 2}{2t + 3} \right)$$

The slope of tangent, $\frac{dy}{dx}$ at (2, -1) = $\left(\frac{dy}{dt} \right)_{(-2,1)}$

$$\text{when } x = 2, t^2 + 3t - 8 = 2 \Rightarrow t^2 + 3t - 10 = 0$$

$$\Rightarrow t^2 + 5t - 2t - 10 = 0 \Rightarrow t(t + 5) - 2(t + 5) = 0$$

$$\Rightarrow (t - 2)(t + 5) = 0 \Rightarrow t = 2, \text{ or } -5$$

$$\therefore \frac{dy}{dx} = \left(\frac{4(2) - 2}{2(2) + 3} \right) = \frac{6}{7}$$

When $x = 2$, $y = -1$

$$\therefore -1 = 2t^2 - 2t - 5 \Rightarrow 2t^2 - 2t - 4 = 0 \Rightarrow 2t^2 - 4t + 2t - 4 = 0$$

$$\Rightarrow 2t(t - 2) + 2(t - 2) = 0 \Rightarrow t = 2, -1$$

So the common value of $t = 2$

$$\therefore \text{Slope of tangent at } (2, -1) = \boxed{\frac{6}{7}}$$

$$\text{Length of tangent} = \sqrt{\left(\frac{y_1}{m}\right)^2 + (y_1)^2}$$

$$\text{Here, } y_1 = -1 \text{ and } m = \frac{6}{7}$$

$$\therefore \text{Length} = \sqrt{\left(\frac{7}{6}\right)^2 + (-1)^2} = \sqrt{\frac{49+36}{6}}$$

112. The function $f(x) = 2 \ln(x-2) - x^2 + 4x + 1$ increases in the intervals :

फलन $f(x) = 2 \log(x-2) - x^2 + 4x + 1$ किस अन्तराल में वर्द्धमान है :

- (A) (1, 2) (B) (2, 3) (C) $\left[\frac{5}{2}, 3\right]$ (D) (2, 4)

Ans. BC

Sol. $f'(x) > 0$ for monotonically increasing function.

$$\text{Hence } \frac{2}{x-2} - 2x + 4 > 0$$

or

$$\frac{2}{x-2} > 2x - 4$$

or

$$\frac{1}{x-2} > x - 2$$

or

$$(x-2)^2 < 1$$

$$-1 < x - 2 < 1$$

or

$$1 < x < 3$$

Hence $X \in (1, 3)$

However $x - 2 > 0$ since input of logarithmic function cannot be negative

$$(x - 2) > 0 \text{ or } x > 2$$

Hence $x \in (2, 3)$

113. If $f(x) = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$, then $f(x)$:

- (A) increases in $[0, \infty)$ (B) decreases in $[0, \infty)$
(C) neither increases nor decreases in $[0, \infty)$ (D) increases in $(-\infty, \infty)$

यदि $f(x) = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$ हो, तो फलन $f(x)$:

- (A) अन्तराल $[0, \infty)$ में वर्द्धमान है (B) अन्तराल $[0, \infty)$ में ह्रासमान है
(C) अन्तराल $[0, \infty)$ में न वर्द्धमान न ही ह्रासमान है (D) अन्तराल $(-\infty, \infty)$ में वर्द्धमान है

Ans. AD

Sol.
$$f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{x^2+1}} = 1 - \frac{1}{1+x^2} + 1 - \frac{1}{\sqrt{x^2+1}}$$
$$= \frac{x^2}{1+x^2} + \left(1 - \frac{1}{\sqrt{x^2+1}}\right) \geq 0$$

114. Let $g(x) = 2f(x/2) + f(1-x)$ and $f'(x) < 0$ in $0 \leq x \leq 1$ then $g(x)$:

- (A) decreases in $\left[0, \frac{2}{3}\right)$ (B) decreases $\left(\frac{2}{3}, 1\right]$
(C) increases in $\left[0, \frac{2}{3}\right)$ (D) increases in $\left(\frac{2}{3}, 1\right]$

यदि अन्तराल $0 \leq x \leq 1$ में $g(x) = 2f(x/2) + f(1-x)$ एवं $f'(x) < 0$ हो, तो $g(x)$ है :

- (A) $\left[0, \frac{2}{3}\right)$ में ह्रासमान (B) $\left(\frac{2}{3}, 1\right]$ में ह्रासमान
(C) $\left[0, \frac{2}{3}\right)$ में वर्द्धमान (D) $\left(\frac{2}{3}, 1\right]$ में वर्द्धमान

Ans. BC

Sol. $g(x) = 2f\left(\frac{x}{2}\right) + f(1-x)$ and

$$g'(x) = f'(x/2) - f'(1-x)$$

Now $g(x)$ is increasing if $g'(x) \geq 0$

$$f'\left(\frac{x}{2}\right) \geq f'(1-x)$$

$[\because f'(x) < 0 \text{ i.e. } f(x) \text{ is decreasing}]$

$$\Rightarrow \frac{x}{2} \leq 1-x \quad \Rightarrow \quad x \leq 2-2x$$

$$\Rightarrow 3x \leq 2 \quad \Rightarrow \quad x \leq 2/3 \quad \Rightarrow \quad 0 \leq x \leq \frac{2}{3}$$

$\Rightarrow g(x)$ increases in $0 \leq x \leq 2/3$

and $g'(x) \leq 0$ for decreasing

$$\Rightarrow f'\left(\frac{x}{2}\right) \leq f'(1-x) \quad \Rightarrow \quad \frac{x}{2} \geq 1-x$$

$$\Rightarrow x \geq 2/3$$

$$\Rightarrow 2/3 \leq x \leq 1$$

115. If $f(x) = \tan^{-1}x - (1/2) \ln x$. Then :

(A) the greatest value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\pi/6 + (1/4) \ln 3$

(B) the least value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\pi/3 - (1/4) \ln 3$

(C) $f(x)$ decreases on $(0, \infty)$

(D) $f(x)$ increases on $(-\infty, 0)$

यदि $f(x) = \tan^{-1}x - (1/2) \ln x$. तो :

(A) $f(x)$ का $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ में महत्तम मान $\pi/6 + (1/4) \ln 3$ होगा

(B) $f(x)$ का $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ में न्यूनतम मान $\pi/3 - (1/4) \ln 3$ होगा

(C) $f(x)$ अन्तराल $(0, \infty)$ में ह्रासमान है

(D) $f(x)$ अन्तराल $(-\infty, 0)$ में वर्धमान है

Ans. ABC

Sol. $f'(x) = \frac{1}{1+x^2} - \frac{1}{2x}, x > 0$

$$= \frac{-(x-1)^2}{2x(1+x^2)} \leq 0 \quad \forall x > 0.$$

$f(x)$ is decreasing $\forall x > 0$.

On $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$, greatest value is

$$f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} - \frac{1}{2} \ln\left(\frac{1}{\sqrt{3}}\right) \text{ and least value is}$$

$$f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln \sqrt{3}.$$

116. If $f(x) = \log(x-2) - \frac{1}{x}$, then :

(A) $f(x)$ is M.I. for $x \in (2, \infty)$

(B) $f(x)$ is M.I. for $x \in [-1, 2]$

(C) $f(x)$ is always concave downwards

(D) $f^{-1}(x)$ is M.I. wherever defined

यदि $f(x) = \log(x-2) - \frac{1}{x}$, तो :

(A) $x \in (2, \infty)$ के लिए $f(x)$ एकदिष्ट वर्द्धमान है

(B) $x \in [-1, 2]$ के लिए $f(x)$ एकदिष्ट वर्द्धमान है

(C) $f(x)$ हमेशा नीचे की ओर अवतल है

(D) जहाँ भी $f^{-1}(x)$ परिभाषित है, एकदिष्ट वर्द्धमान है

Ans. ACD

Sol. $f(x) = \ln(x-2) - \frac{1}{x}$

$$f'(x) = \frac{1}{x-2} + \frac{1}{x^2} = \frac{x^2 + x - 2}{x^2(x-2)} = \frac{x^2 + 2x - x - 2}{x^2(x-2)}$$

$$= \frac{(x+2)(x-1)}{x^2(x-2)}.$$

As $\ln(x-2)$ is defined when $x > 2$

$\Rightarrow f(x)$ is M.I. for $x \in (2, \infty)$

$\Rightarrow f^{-1}(x)$ is M.I. wherever defined

$$\text{Also } f'(x) = \frac{-1}{(x-2)^2} - \frac{2}{x^3} < 0$$

$\Rightarrow f(x)$ is always concave downward

117. The function $f(x) = x^{1/3}(x-1)$:

- (A) has 2 inflection points
- (B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$
- (C) is concave down in $(-1/2, 0)$
- (D) Area enclosed by the curve lying in the fourth quadrant is $9/28$

फलन $f(x) = x^{1/3}(x-1)$ के लिये :

- (A) 2 नति परिवर्तन बिन्दु है
- (B) $x > 1/4$ के लिये एकदिष्ट वर्धमान और $x < 1/4$ में एकदिष्ट ह्यसमान है
- (C) $(-1/2, 0)$ में उत्तलता नीचे की ओर होगी
- (D) चतुर्थ चतुर्थांश में परिबद्ध क्षेत्र का क्षेत्रफल $9/28$ है

Ans. ABCD

Sol. $y = x^{1/3}(x-1)$

$$\frac{dy}{dx} = \frac{4}{3}x^{1/3} - \frac{1}{3} \cdot \frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}} [4x - 1]$$

hence f is $\uparrow$ for $x > \frac{1}{4}$
 and $f \downarrow$ for $x < \frac{1}{4}$] $x^{2/3}$ is always positive and $x = 1/4$ the curves has a local minima

now $f'(x) = \frac{4}{3}x^{1/3} - \frac{1}{3} \cdot x^{-2/3}$ (non existent at $x=0$, vertical tangent)

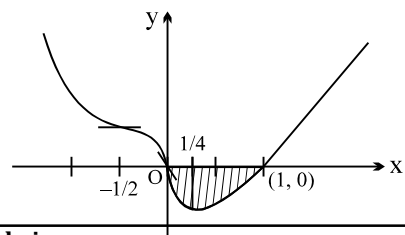
$$f''(x) = \frac{4}{9} \cdot \frac{1}{x^{2/3}} + \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{x^{5/3}}$$

$$= \frac{2}{9x^{2/3}} \left[2 + \frac{1}{x} \right] = \frac{2}{9x^{2/3}} \left[\frac{2x+1}{x} \right]$$

$\therefore f''(x) = 0$ at $x = -\frac{1}{2}$ (inflection point)

graph of $f(x)$ is as

$$A = \int_0^1 (x^{4/3} - x^{1/3}) dx = \left[\frac{3}{7}x^{7/3} - \frac{3}{4}x^{4/3} \right]_0^1$$



$$= \left| \frac{3}{7} - \frac{3}{4} \right| = 3 \left| \frac{4-7}{28} \right| = \frac{9}{28} \Rightarrow (D)$$

118. $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true?

- (A) $f(x)$ has a relative minimum on $(-3, 4)$
 (B) $f(x)$ has a minimum on $[-3, 4]$
 (C) $f(x)$ is concave upwards on $[-3, 4]$
 (D) if $f(3) = f(4)$ then $f(x)$ has a critical point on $[-3, 4]$

यदि $f''(x) > 0; \forall x \in [-3, 4]$, तो निम्न में से कौनसा/से सदैव सत्य होंगे?

- (A) $f(x); (-3, 4)$ में सापेक्षतः निम्निष्ठ है (B) $f(x); [-3, 4]$ में निम्निष्ठ देगा
 (C) $f(x); [-3, 4]$ में ऊपर की ओर अवतल है (D) यदि $f(3) = f(4)$ तो $[-3, 4]$ में $f(x)$ का क्रान्तिक बिन्दु होगा

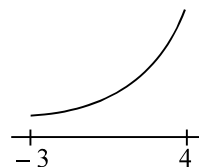
Ans. BCD

- Sol.** (A) $f(x)$ has no relative minimum on $(-3, 4)$
 (B) $f(x)$ is continuous function on $[-3, 4]$
 $\Rightarrow f(x)$ has min. and max. on $[-3, 4]$ by IVT
 (C) $f''(x) > 0 \Rightarrow f(x)$ is concave upwards on $[-3, 4]$
 (D) $f(3) = f(4)$

By Rolle's theorem

$$\exists c \in (3, 4), \text{ where } f'(c) = 0$$

$$\Rightarrow \exists \text{ critical point on } [-3, 4]$$



119. The function $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$

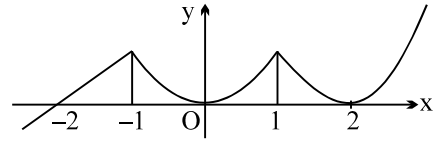
- (A) is continuous for all $x \in \mathbb{R}$ (B) is continuous but not differentiable $\forall x \in \mathbb{R}$
 (C) is such that $f'(x)$ change its sign exactly twice (D) has two local maxima and two local minima

यदि फलन $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$

- (A) $\forall x \in \mathbb{R}$ सतत् है (B) $\forall x \in \mathbb{R}$ सतत् है लेकिन अवकलनीय नहीं
 (C) $f'(x)$ का चिन्ह दो बार बदलता है (D) $f(x)$ के दो स्थानिय निम्निष्ठ और दो स्थानिय उच्चिष्ठ बिन्दु है

Ans. ABD

Sol. f is obvious continuous $\forall x \in \mathbb{R}$ and not derivable at -1 and 1
 $f'(x)$ changes sign 4 times at $-1, 0, 1, 2$
 local maxima at 1 and -1
 local minima at $x = 0$ and 2



120. Consider the function $f(x) = \left[\cos \left(\tan^{-1} \left(\sin (\cot^{-1} x) \right) \right) \right]^2$. Which of the following is correct?
- (A) range of f is $(0, 1)$
 (B) f is even
 (C) $f'(0) = 0$
 (D) the line $y = 1$ is asymptotes to the graph $y = f(x)$

माना $f(x) = \left[\cos \left(\tan^{-1} \left(\sin (\cot^{-1} x) \right) \right) \right]^2$. निम्न में से सत्य कथन है?

- (A) f का परिसर $(0, 1)$ (B) f सम है
 (C) $f'(0) = 0$ (D) $y = 1$ रेखा, $y = f(x)$ की अनन्तस्पर्शी है

Ans. BCD

Sol. Domain is $x \in \mathbb{R}$

Also $f(x) = \left[\cos \left(\tan^{-1} (\sin \theta) \right) \right]^2$ where $\cot \theta = x$

$$= \left[\cos \left(\tan^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \right) \right]^2 = (\cos \phi)^2 \text{ where } \tan \phi = \frac{1}{\sqrt{1+x^2}}$$

$$= \left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}} \right)^2$$

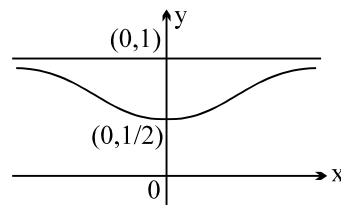
$$g(x) = \frac{1+x^2}{2+x^2} = 1 - \frac{1}{2+x^2}$$

$$\text{range is } \left[\frac{1}{2}, 1 \right); f'(x) = \frac{2x}{(2+x^2)^2}$$

$$\text{hence } f'(0) = 0$$

$$\text{also } \lim_{x \rightarrow \infty} f(x) = 1$$

hence (B), (C), (D)



121. The function $f(x) = \int_{-1}^x t(e^t - 1)(t-1)(t-2)^3(t-3)^5 dt$ has a local minimum at $x =$

फलन $f(x) = \int_{-1}^x t(e^t - 1)(t-1)(t-2)^3(t-3)^5 dt$, x के किस भाग पर स्थानिय निम्निष्ठ है :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. BD

Sol. (A) may be false since if $0 < f(x) < 1$, $f^2(x) < f(x)$ and $\int_a^b f(x) dx > \int_a^b f^2(x) dx$

(B) may be false since if $f(x) < 0$, $\frac{d}{dx} f^2(x) = 2f(x)f'(x) < 0$ when $f'(x) > 0$ and so $f^2(x)$ is decreasing while $f(x)$ is increasing

(C) may be false since a function can be negative and increasing.

(D) may be false since a function may not be differentiable at $x = c$ for which it attains its minimum.

122. Consider a real valued continuous function $f(x)$ defined on the interval $[a, b]$. Which of the following statements does not hold(s) good?

(A) If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x) dx \leq \int_a^b f^2(x) dx$

(B) If $f(x)$ is increasing on $[a, b]$, then $f^2(x)$ is increasing on $[a, b]$.

(C) If $f(x)$ is increasing on $[a, b]$, then $f(x) \geq 0$ on (a, b) .

(D) If $f(x)$ attains a minimum at $x = c$ where $a < c < b$, then $f'(c) = 0$

माना $f(x)$, $[a, b]$ में परिभाषित वास्तविक मान वाला सम फलन है तो निम्न में से असत्य कथन है ?

(A) यदि $f(x) \geq 0$, $[a, b]$ तो $\int_a^b f(x) dx \leq \int_a^b f^2(x) dx$

(B) यदि $f(x)$, $[a, b]$ में वर्धमान है तो $f^2(x)$ भी $[a, b]$ में वर्धमान होगा

(C) यदि $f(x)$, $[a, b]$ में वर्धमान है तो $f(x) \geq 0$; $x \in (a, b)$

(D) यदि $f(x)$, $x = c$ पर निम्निष्ठ है जहाँ $a < c < b$ तो $f'(c) = 0$

Ans. ABCD

Sol.

123. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ (a is a finite quantity), where $[\cdot]$ denotes greatest integer function and $f(x)$ is a non constant continuous function, then :

- (A) $\lim_{x \rightarrow a} f(x)$ is an integer (B) $\lim_{x \rightarrow a} f(x)$ need not be an integer
(C) $f(x)$ has a local minimum at $x = a$ (D) $f(x)$ has a local maximum at $x = a$

यदि $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ (a परिमित संख्या है), जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है यदि $f(x)$ एक नियतांक सतत फलन नहीं है, तो :

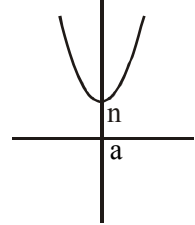
- (A) $\lim_{x \rightarrow a} f(x)$ एक पूर्णांक होगा (B) $\lim_{x \rightarrow a} f(x)$ एक पूर्णांक नहीं होगा
(C) $f(x)$; $x = a$ पर स्थानीय निम्निष्ठ है (D) $x = a$ पर $f(x)$ का स्थानीय उच्चिष्ठ मान प्राप्त होगा

Ans. AC

Sol. Clearly $\lim_{x \rightarrow a} [f(x)]$ is an integer and LHL and RHL should be same for existence of $\lim_{x \rightarrow a} f(x)$

Let $\lim_{x \rightarrow a} [f(x)] = n$ ($n \in \mathbb{I}$)

Clearly LHL and RHL both should be just greater than n as $f(x)$ is cont. at $x = a$
 $\therefore f(x)$ has local minimum at $x = a$



124. Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is :

- (A) even and is strictly increasing in $(0, \infty)$
(B) odd and is strictly decreasing in $(-\infty, \infty)$
(C) odd and is strictly increasing in $(-\infty, \infty)$
(D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

माना फलन $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ इस प्रकार है कि $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$ तो g एक :

- (A) सम तथा $(0, \infty)$ में निरन्तर वर्द्धमान फलन है
(B) विषम तथा $(-\infty, \infty)$ में निरन्तर ह्रासमान फलन है
(C) विषम तथा $(-\infty, \infty)$ में निरन्तर वर्द्धमान फलन है
(D) ना ही सम ना ही विषम है, लेकिन $(-\infty, \infty)$ में निरन्तर वर्द्धमान फलन है

Ans. C

Sol. $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$

$g(u) = \tan^{-1}(e^u) - \cot^{-1}(e^u)$

$$g(u) = \tan^{-1}(e^u) - \tan^{-1}(e^{-u}) \quad \text{odd function.}$$

$$g'(u) = \frac{e^u}{1+e^{2u}} + \frac{e^{-u}}{1+e^{-2u}} > 0 \quad \text{Strictly increasing function}$$

125. For the function $f(x) = x \cos \frac{1}{x}$, $x \geq 1$:

[JEE-2009]

(A) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$

(B) $\lim_{x \rightarrow \infty} f'(x) = 1$

(C) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$

(D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

फलन $f(x) = x \cos \frac{1}{x}$, $x \geq 1$, के लिये :

[JEE-2009]

(A) अन्तराल $[1, \infty)$ में x का कम से कम एक मान होगा कि $f(x+2) - f(x) < 2$

(B) $\lim_{x \rightarrow \infty} f'(x) = 1$

(C) अन्तराल $[1, \infty)$ में x के सभी मानों के लिये, $f(x+2) - f(x) > 2$ होगा

(D) $f'(x)$ अन्तराल $[1, \infty)$ में घटसमान है

Ans. BCD

Sol. $f(x) = x \cos \frac{1}{x}$, $x \geq 1$

$$\Rightarrow f'(x) = \frac{1}{x} \sin \frac{1}{x} + \cos \frac{1}{x}$$

$$\Rightarrow f'(x) = -\frac{1}{x^3} \cos \left(\frac{1}{x} \right)$$

$$\text{Now } \lim_{x \rightarrow \infty} f'(x) = 0 + 1 = 1 \Rightarrow \text{option 'B' is correct}$$

$$x \in [1, \infty) \Rightarrow \frac{1}{x} \in (0, 1]$$

$$\Rightarrow f'(x) < 0 \Rightarrow \text{option 'D' is correct}$$

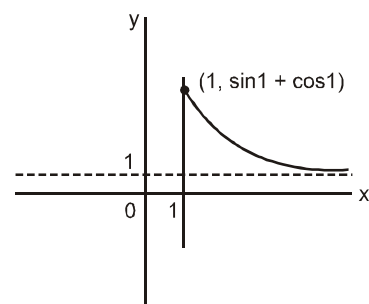
$$\text{As } f(1) = \sin 1 + \cos 1 > 1$$

$$f(x) \text{ is strictly decreasing and } \lim_{x \rightarrow \infty} f(x) = 1$$

so graph of $f(x)$ is as below

Now in $[x, x+2]$, $x \in [1, \infty)$, $f(x)$ is continuous and differentiable

$$\text{so by LMVT, } f'(x) = \frac{f(x+2) - f(x)}{2}$$



as $f'(x) > 1$ for all $x \in [1, \infty)$

$$\Rightarrow \frac{f(x+2) - f(x)}{2} > 1 \Rightarrow f(x+2) - f(x) > 2$$

for all $x \in [1, \infty)$

MATCH THE COLUMN TYPE QUESTION(S)

126. Match the following column :

Column – I

(A) $f(x) = \frac{\sin x}{e^x}, x \in [0, \pi]$

(B) $f(x) = \operatorname{sgn}((e^x - 1) \ln x), x \in \left[\frac{1}{2}, \frac{3}{2}\right]$

(C) $f(x) = (x-1)^{2/5}, x \in [0, 3]$

(D) $f(x) = \begin{cases} x \left(\frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \right), & x \in [-1, 1] - \{0\} \\ 0, & x = 0 \end{cases}$

Column – II

(P) Condition of Rolle's theorem are satisfied

(Q) Condition of LMVT are satisfied

(R) At least one condition of Rolle's theorem is not satisfied

(S) At least one condition of LMVT is not satisfied.

निम्न कॉलम को सुमेलित कीजिए :

Column – I

(A) $f(x) = \frac{\sin x}{e^x}, x \in [0, \pi]$

(B) $f(x) = \operatorname{sgn}((e^x - 1) \ln x), x \in \left[\frac{1}{2}, \frac{3}{2}\right]$

(C) $f(x) = (x-1)^{2/5}, x \in [0, 3]$

(D) $f(x) = \begin{cases} x \left(\frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \right), & x \in [-1, 1] - \{0\} \\ 0, & x = 0 \end{cases}$

Column – II

(P) रोले प्रमेय के प्रतिबन्ध सन्तुष्ट होते हैं

(Q) LMVT के प्रतिबन्ध सन्तुष्ट होते हैं

(R) रोले प्रमेय की कम से कम एक प्रतिबन्ध सन्तुष्ट नहीं होता है

(S) LMVT का कम से कम एक प्रतिबन्ध सन्तुष्ट नहीं होता है

Ans. (A) P,Q (B) R,S (C) R,S (D) R, S

Sol. (A) $f(x)$ is continuous and differentiable $f(0) = f(\pi)$
Hence condition in Rolle's theorem and LMVT are satisfied.

(B) $f(1^-) = -1, f(1) = 0, f(1^+) = 1$

$f(x)$ is not continuous at $x = 1$, belonging to $\left[\frac{1}{2}, \frac{3}{2}\right]$

Hence, at least one condition in LMVT and Rolle's theorem is not satisfied

(C) $f'(x) = \frac{2}{5}(x-1)^{-3/5}, x \neq 1$

At $x = 1$, $f(x)$ is not differentiable.

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

(D) At $x = 0$

$$\text{L.H.D.} = \lim_{x \rightarrow 0^+} \frac{x \left(\frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \right) - 0}{x - 0} = \frac{0 - 1}{0 + 1} = -1$$

R.H.D. = 1

At $x = 0$, $f(x)$ is not differentiable

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

127. In the following $[x]$ denotes the greatest integer less than or equal to x . Match the functions in Column I with the properties in Column II :

Column – I

- (A) $x |x|$
 (B) $\sqrt{|x|}$
 (C) $x + [x]$
 (D) $|x - 1| + |x + 1|$

Column – II

- (P) continuous in $(-1, 1)$
 (Q) differentiable in $(-1, 1)$
 (R) strictly increasing in $(-1, 1)$
 (S) non differentiable at least at one point in $(-1, 1)$

स्तम्भ का मिलान कीजिये : निम्न में $[x]$, x के महत्तम पूर्णांक फलन को दर्शाता है। स्तम्भ– I में फलन है तथा स्तम्भ–II उनके गुणधर्म :

Column – I

- (A) $x |x|$
 (B) $\sqrt{|x|}$
 (C) $x + [x]$
 (D) $|x - 1| + |x + 1|$

Column – II

- (P) $(-1, 1)$ में सतत्
 (Q) $(-1, 1)$ में अवकलनीय
 (R) $(-1, 1)$ में वर्द्धमान
 (S) $(-1, 1)$ में कम से कम एक बिन्दु पर अवकलनीय नहीं

Ans. (A) P,Q,R (B) P,S (C) R,S (D) P, Q

Sol.